

ImPRESS Statistical Analysis Mock-up report Report

Baseline Review for the Newly Diagnosed Glioblastoma Cohort

ImPRESS Statistical Team

23 January 2026

Contents

Introduction	2
Randomisation Status	3
Cohort Scope	3
Baseline Characteristics	3
Primary Efficacy Analyses (MRI)	10
Candidate dose-response models	10
Mean relative cerebral blood flow (PTA-ROI)	10
Mean relative cerebral blood flow (ROI2)	15
Mean relative solid stress (PTA-ROI)	20
Mean relative solid stress (ROI2)	25
Other MRI Endpoints up to Day 43	30
Mean DSC K2 leakage	30
Mean relative DSC extravasation	33
Mean relative apparent diffusion coefficient	35
Mean relative cerebral blood	37
Mean relative direction value	39
Mean relative fractional anisotropy	41
Mean relative restricted spectrum imaging	43
Mean relative transit time	45
Mean relative vessel caliber	47
Mean relative vessel size index	49
Mean relative vortex area	51
Mean tissue elasticity	53
Mean tissue relative viscosity	55
Mean tissue stiffness	57
Mean tissue viscosity	59
Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3	61
Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3	63
MRI Endpoints at Cycle 11 (Day 141)	65
Mean DSC K2 leakage	65
Mean relative DSC extravasation	67
Mean relative apparent diffusion coefficient	68
Mean relative cerebral blood	69

Mean relative cerebral blood flow (PTA-ROI)	70
Mean relative cerebral blood flow (ROI2)	71
Mean relative direction value	72
Mean relative fractional anisotropy	73
Mean relative restricted spectrum imaging	74
Mean relative solid stress (PTA-ROI)	75
Mean relative solid stress (ROI2)	76
Mean relative transit time	77
Mean relative vessel caliber	78
Mean relative vessel size index	79
Mean relative vortex area	80
Mean tissue elasticity	81
Mean tissue relative viscosity	82
Mean tissue stiffness	83
Mean tissue viscosity	84
Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3	85
Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3	86
Neurologic Performance	87
Karnofsky Performance Scale	87
ECOG Score	90
Total NANO score	93
Quality of Life Outcomes	96
EORTC QLQ-C30 score	96
EORTC QLQ-BN20 score	98
Steroid Use	100
Steroid use	100
RANO Responses	103
State of disease according to the RANO criteria	103
Best overall radiographic response	104
Time-to-Event Endpoints	105
Progression-free survival	106
6-month progression-free survival	108
Overall survival	110
Overall survival at 12 months	112
Overall survival at 24 months	114

Introduction

The ImPRESS study is a prospective investigation of losartan as an adjunctive therapy for patients with glioblastoma. According to Protocol Version 7.0 (17-Dec-2024), the trial collects comprehensive demographic, clinical, imaging, and quality-of-life information prior to treatment initiation. The statistical objectives for the interim analyses focus on characterising baseline status, neurological function, and tumour burden before randomisation to evaluate comparability across planned treatment arms.

This report summarises the currently available baseline data with emphasis on clinical status, imaging assessments, and prior treatments that may influence subsequent outcomes. The material is intended to provide early insight into data quality and cohort composition ahead of formal efficacy or safety analyses.

Randomisation Status

The present dataset is built on *sham randomisation* generated from the most recent export (see `_targets.R` workflow). No true treatment assignments have been unlocked. All findings in this document should therefore be interpreted as preliminary and descriptive only. Final efficacy comparisons will not be conducted until the database is locked and genuine randomisation codes are released.

Cohort Scope

The results herein are restricted to participants in cohort code 2 as configured in `config/cfg.yml`. Subjects from other cohorts are excluded from all summaries, figures, and tables in this draft. A combined-cohort analysis can be delivered once recruitment and cleaning are complete for all strata.

Baseline Characteristics

This section summarises baseline characteristics for the newly diagnosed glioblastoma cohort using the `adb1` ADaM dataset.

Demographics and Baseline characteristics

	all obs
Age (years)	
n	54
Mean (SD)	52.6 (10.2)
Median	55.0
Min - Max	26.0 - 69.0
Sex	
n	54
Female	29 (53.7%)
Male	25 (46.3%)
Cohort	
n	54
Newly diagnosed glioblastoma	54 (100%)
Country	
n	54
NOR	54 (100%)
Childbearing potential?	
n	29
No	13 (44.8%)
Yes	16 (55.2%)
Diagnosis	
n	54

Demographics and Baseline characteristics

	all obs
Newly diagnosed glioblastoma	54 (100%)
Time from surgery to randomisation (days)	
n	54
Mean (SD)	27.4 (7.3)
Median	25.5
Min - Max	13.0 - 48.0
IDH1/2 mutation	
n	54
Negative	47 (87%)
Positive	7 (13%)
MGMT status	
n	54
Negative	30 (55.6%)
Positive	24 (44.4%)
Pregnancy test result	
n	13
Negative	13 (100%)
ECOG Score	
n	54
Mean (SD)	0.3 (0.5)
Median	0.0
Min - Max	0.0 - 2.0
ECOG Score (categorical)	
n	54
ECOG 0	40 (74.1%)
ECOG 1	13 (24.1%)
ECOG 2	1 (1.9%)
Karnofsky Performance Scale	
n	54
Mean (SD)	95.0 (8.0)
Median	100.0
Min - Max	60.0 - 100.0
Karnofsky Performance Scale (categorical)	
n	54

Demographics and Baseline characteristics

	all obs
KPS 100	34 (63%)
KPS 60	1 (1.9%)
KPS 80	4 (7.4%)
KPS 90	15 (27.8%)
Total NANO score	
n	54
Mean (SD)	1.0 (1.3)
Median	0.0
Min - Max	0.0 - 4.0
EORTC QLQ-C30 score	
n	53
Mean (SD)	56.3 (8.6)
Median	54.0
Min - Max	41.0 - 77.0
EORTC QLQ-BN20 score	
n	53
Mean (SD)	30.8 (6.6)
Median	30.0
Min - Max	20.0 - 49.0
Systolic blood pressure	
n	54
Mean (SD)	130.0 (15.4)
Median	128.5
Min - Max	106.0 - 180.0
Diastolic blood pressure	
n	54
Mean (SD)	81.9 (7.1)
Median	82.0
Min - Max	67.0 - 97.0
Pulse Rate	
n	54
Mean (SD)	80.7 (15.7)
Median	78.0
Min - Max	54.0 - 114.0

Demographics and Baseline characteristics

	all obs
Weight (kg)	
n	54
Mean (SD)	78.8 (13.2)
Median	78.3
Min - Max	48.2 - 116.0
BMI	
n	51
Mean (SD)	25.0 (4.2)
Median	23.9
Min - Max	18.6 - 39.9
Height (cm)	
n	51
Mean (SD)	176.5 (8.8)
Median	175.0
Min - Max	158.0 - 194.0
Has the patient received chemotherapy	
n	54
No	54 (100%)
Has the patient received immunotherapy	
n	54
No	54 (100%)
Has the patient received radiation therapy to the brain?	
n	54
No	54 (100%)
CNS surgery	
n	54
No	3 (5.6%)
Yes	51 (94.4%)
Steroids at time of MRI	
n	54
No	42 (77.8%)
Yes	12 (22.2%)
Mean relative cerebral blood flow (PTA-ROI)	
n	54

Demographics and Baseline characteristics

	all obs
Mean (SD)	1.5 (0.4)
Median	1.4
Min - Max	0.7 - 2.4
Mean relative cerebral blood flow (ROI2)	
n	47
Mean (SD)	1.1 (0.5)
Median	1.0
Min - Max	0.2 - 2.2
Mean relative solid stress (PTA-ROI)	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1
Mean relative solid stress (ROI2)	
n	32
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.5 - 1.4
Mean relative apparent diffusion coefficient	
n	52
Mean (SD)	1.4 (0.2)
Median	1.4
Min - Max	1.0 - 2.0
Mean relative fractional anisotropy	
n	52
Mean (SD)	0.5 (0.1)
Median	0.5
Min - Max	0.3 - 1.0
Mean relative restricted spectrum imaging	
n	50
Mean (SD)	0.5 (0.2)
Median	0.5
Min - Max	0.2 - 1.1
Mean DSC K2 leakage	

Demographics and Baseline characteristics

	all obs
n	54
Mean (SD)	0.2 (1.6)
Median	0.4
Min - Max	-3.9 - 4.5
Mean relative cerebral blood	
n	54
Mean (SD)	1.7 (0.5)
Median	1.7
Min - Max	1.0 - 2.8
Mean relative transit time	
n	54
Mean (SD)	1.4 (0.3)
Median	1.4
Min - Max	0.9 - 2.3
Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3	
n	54
Mean (SD)	6.3 (6.5)
Median	4.3
Min - Max	0.0 - 26.3
Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3	
n	54
Mean (SD)	13.5 (14.3)
Median	8.5
Min - Max	0.1 - 57.4
Mean tissue elasticity	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.2
Mean tissue relative viscosity	
n	41
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.0 - 1.1

Demographics and Baseline characteristics

	all obs
Mean tissue stiffness	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1
Mean tissue viscosity	
n	41
Mean (SD)	0.7 (0.2)
Median	0.7
Min - Max	-0.0 - 1.1
Mean relative DSC extravasation	
n	28
Mean (SD)	1.2 (0.8)
Median	1.0
Min - Max	0.7 - 5.1
Mean relative direction value	
n	28
Mean (SD)	0.4 (1.4)
Median	0.3
Min - Max	-2.7 - 5.4
Mean relative vessel caliber	
n	28
Mean (SD)	1.0 (0.3)
Median	1.0
Min - Max	0.5 - 1.9
Mean relative vessel size index	
n	28
Mean (SD)	1.4 (0.6)
Median	1.2
Min - Max	0.6 - 2.6
Mean relative vortex area	
n	28
Mean (SD)	1.0 (0.1)
Median	1.0

Demographics and Baseline characteristics

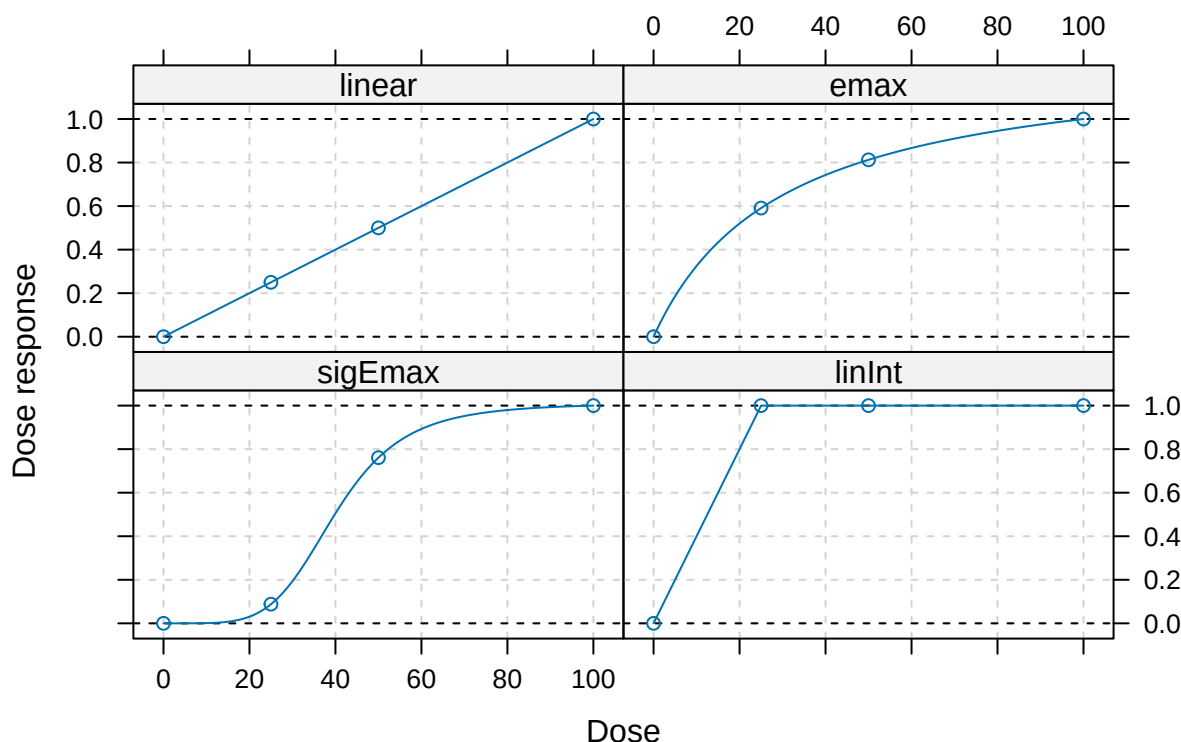
	all obs
Min - Max	0.8 - 1.1

Primary Efficacy Analyses (MRI)

The primary variable is analysed with the MCP-Mod methodology as described in the protocol and SAP. We pre-specify four candidate dose-response models (linear, Emax, sigmoid Emax, logistic) and derive optimal contrasts using the estimated variance-covariance from the mixed model. The MCP step tests for a dose-response signal across candidate models with multiplicity control; if significant, the Mod step selects the best-fitting model (by AIC) and reports ED values. Results below are grouped by MRI endpoint (primary, key secondary, sensitivity). All analyses are based on data up to and including Day 43 (Week 6).

Candidate dose-response models

Figure 1: Pre-specified candidate dose-response models



Mean relative cerebral blood flow (PTA-ROI)

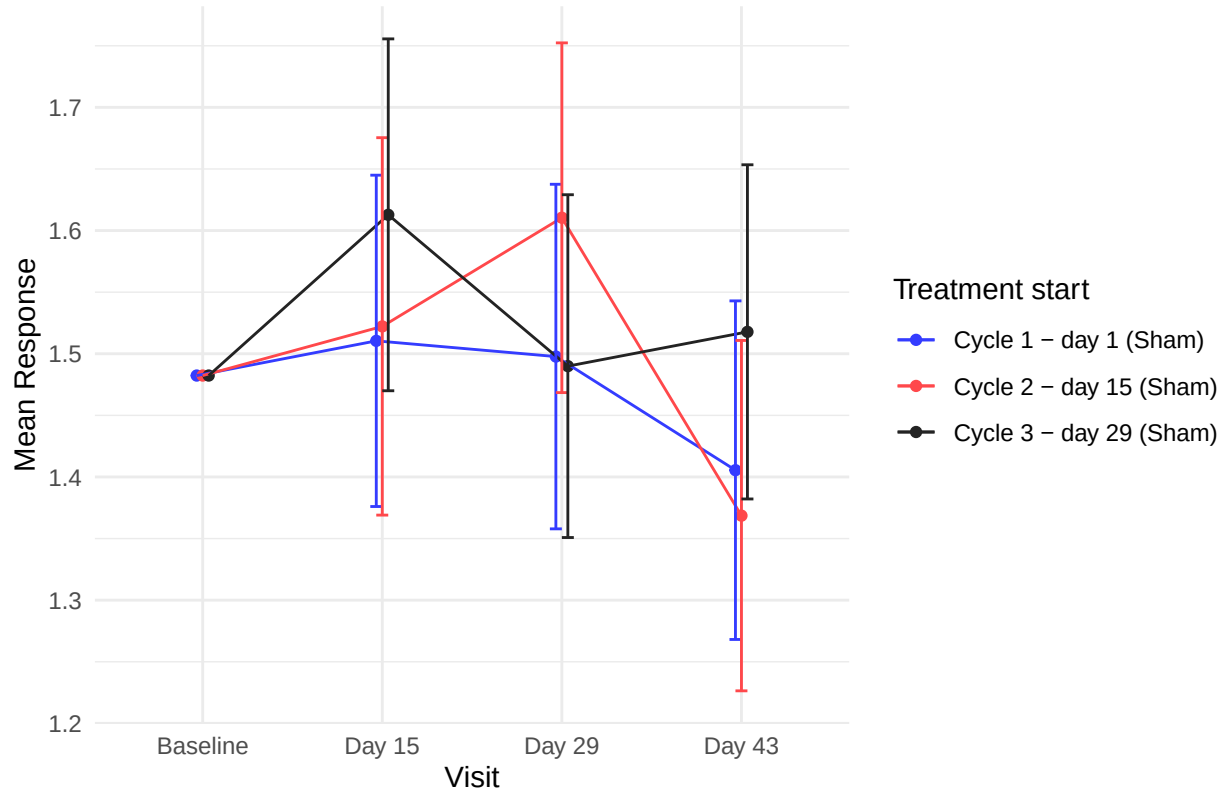
The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot

Mean Response over Time for Mean relative cerebral blood flow (PTA-ROI)



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.659	-0.825	-0.619	-0.866
25 mg	-0.171	0.057	-0.342	0.297
50 mg	0.106	0.278	0.342	0.286
100 mg	0.724	0.490	0.619	0.284

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
sigEmax	1.11	0.22
emax	1.08	0.23
linear	1.05	0.24
linInt	0.92	0.28

Fitted model parameters (AIC/ED)

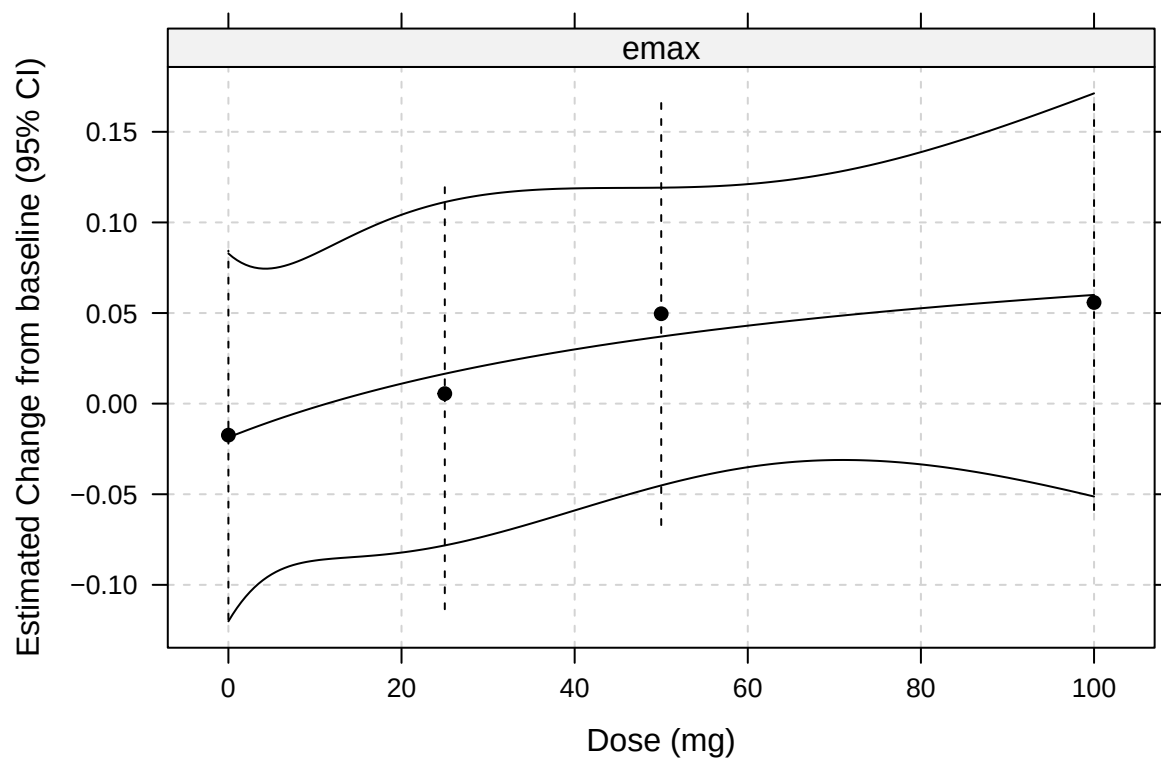
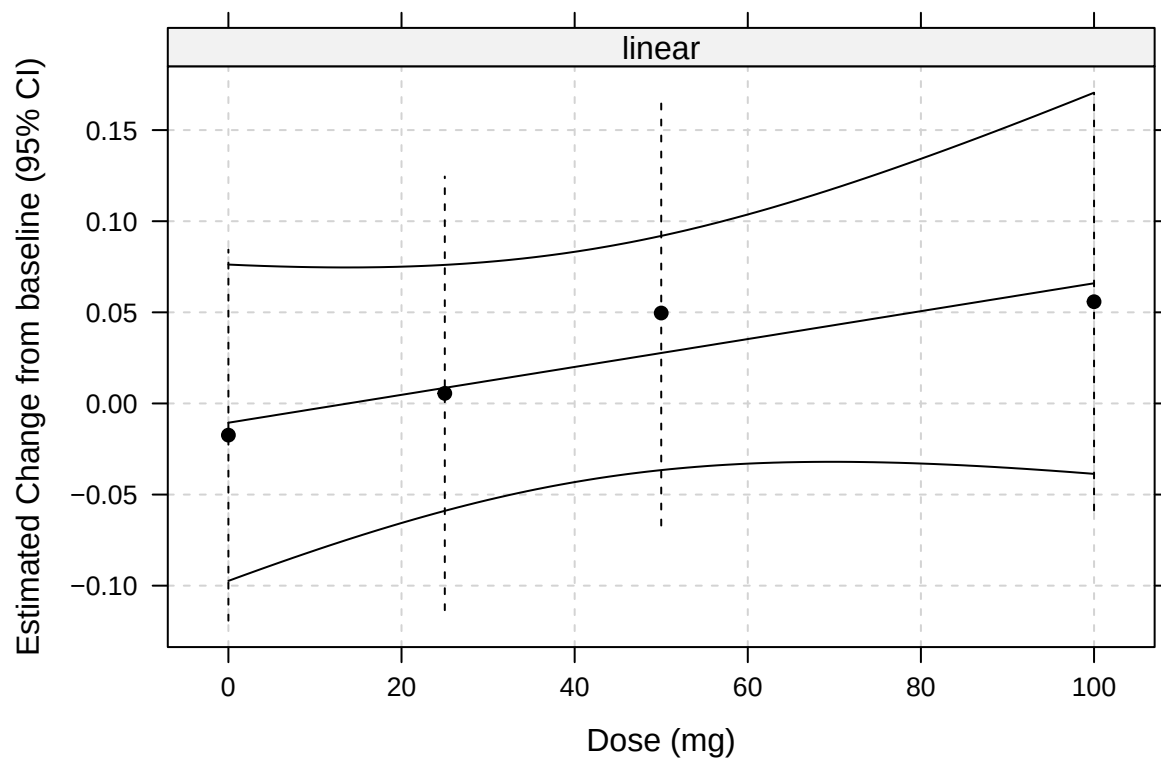
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	4.20	50.00	70.00	90.00

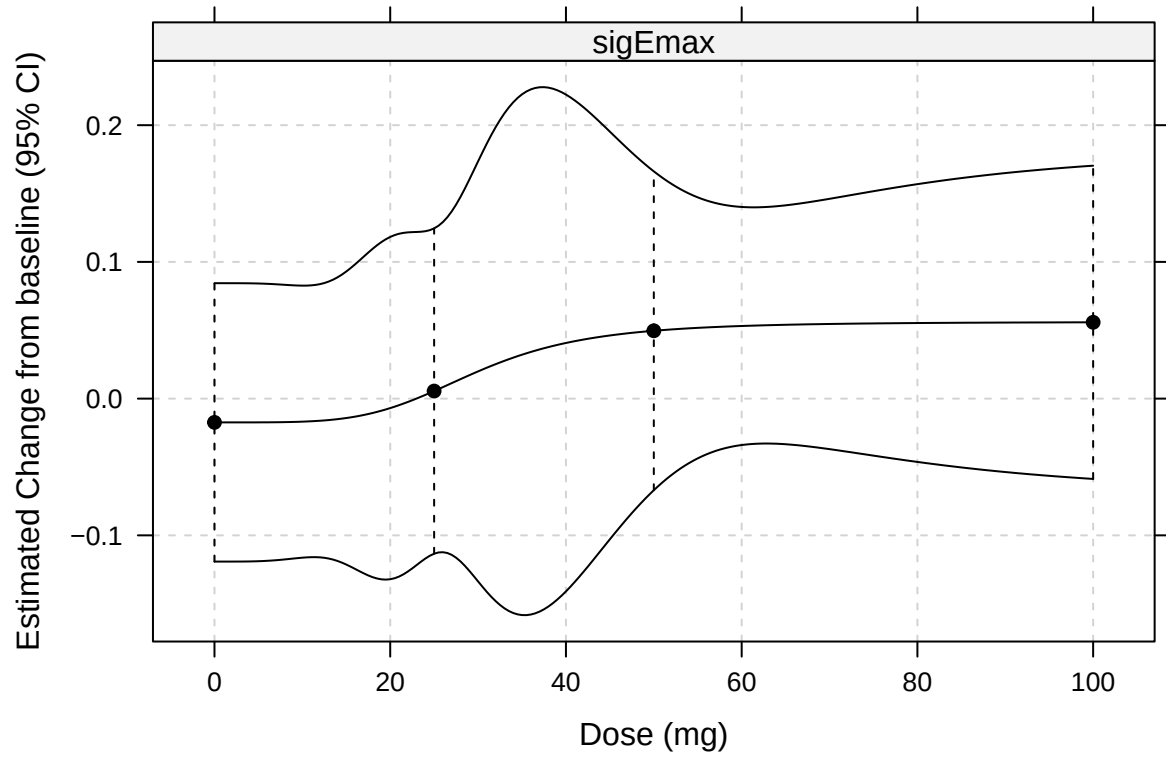
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
emax	6.09	29.21	49.05	78.78
linInt	8.00	32.76	41.06	49.37
sigEmax	8.00	29.75	35.86	48.10

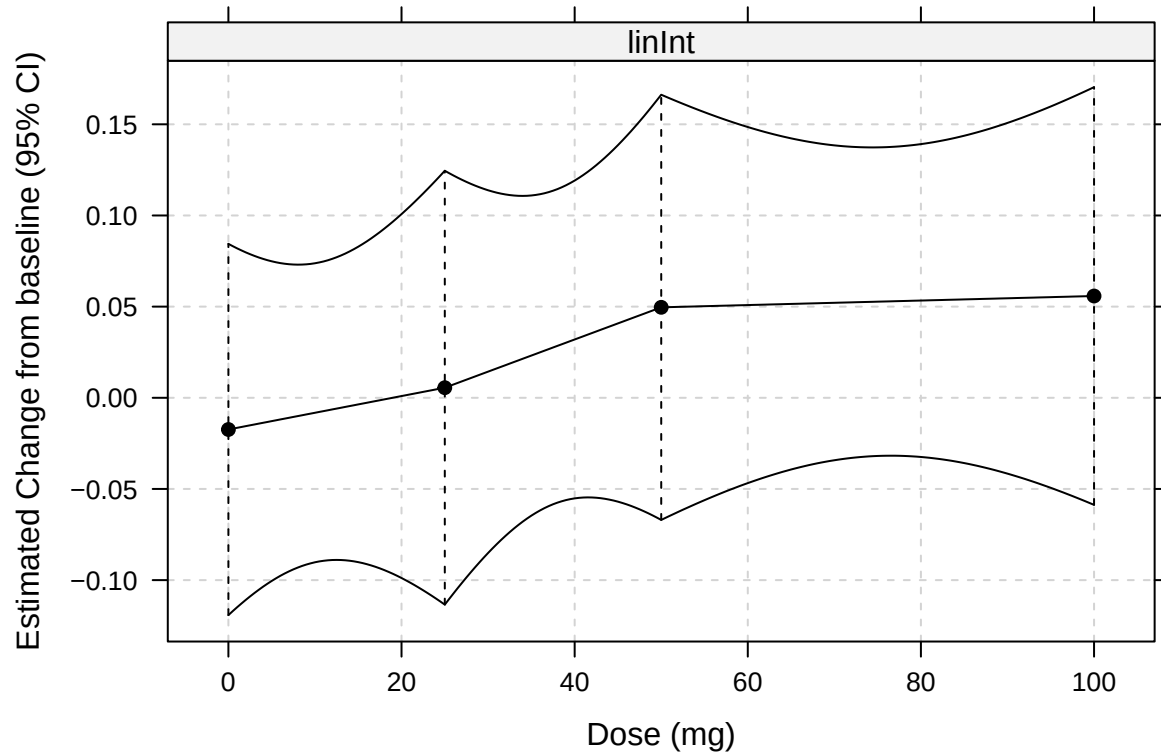
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	-0.02 (-0.12 to 0.09)
25 mg	0.01 (-0.11 to 0.13)
50 mg	0.05 (-0.07 to 0.17)
100 mg	0.06 (-0.06 to 0.17)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.05 (-0.06 to 0.17)

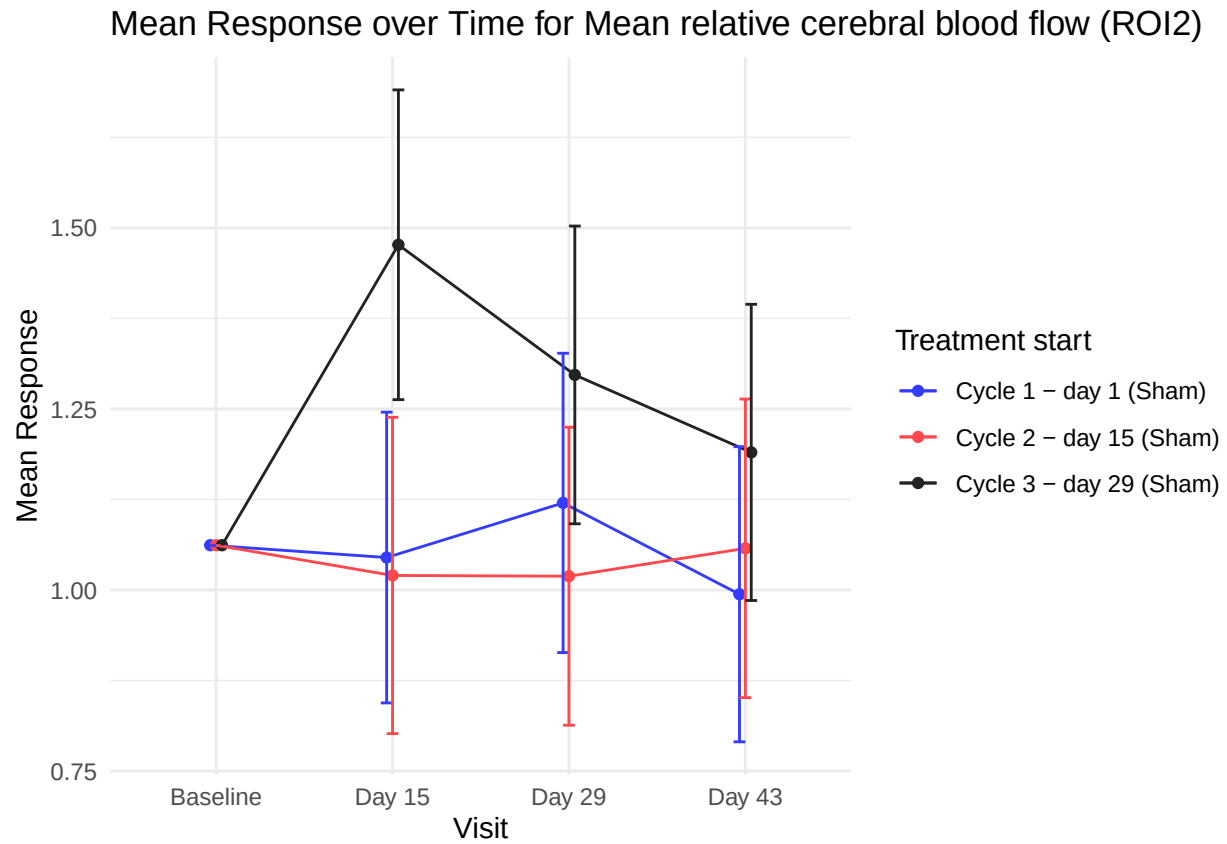
Mean relative cerebral blood flow (ROI2)

The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.710	-0.840	-0.662	-0.864
25 mg	-0.102	0.130	-0.276	0.340
50 mg	0.127	0.245	0.318	0.238
100 mg	0.685	0.466	0.620	0.286

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
sigEmax	0.09	0.61
linear	0.00	0.65
emax	-0.73	0.87
linInt	-1.23	0.95

Fitted model parameters (AIC/ED)

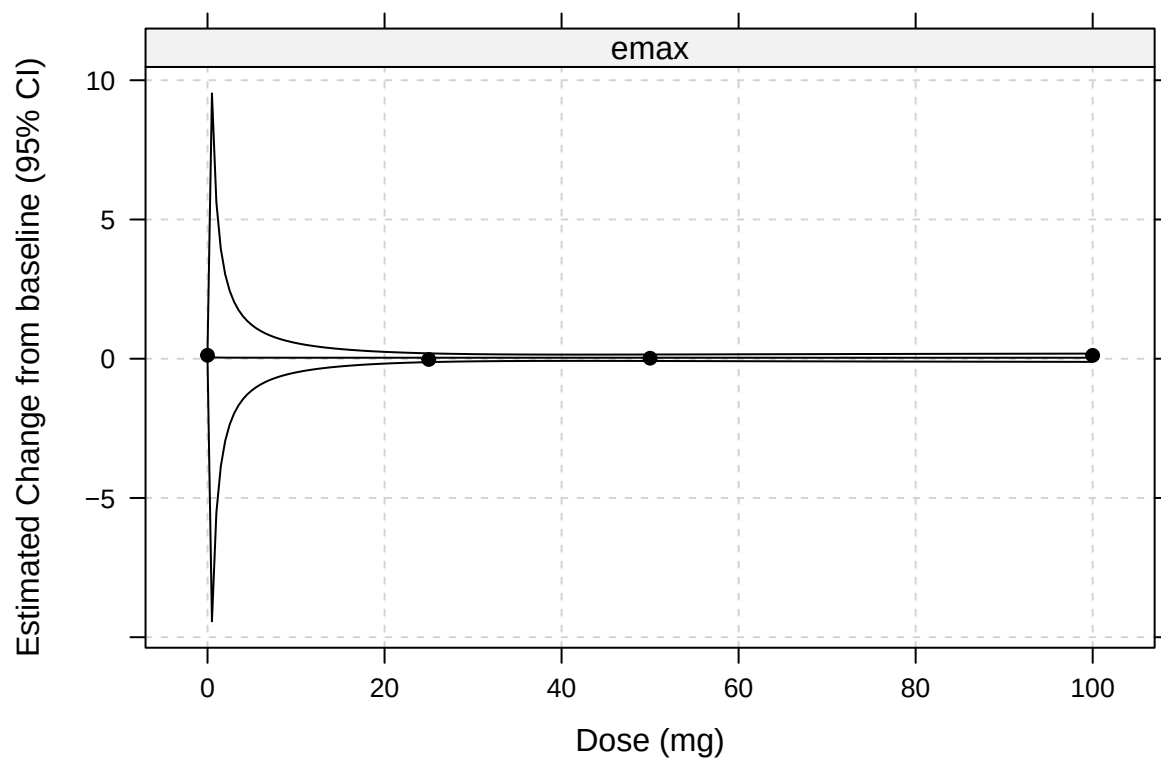
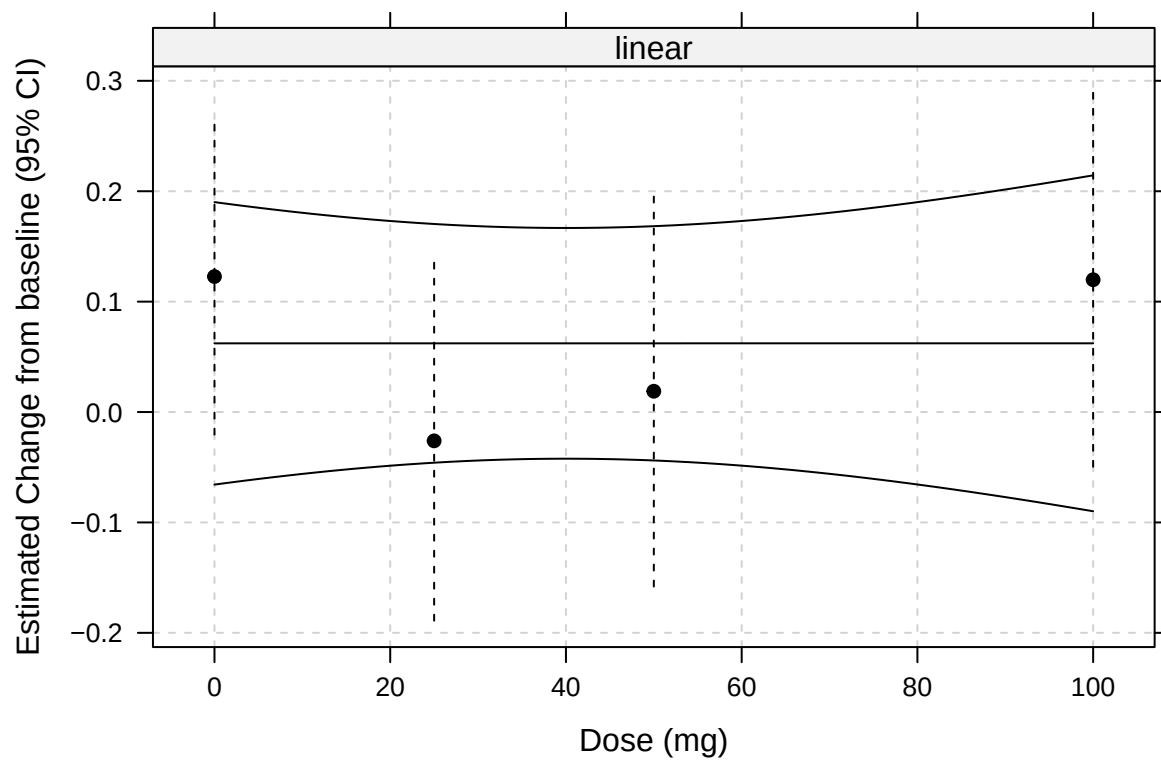
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	7.44	50.00	70.00	90.00

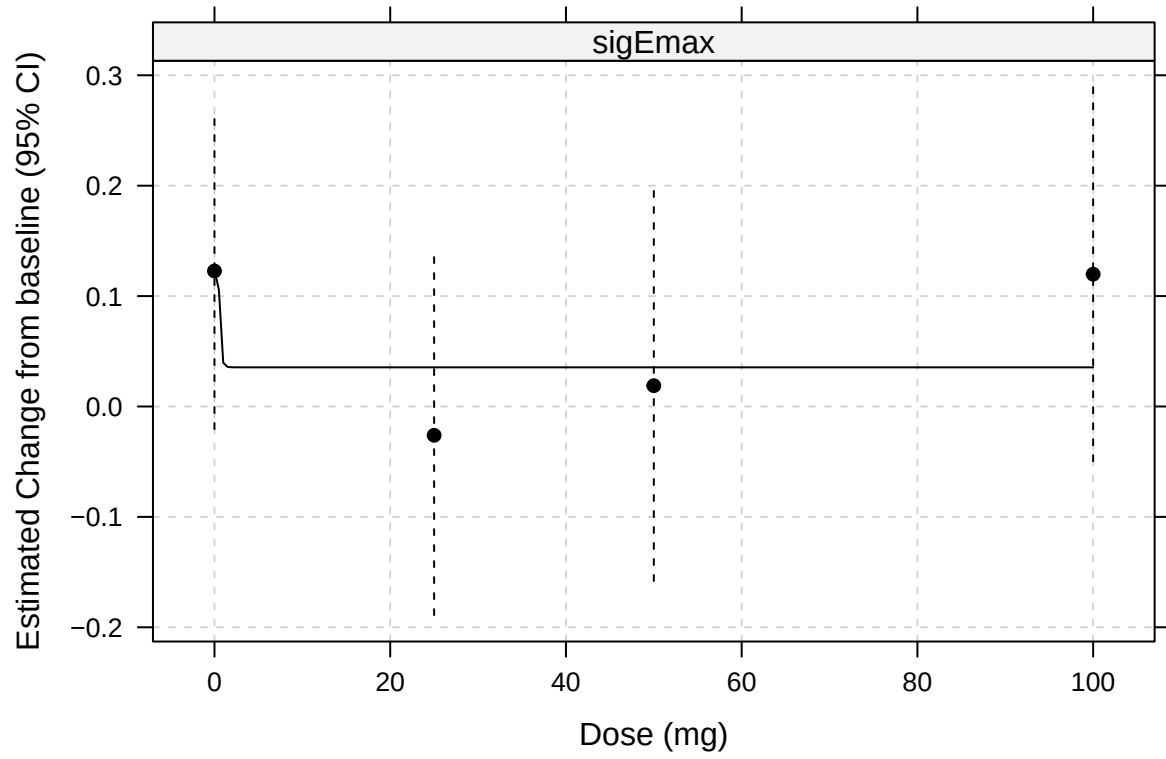
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
emax	7.92	0.10	0.23	0.89
linInt	8.00	12.50	17.50	22.50
sigEmax	9.91	0.62	0.71	0.88

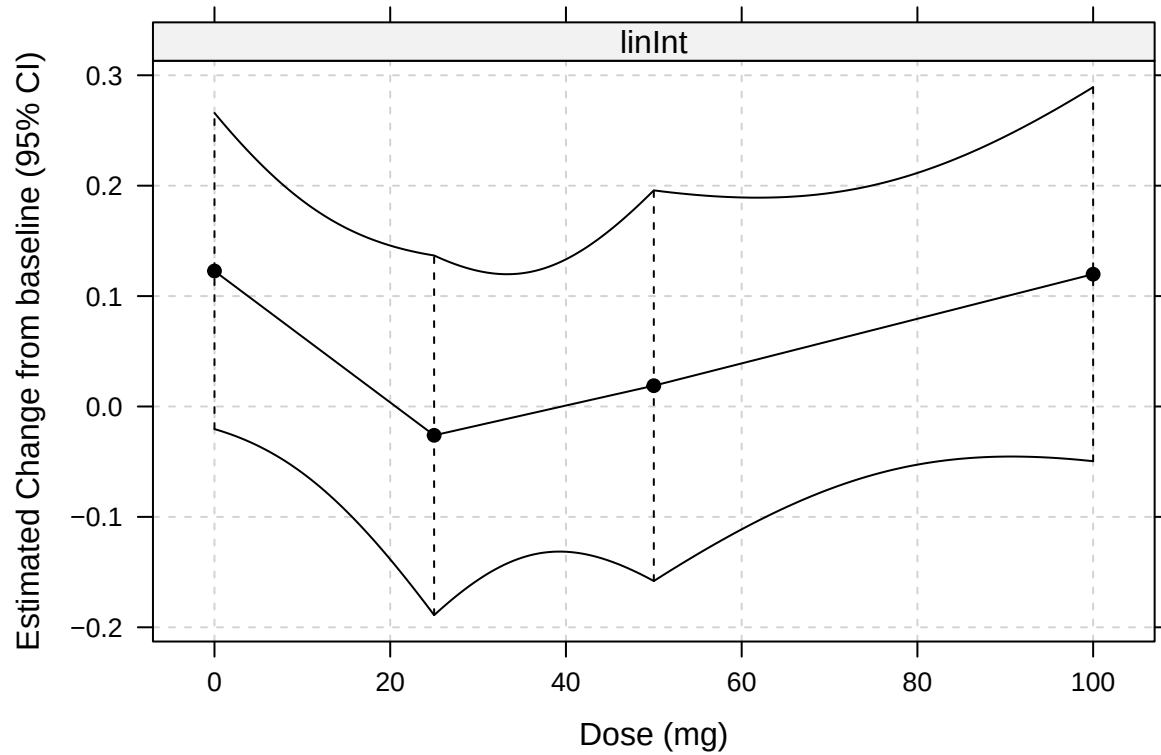
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	0.12 (-0.02 to 0.27)
25 mg	-0.03 (-0.19 to 0.14)
50 mg	0.02 (-0.16 to 0.20)
100 mg	0.12 (-0.05 to 0.29)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.09 (-0.23 to 0.05)

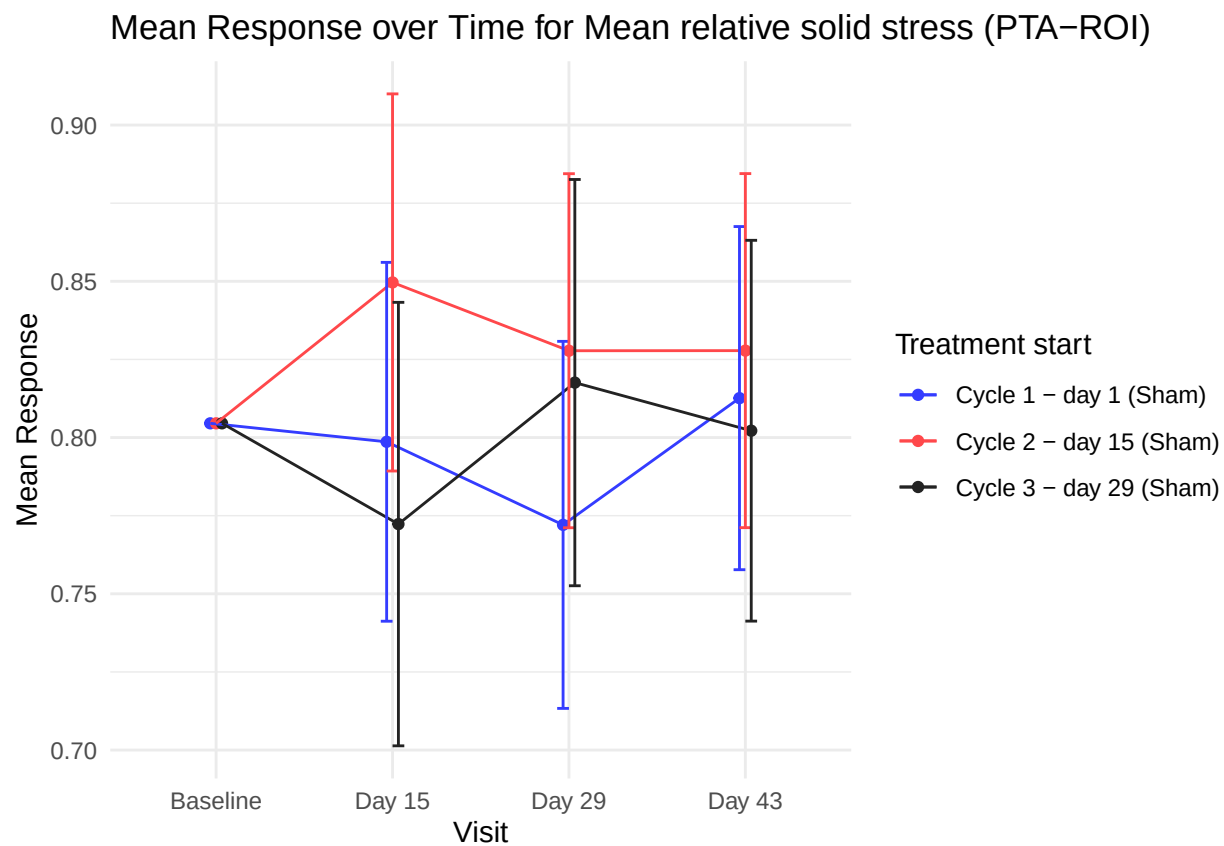
Mean relative solid stress (PTA-ROI)

The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.683	-0.817	-0.648	-0.864
25 mg	-0.138	0.051	-0.295	0.263
50 mg	0.114	0.247	0.317	0.252
100 mg	0.708	0.519	0.627	0.349

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
linear	-0.16	0.7
sigEmax	-0.43	0.79
emax	-0.70	0.86
linInt	-0.95	0.91

Fitted model parameters (AIC/ED)

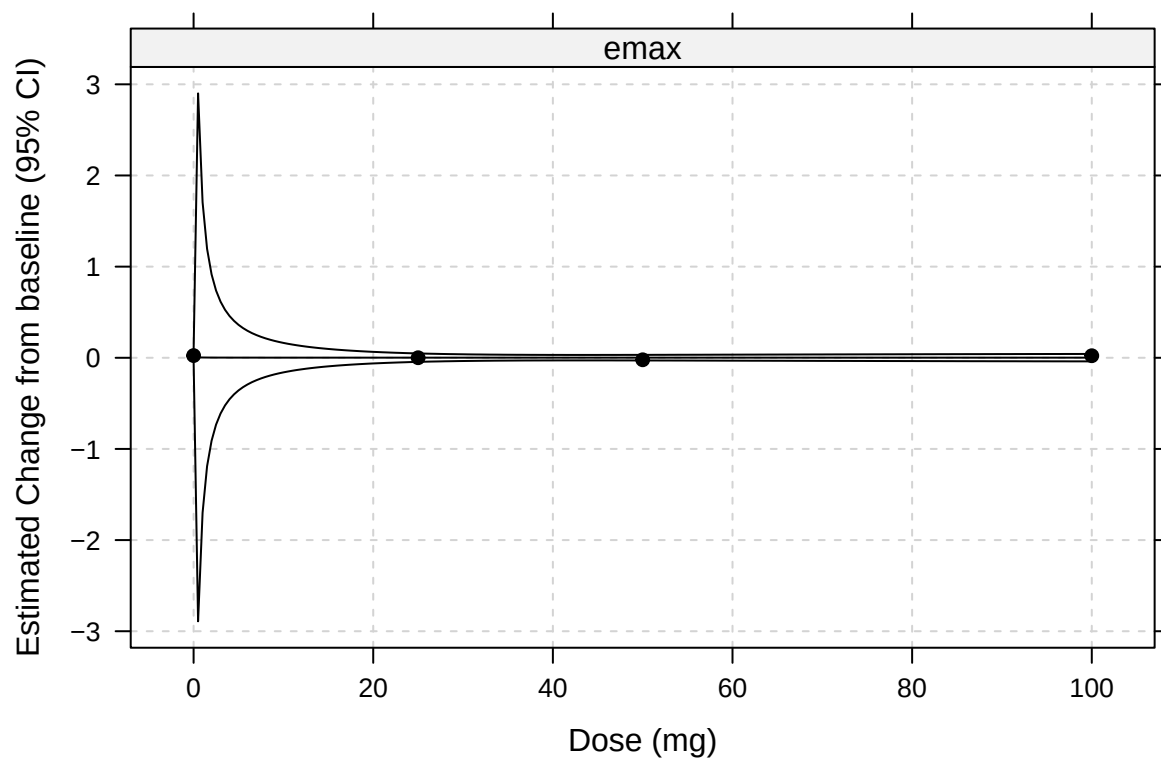
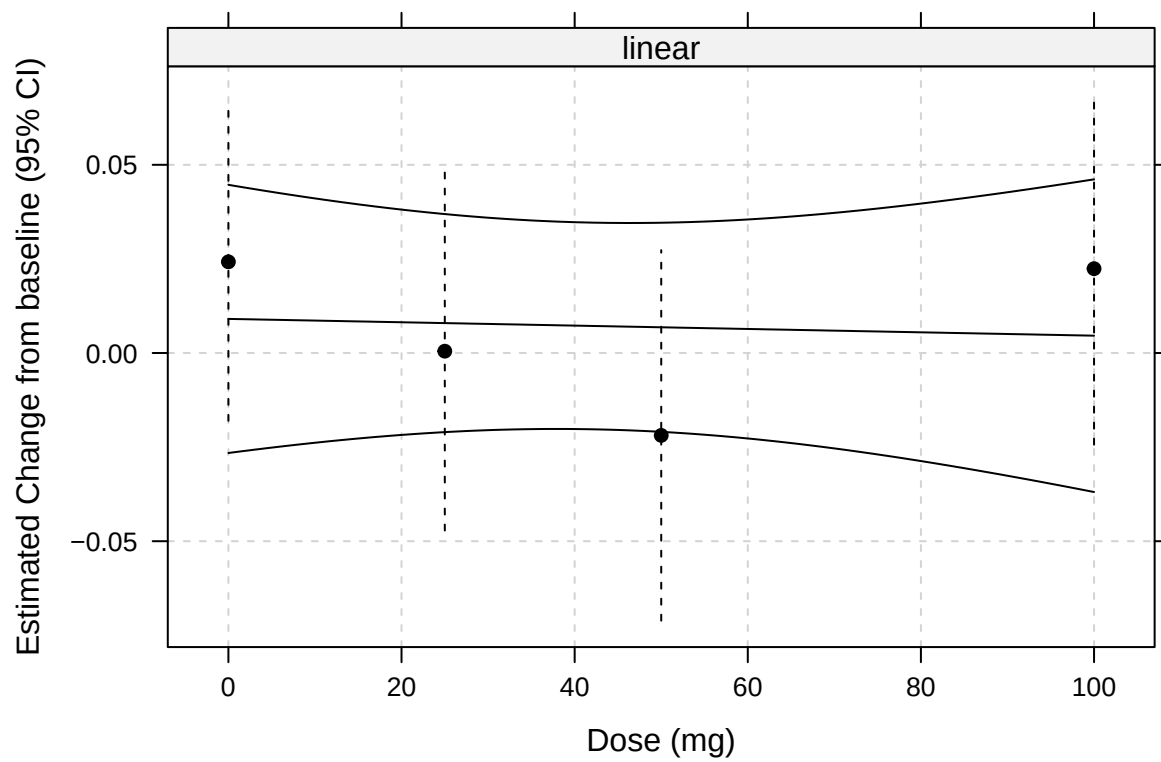
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	6.68	50.00	70.00	90.00

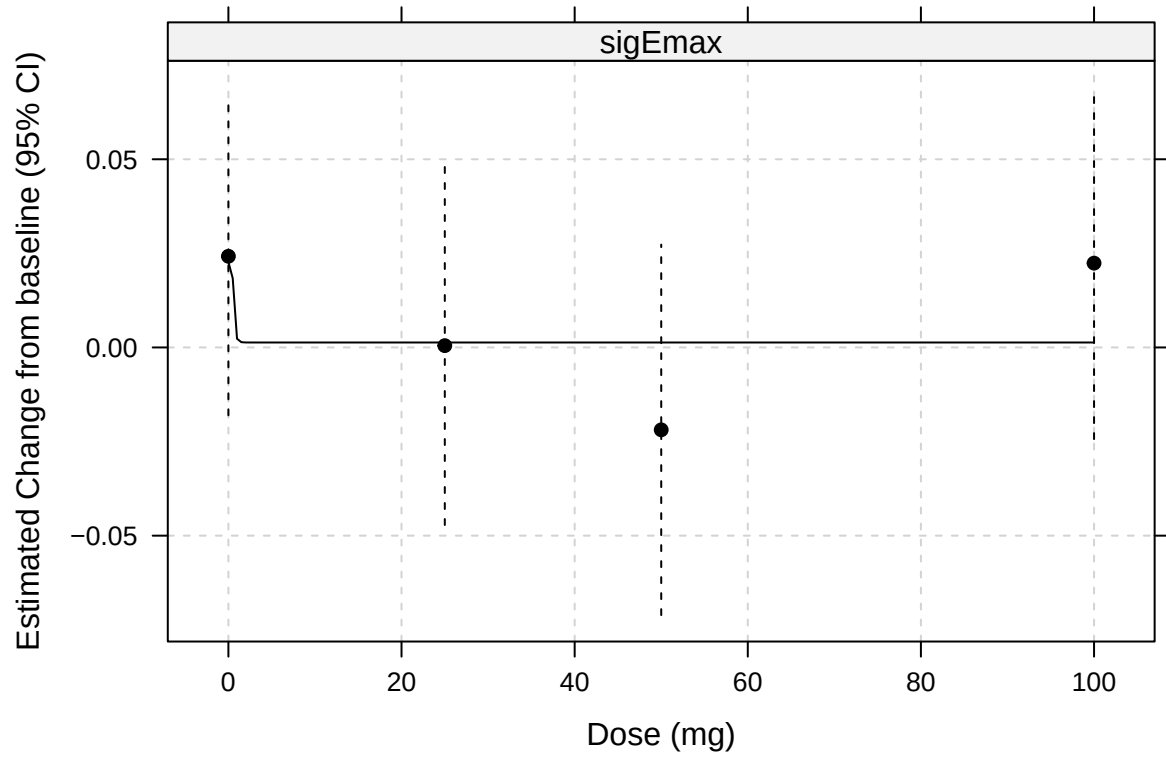
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
emax	7.80	0.10	0.23	0.89
linInt	8.00	24.28	34.54	44.85
sigEmax	9.80	0.62	0.71	0.88

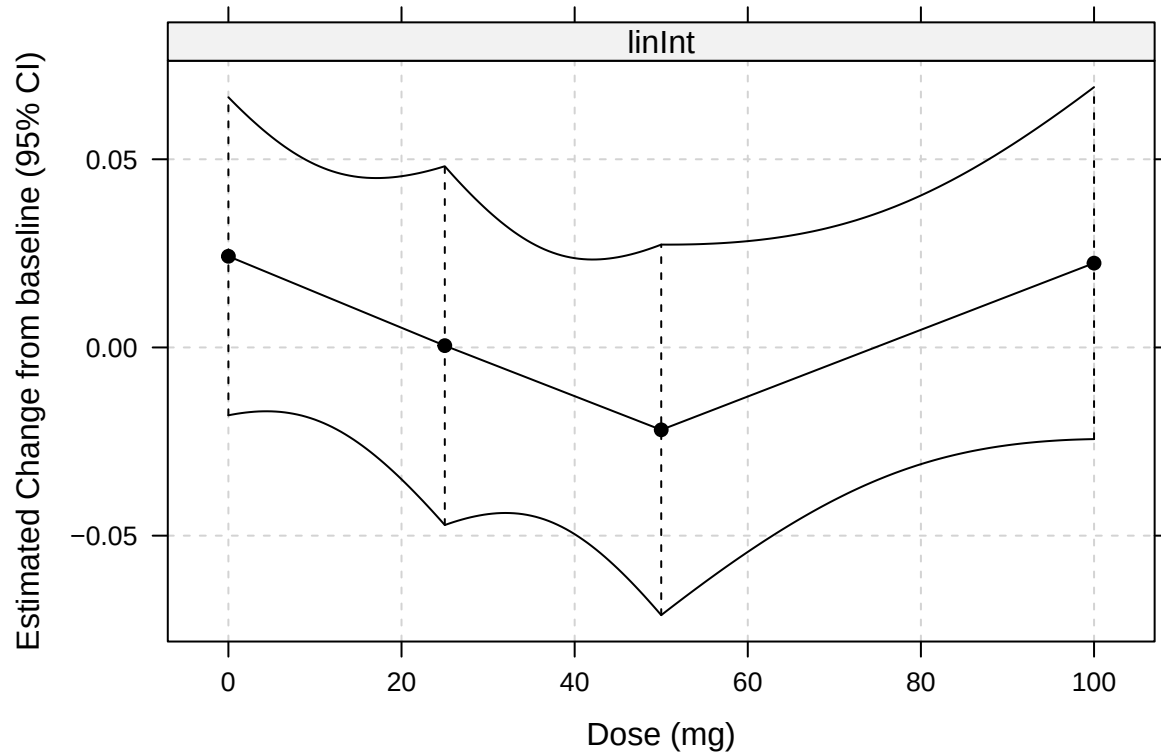
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	0.02 (-0.02 to 0.07)
25 mg	0.00 (-0.05 to 0.05)
50 mg	-0.02 (-0.07 to 0.03)
100 mg	0.02 (-0.03 to 0.07)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.02 (-0.07 to 0.02)

Mean relative solid stress (ROI2)

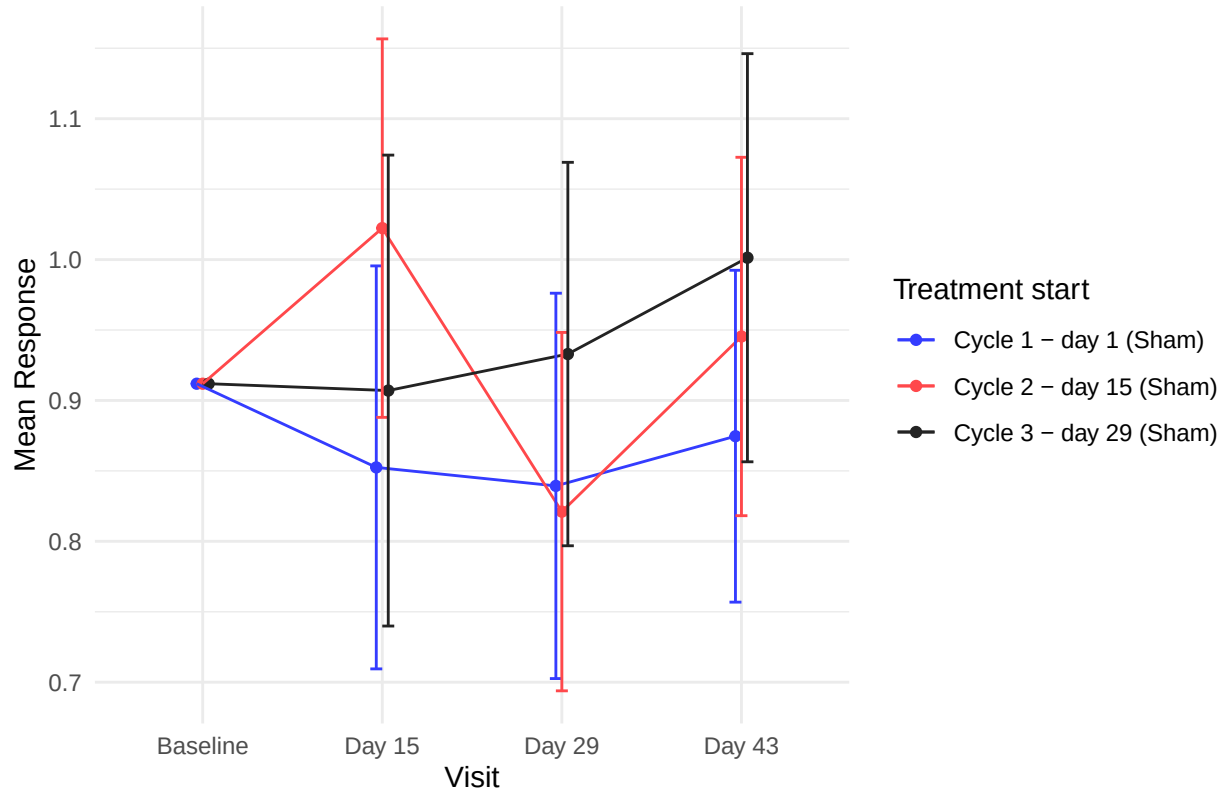
The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot

Mean Response over Time for Mean relative solid stress (ROI2)



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.601	-0.805	-0.538	-0.864
25 mg	-0.242	0.022	-0.427	0.328
50 mg	0.085	0.240	0.305	0.236
100 mg	0.757	0.543	0.659	0.301

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
sigEmax	-0.44	0.81
linear	-0.98	0.93
emax	-1.30	0.97
linInt	-1.58	0.98

Fitted model parameters (AIC/ED)

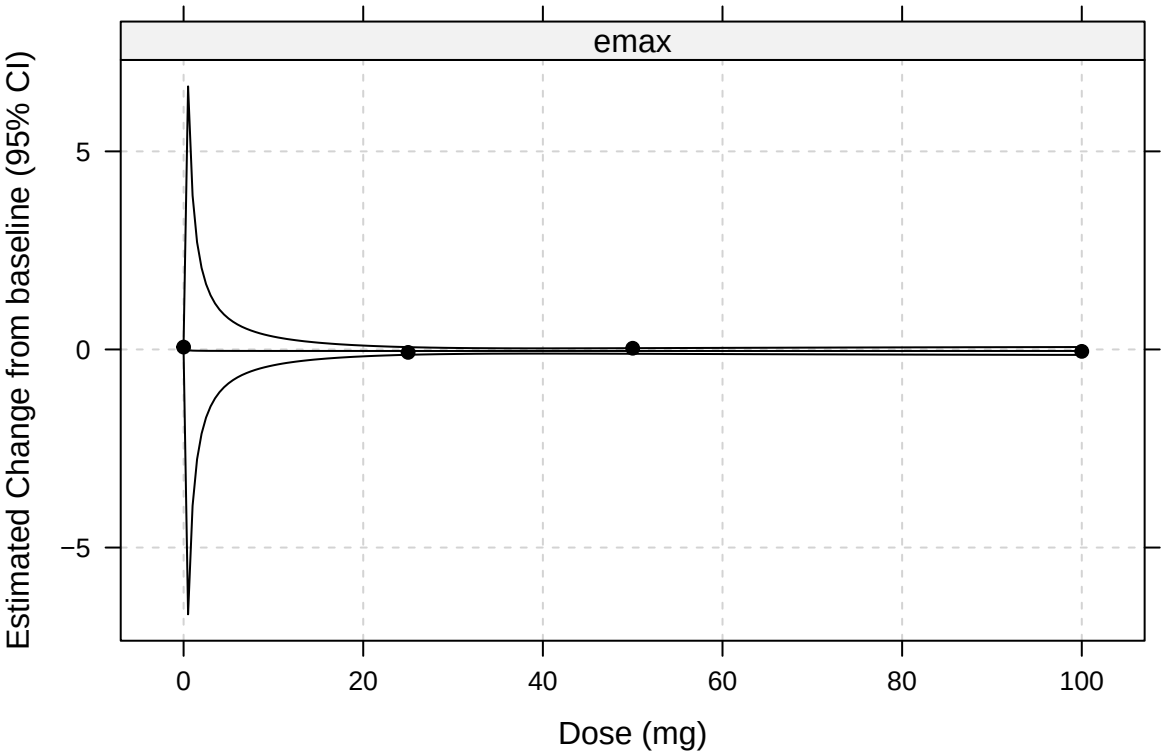
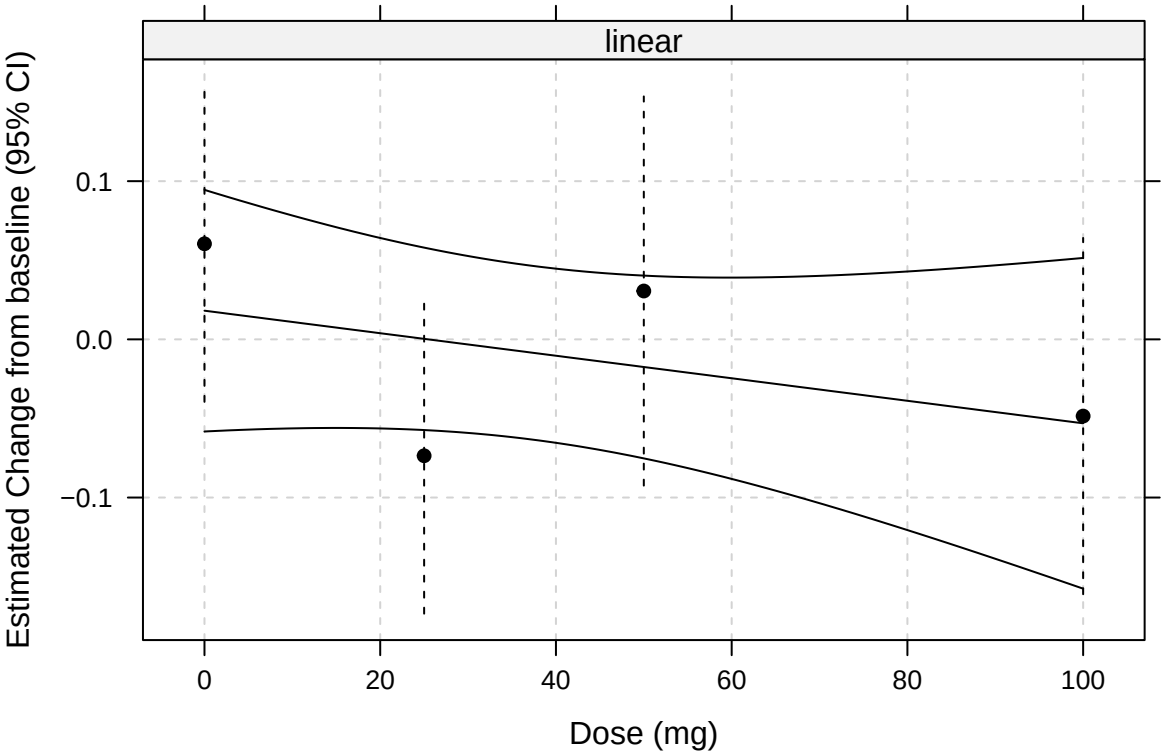
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	7.29	50.00	70.00	90.00

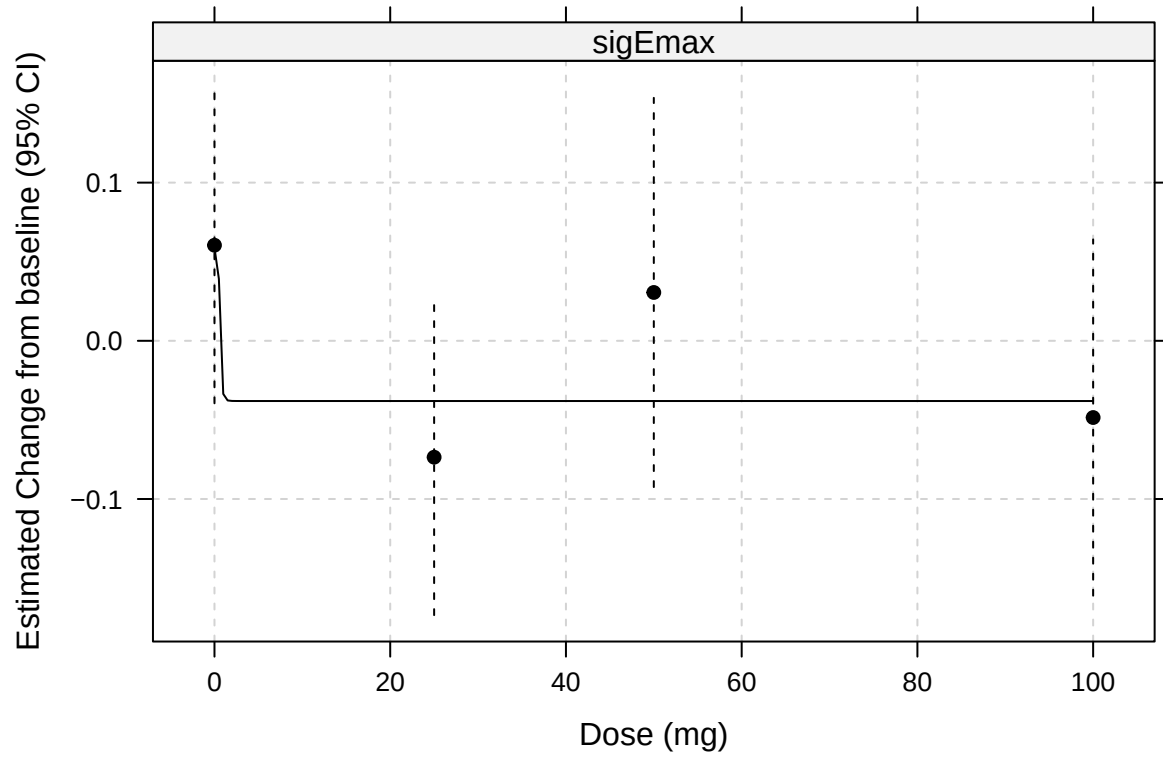
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
emax	7.75	0.10	0.23	0.89
linInt	8.00	12.50	17.50	22.50
sigEmax	9.75	0.62	0.71	0.88

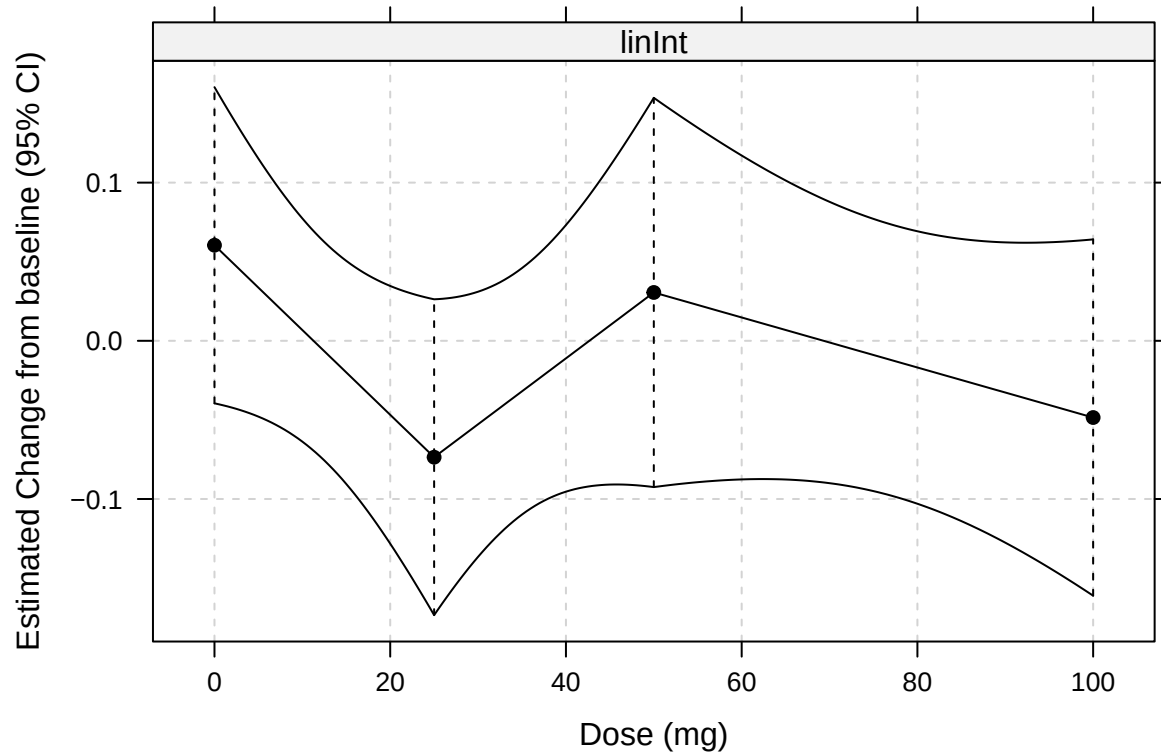
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	0.06 (-0.04 to 0.16)
25 mg	-0.07 (-0.18 to 0.03)
50 mg	0.03 (-0.10 to 0.16)
100 mg	-0.05 (-0.16 to 0.07)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.10 (-0.22 to 0.02)

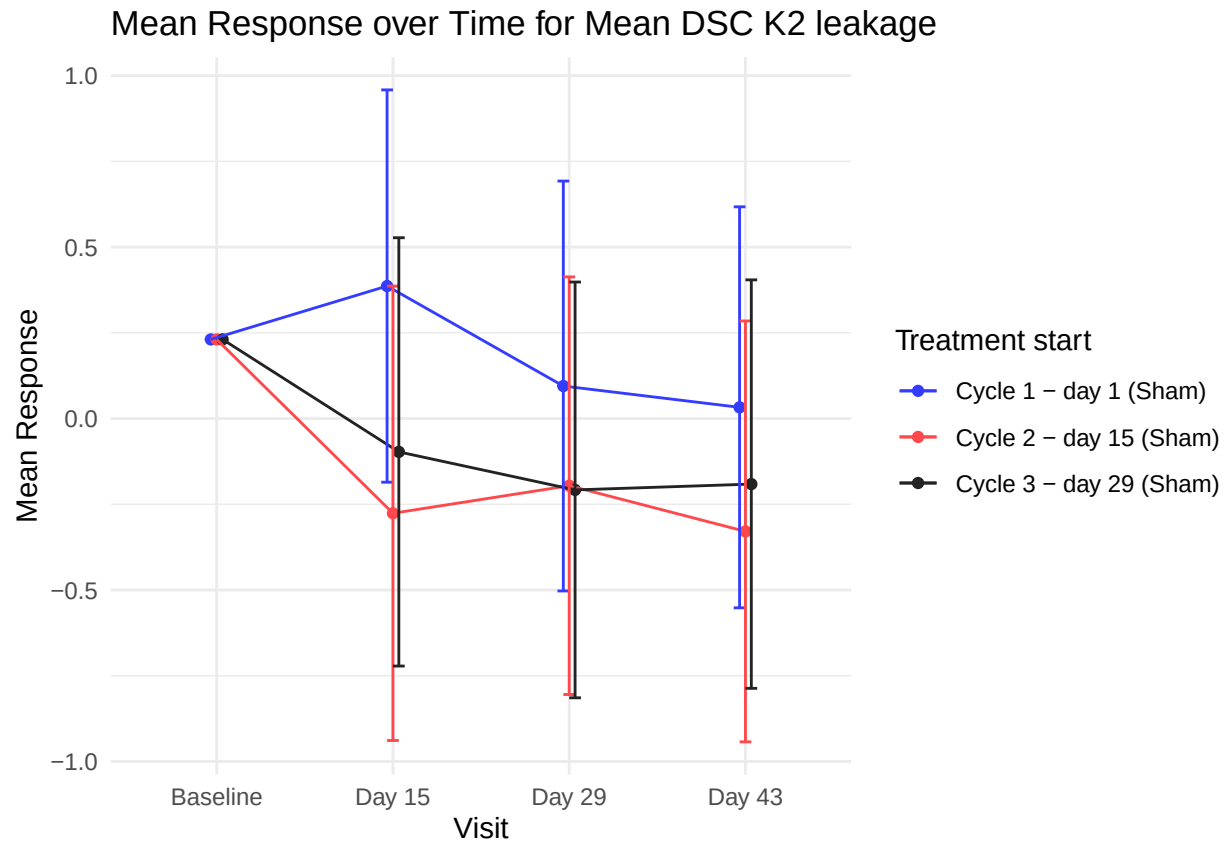
Other MRI Endpoints up to Day 43

For remaining MRI endpoints, we present estimated means by dose, contrasts vs 0 mg, and a plot with confidence intervals.

Mean DSC K2 leakage

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start



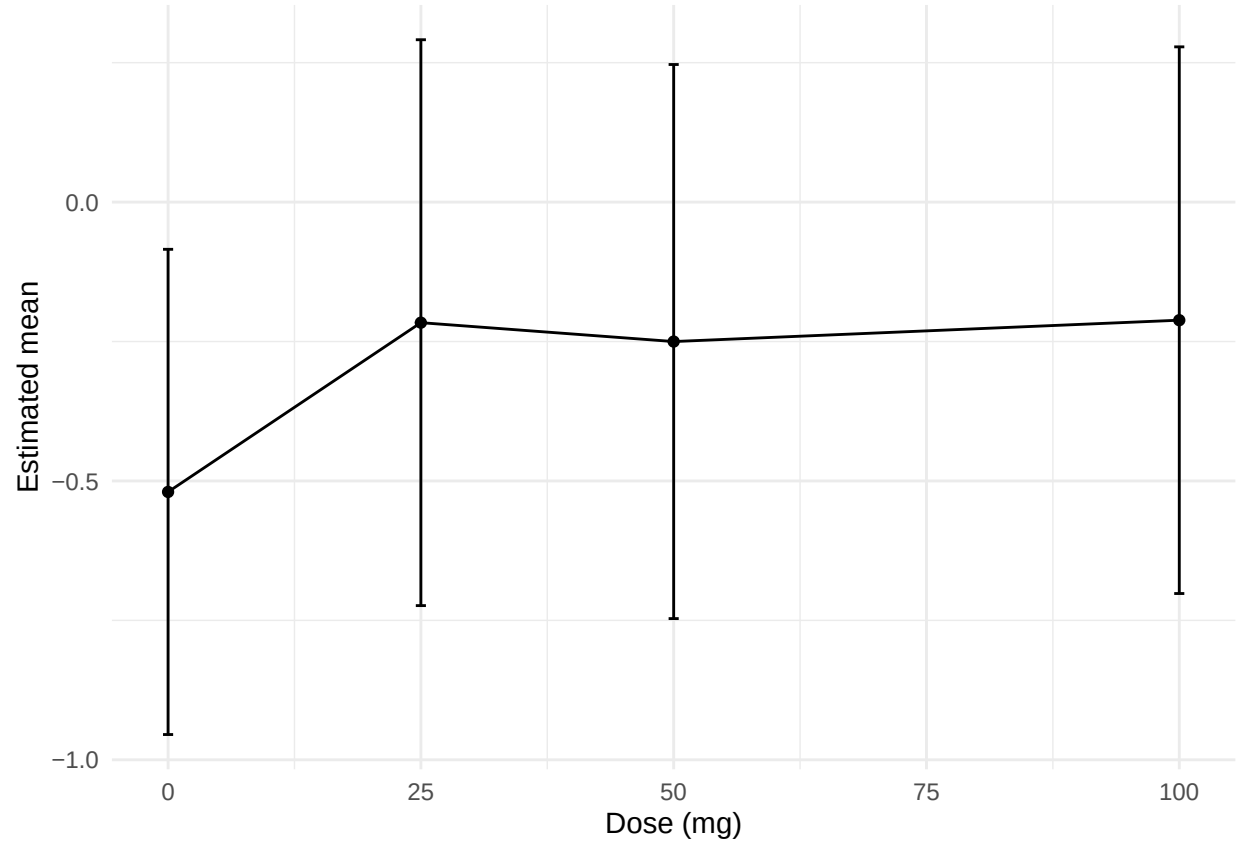
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.52 (-0.95 to -0.08)
25 mg	-0.22 (-0.72 to 0.29)
50 mg	-0.25 (-0.75 to 0.25)
100 mg	-0.21 (-0.70 to 0.28)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.30 (-0.45 to 1.06)
50 mg vs 0 mg	0.27 (-0.49 to 1.03)
100 mg vs 0 mg	0.31 (-0.45 to 1.07)

Plot



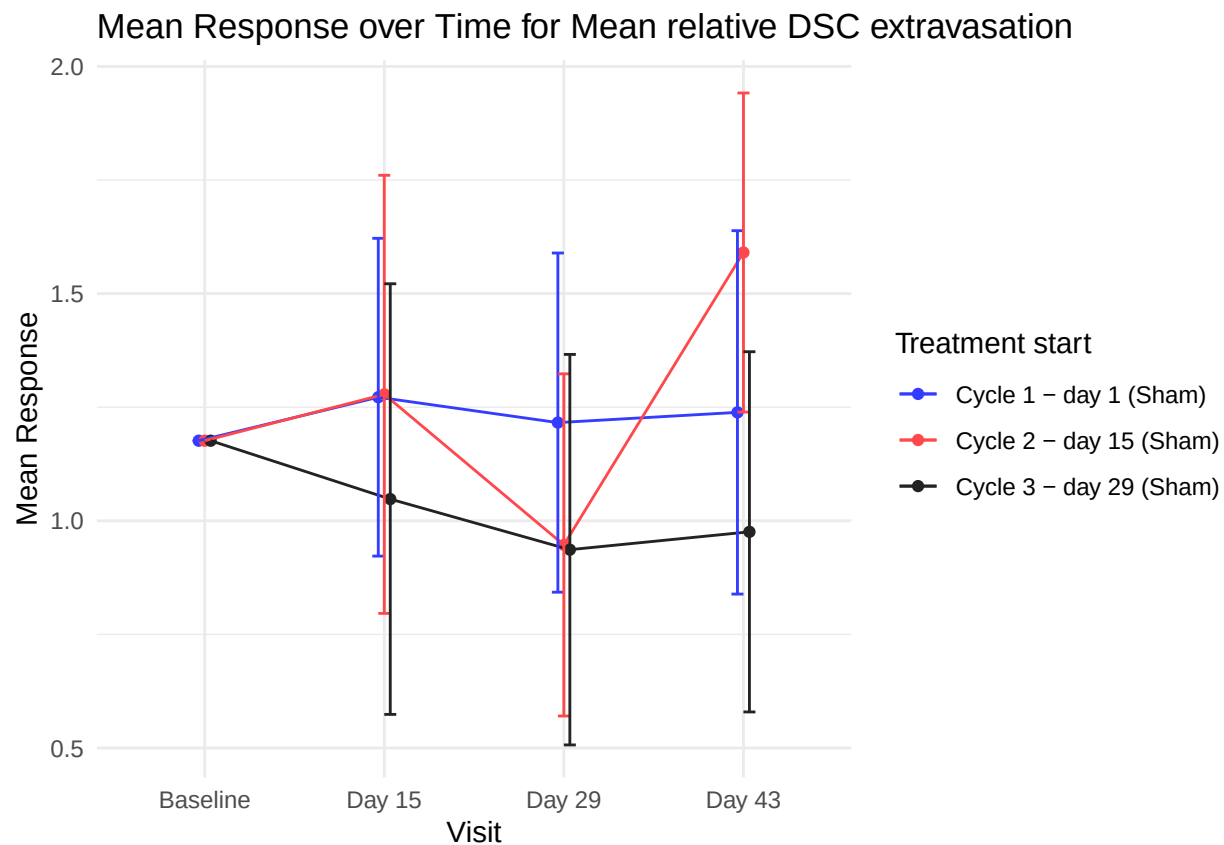
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.30 (-0.19 to 0.78)

Mean relative DSC extravasation

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



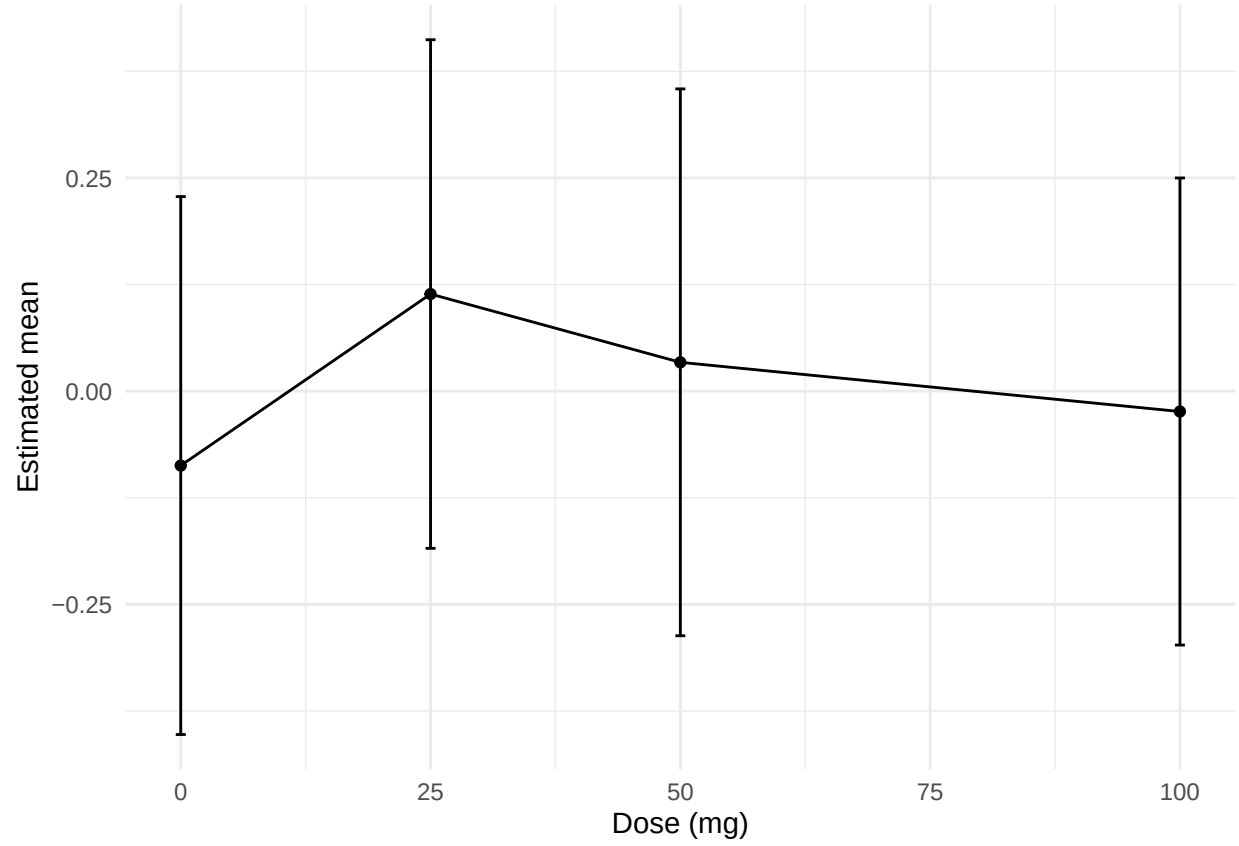
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.09 (-0.40 to 0.23)
25 mg	0.11 (-0.18 to 0.41)
50 mg	0.03 (-0.29 to 0.35)
100 mg	-0.02 (-0.30 to 0.25)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.20 (-0.33 to 0.73)
50 mg vs 0 mg	0.12 (-0.44 to 0.69)
100 mg vs 0 mg	0.06 (-0.47 to 0.60)

Plot



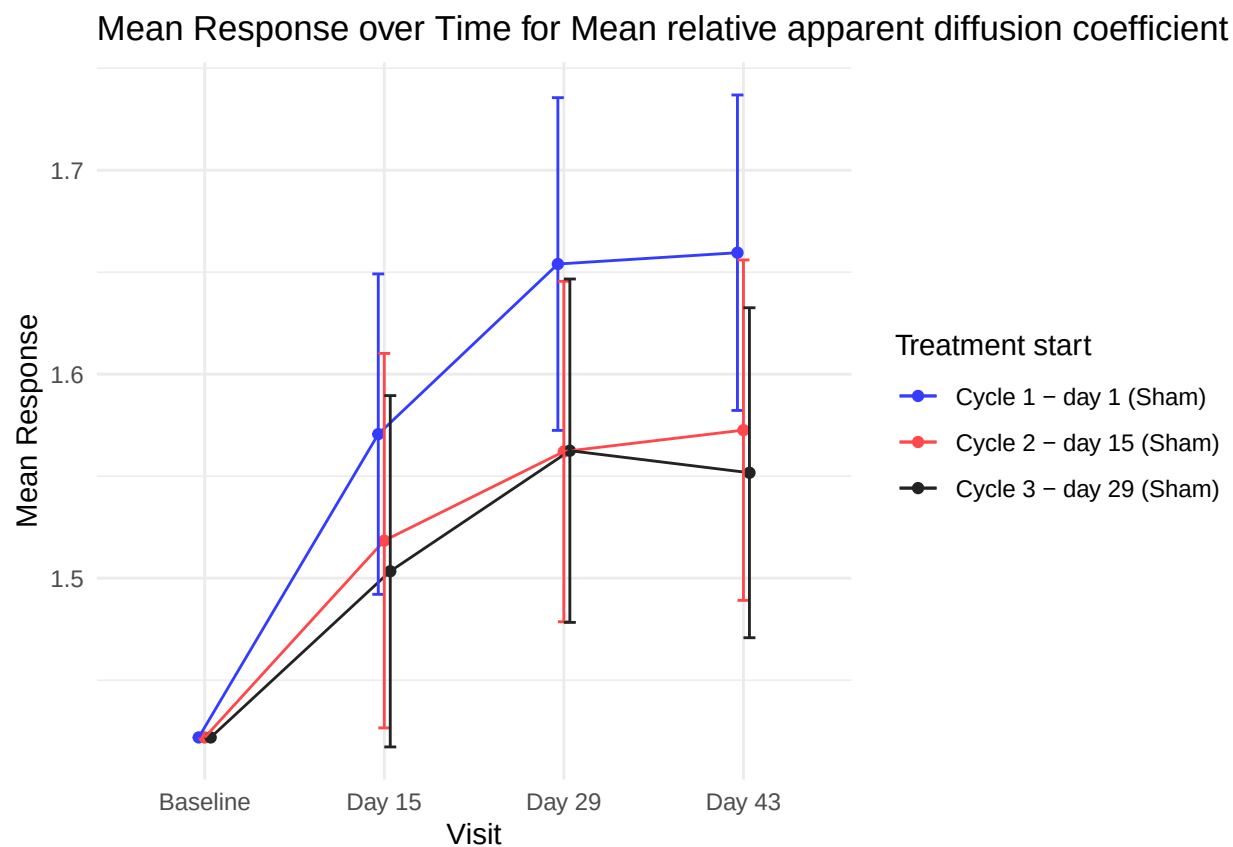
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.13 (-0.24 to 0.50)

Mean relative apparent diffusion coefficient

A total of 51 subjects were included in the analysis for this endpoint.

Line plot by treatment start



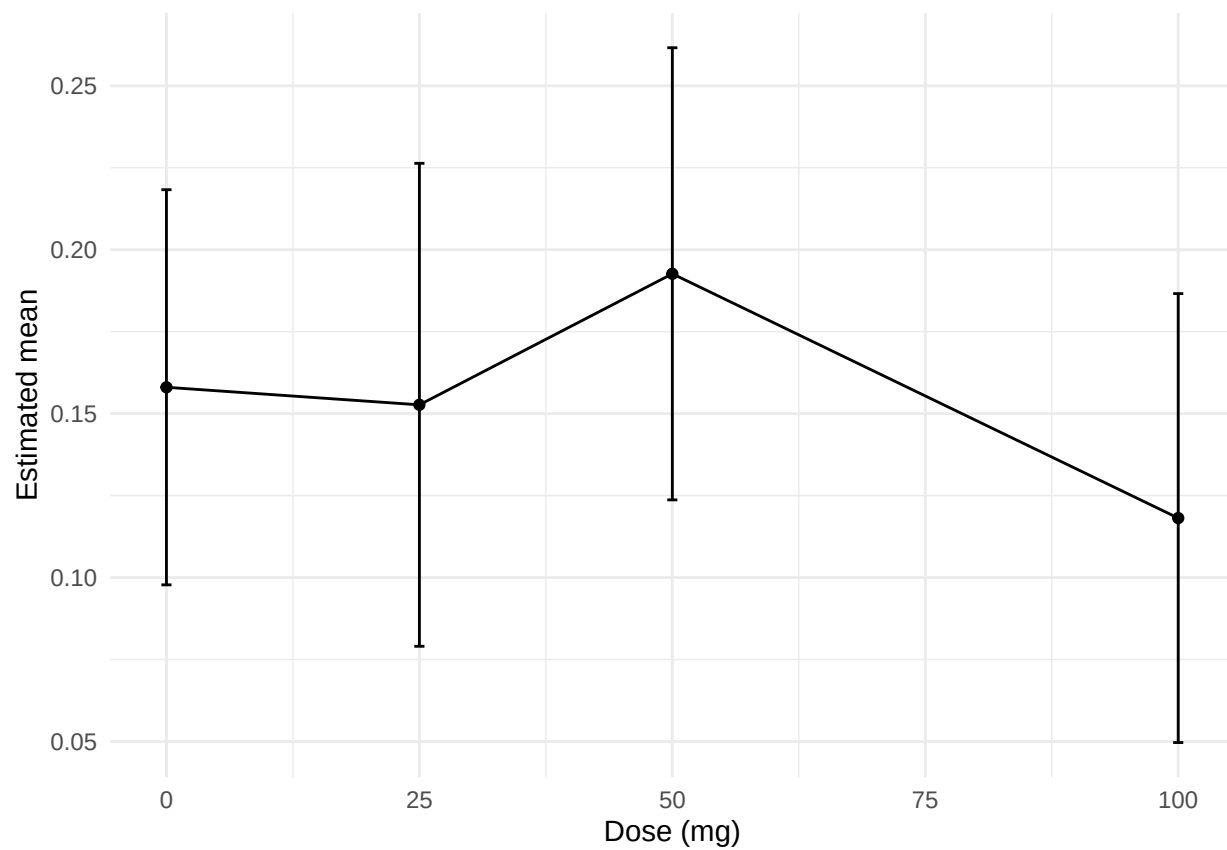
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.16 (0.10 to 0.22)
25 mg	0.15 (0.08 to 0.23)
50 mg	0.19 (0.12 to 0.26)
100 mg	0.12 (0.05 to 0.19)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.01 (-0.11 to 0.10)
50 mg vs 0 mg	0.03 (-0.06 to 0.13)
100 mg vs 0 mg	-0.04 (-0.14 to 0.06)

Plot



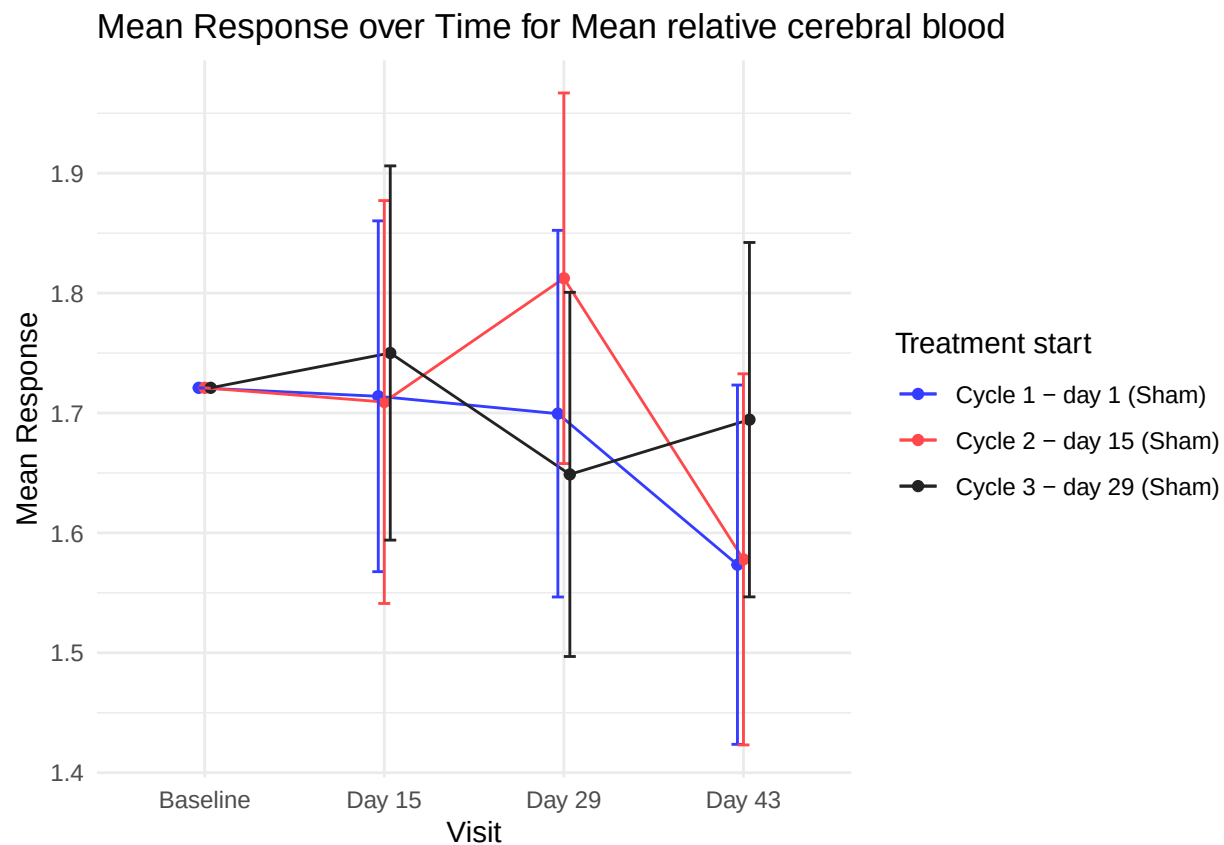
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.00 (-0.07 to 0.06)

Mean relative cerebral blood

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start



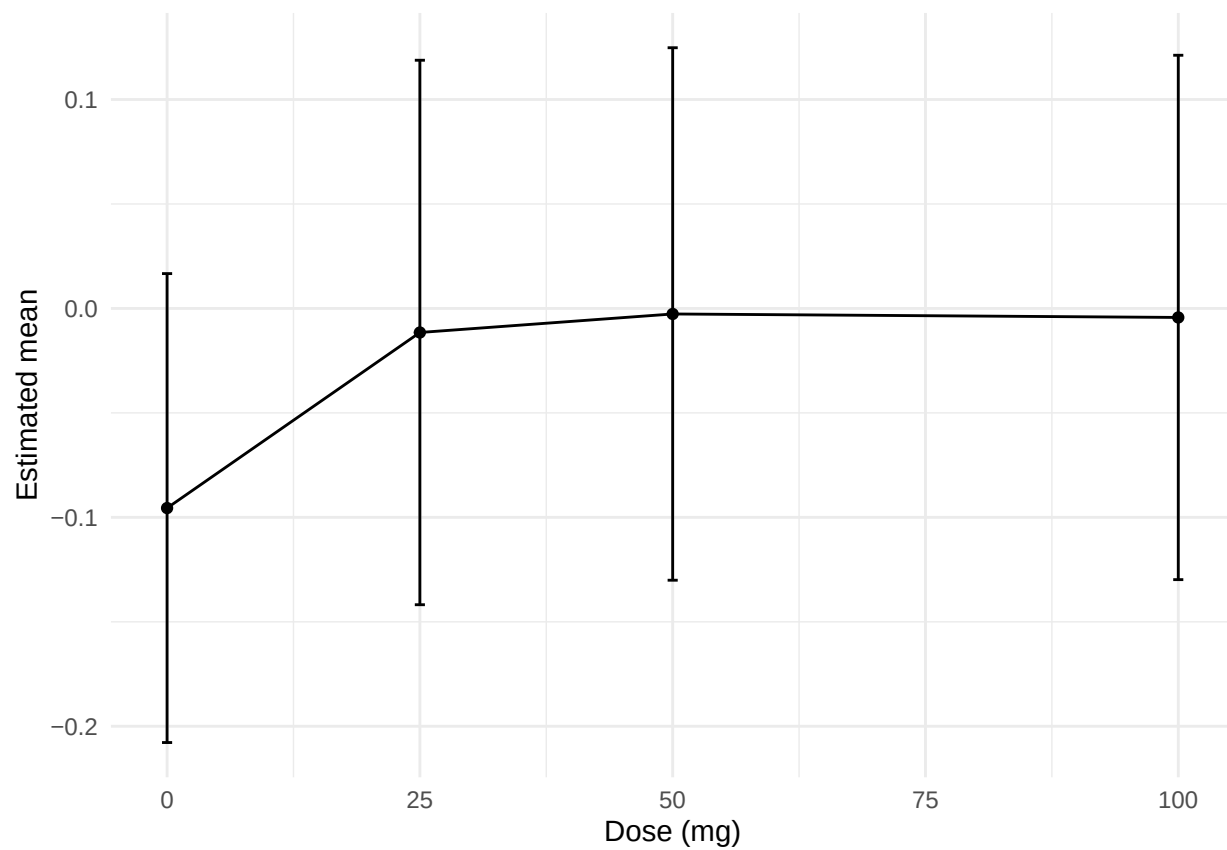
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.10 (-0.21 to 0.02)
25 mg	-0.01 (-0.14 to 0.12)
50 mg	0.00 (-0.13 to 0.12)
100 mg	0.00 (-0.13 to 0.12)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.08 (-0.11 to 0.28)
50 mg vs 0 mg	0.09 (-0.10 to 0.29)
100 mg vs 0 mg	0.09 (-0.10 to 0.29)

Plot



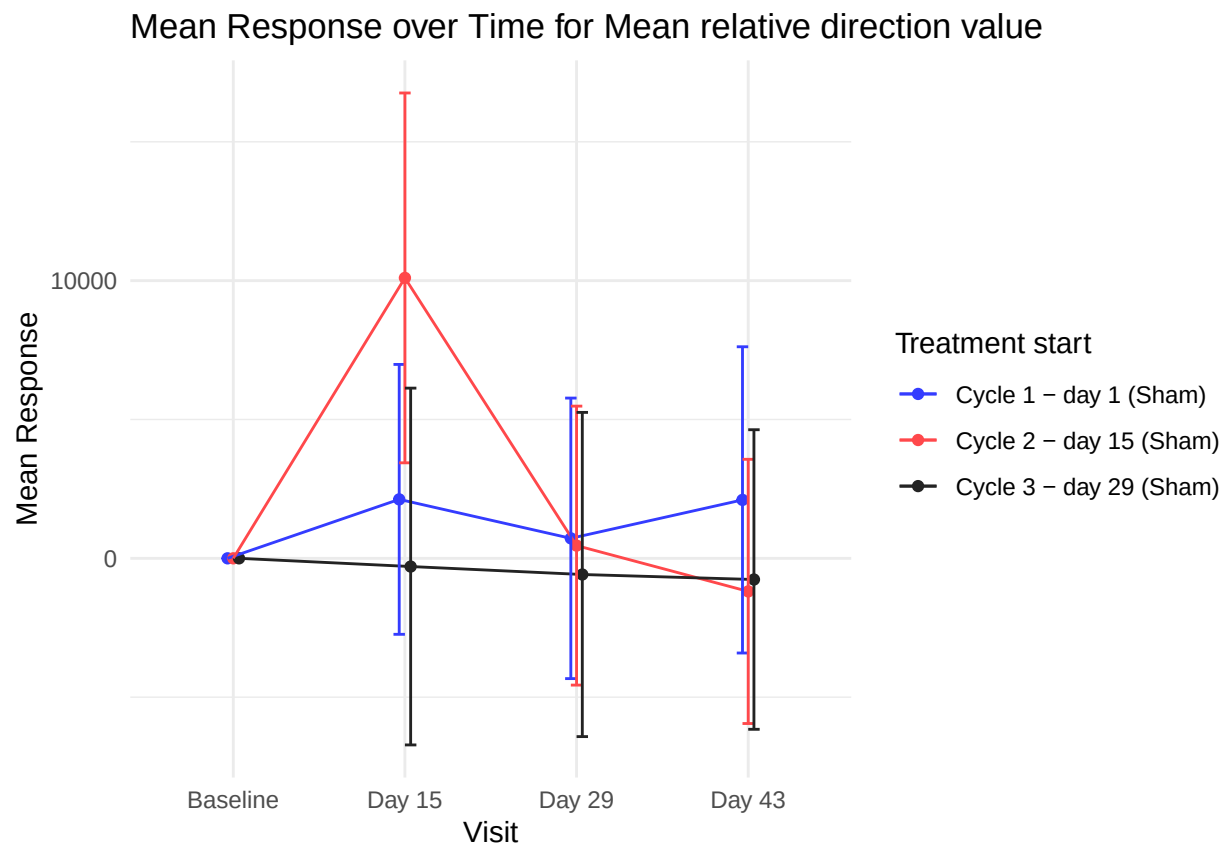
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.09 (-0.04 to 0.22)

Mean relative direction value

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



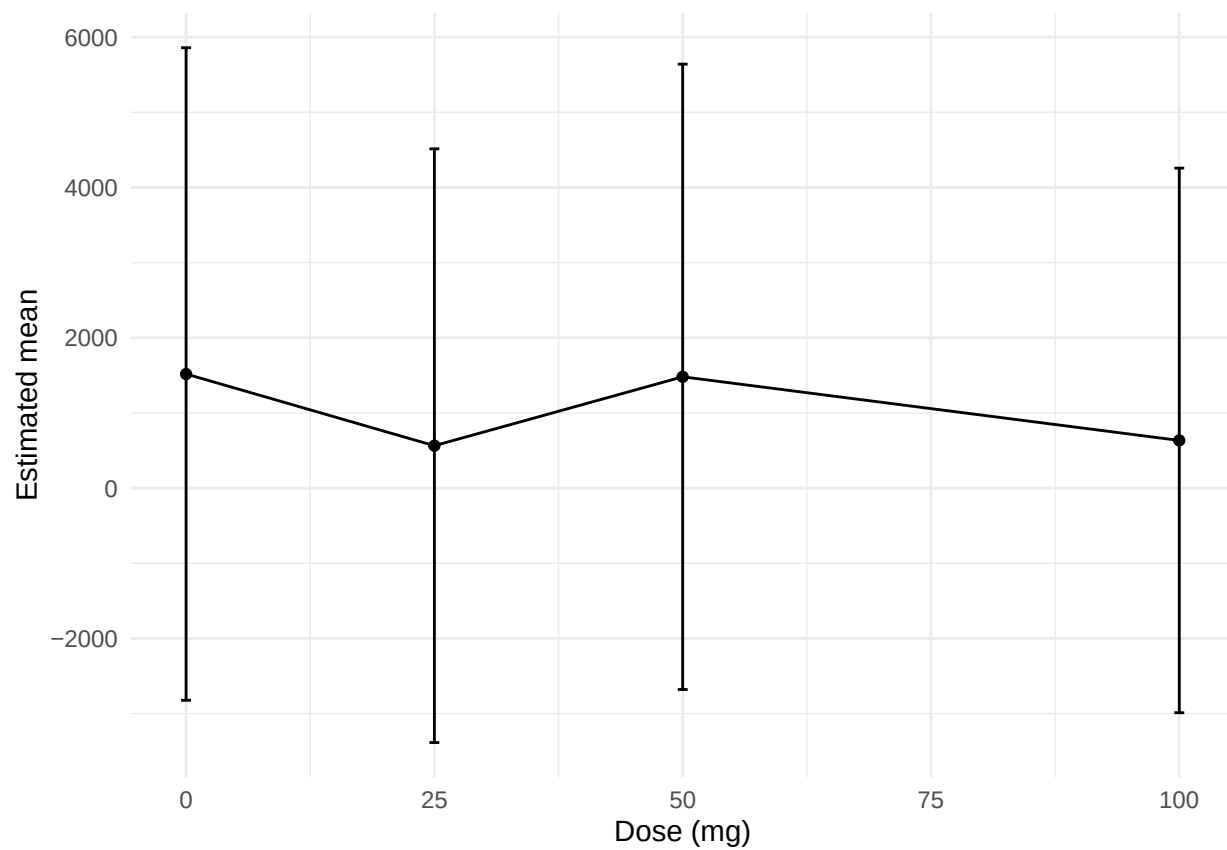
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	1518.29 (-2822.64 to 5859.21)
25 mg	564.43 (-3384.90 to 4513.76)
50 mg	1480.93 (-2679.09 to 5640.96)
100 mg	634.49 (-2988.31 to 4257.29)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-953.86 (-8186.79 to 6279.07)
50 mg vs 0 mg	-37.35 (-7740.97 to 7666.26)
100 mg vs 0 mg	-883.80 (-8104.17 to 6336.58)

Plot



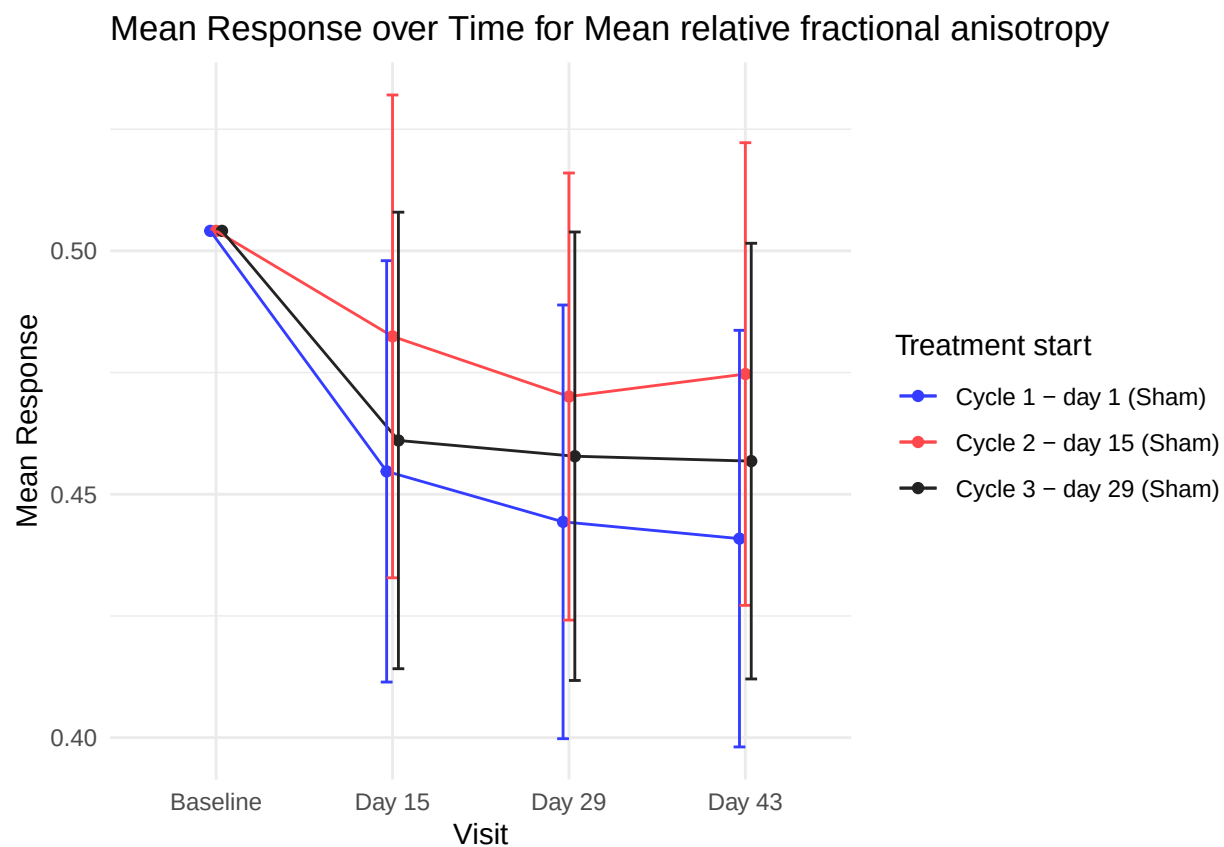
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-705.68 (-5805.17 to 4393.82)

Mean relative fractional anisotropy

A total of 51 subjects were included in the analysis for this endpoint.

Line plot by treatment start



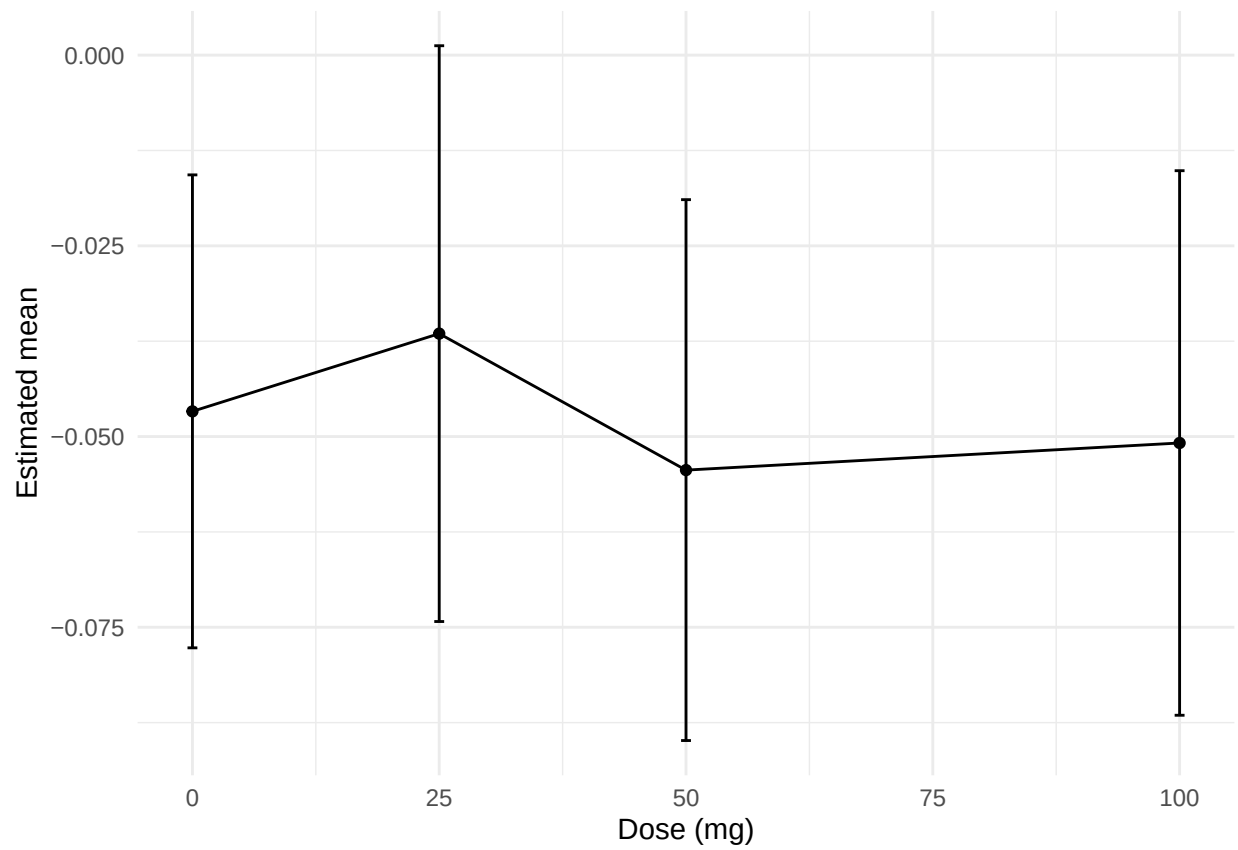
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.05 (-0.08 to -0.02)
25 mg	-0.04 (-0.07 to 0.00)
50 mg	-0.05 (-0.09 to -0.02)
100 mg	-0.05 (-0.09 to -0.02)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.01 (-0.04 to 0.06)
50 mg vs 0 mg	-0.01 (-0.06 to 0.04)
100 mg vs 0 mg	0.00 (-0.05 to 0.04)

Plot



Supplementary analysis: any dose vs 0 mg

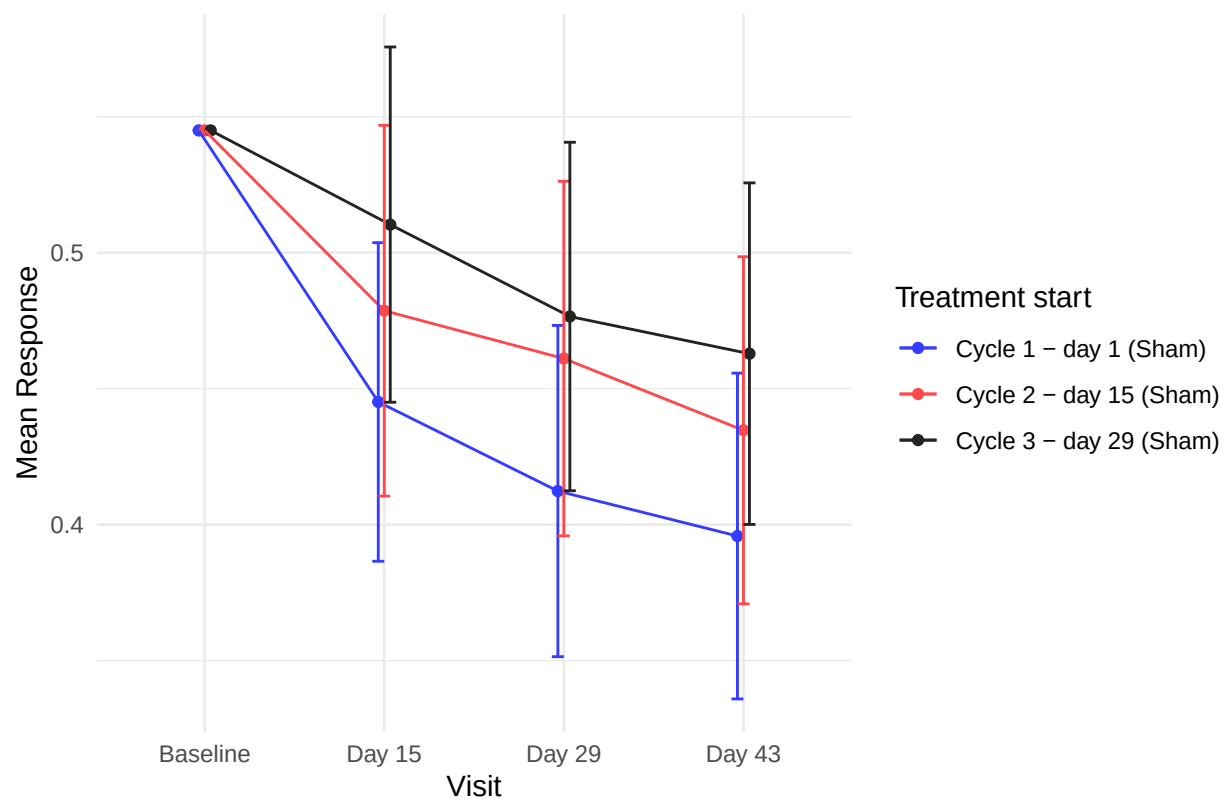
Comparison	Difference (95% CI)
Any dose vs 0 mg	0.00 (-0.03 to 0.03)

Mean relative restricted spectrum imaging

A total of 49 subjects were included in the analysis for this endpoint.

Line plot by treatment start

Mean Response over Time for Mean relative restricted spectrum imaging



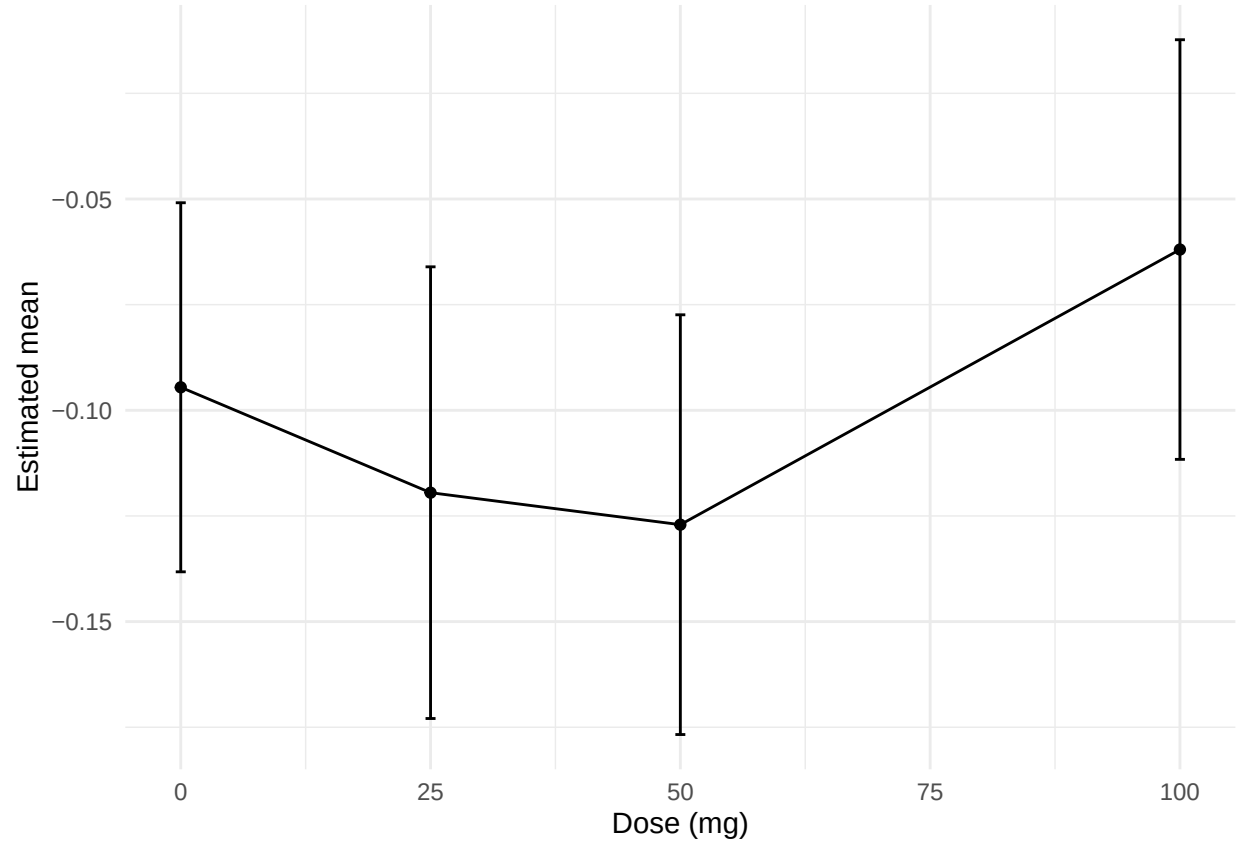
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.09 (-0.14 to -0.05)
25 mg	-0.12 (-0.17 to -0.07)
50 mg	-0.13 (-0.18 to -0.08)
100 mg	-0.06 (-0.11 to -0.01)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.02 (-0.10 to 0.05)
50 mg vs 0 mg	-0.03 (-0.10 to 0.04)
100 mg vs 0 mg	0.03 (-0.04 to 0.10)

Plot



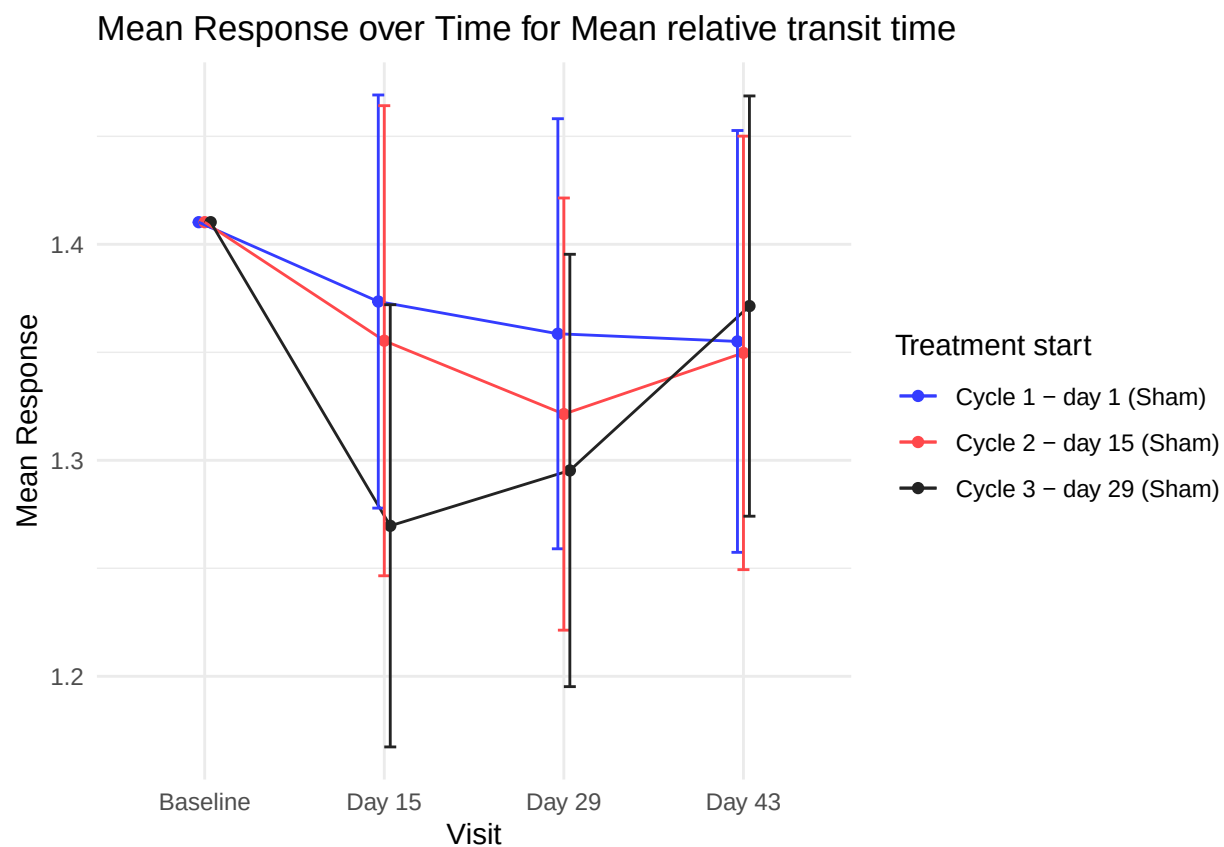
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.01 (-0.05 to 0.04)

Mean relative transit time

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start



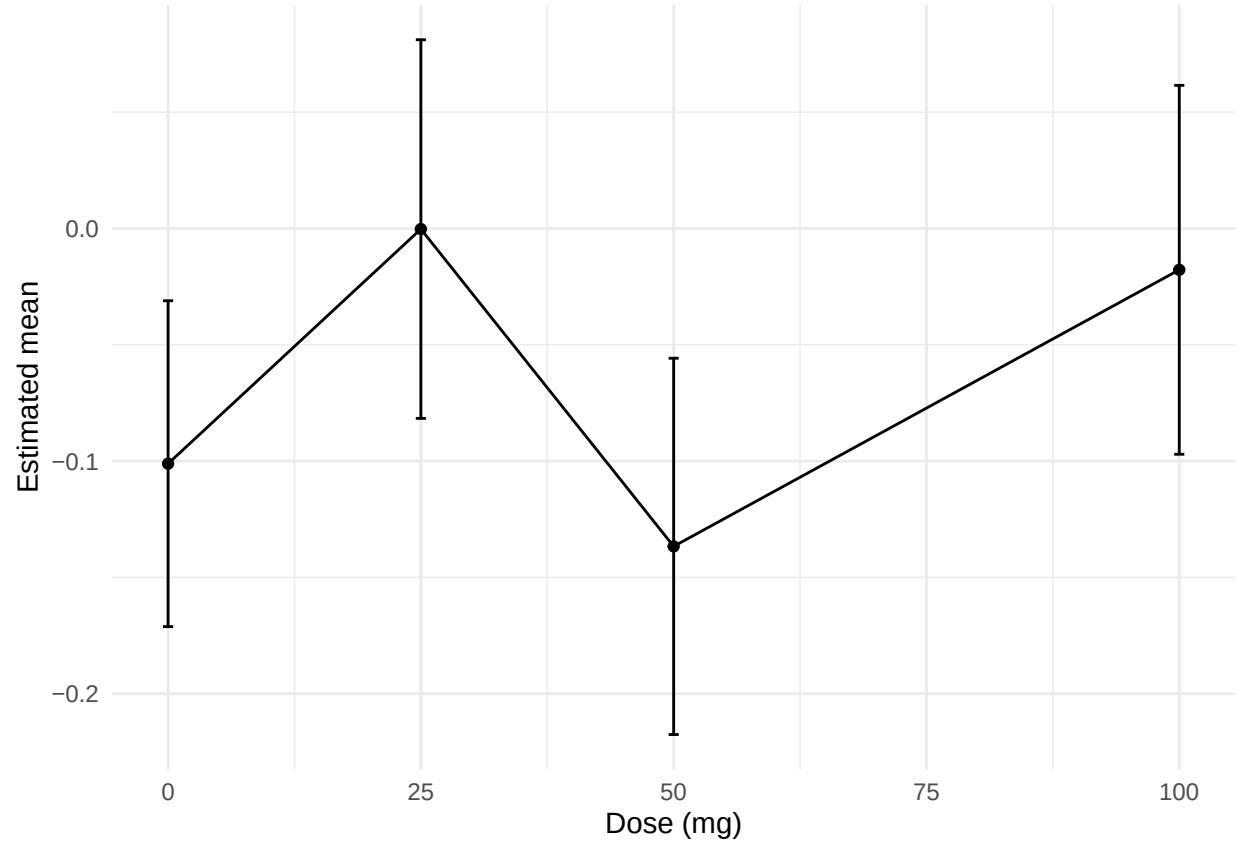
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.10 (-0.17 to -0.03)
25 mg	0.00 (-0.08 to 0.08)
50 mg	-0.14 (-0.22 to -0.06)
100 mg	-0.02 (-0.10 to 0.06)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.10 (-0.02 to 0.22)
50 mg vs 0 mg	-0.04 (-0.16 to 0.09)
100 mg vs 0 mg	0.08 (-0.04 to 0.21)

Plot



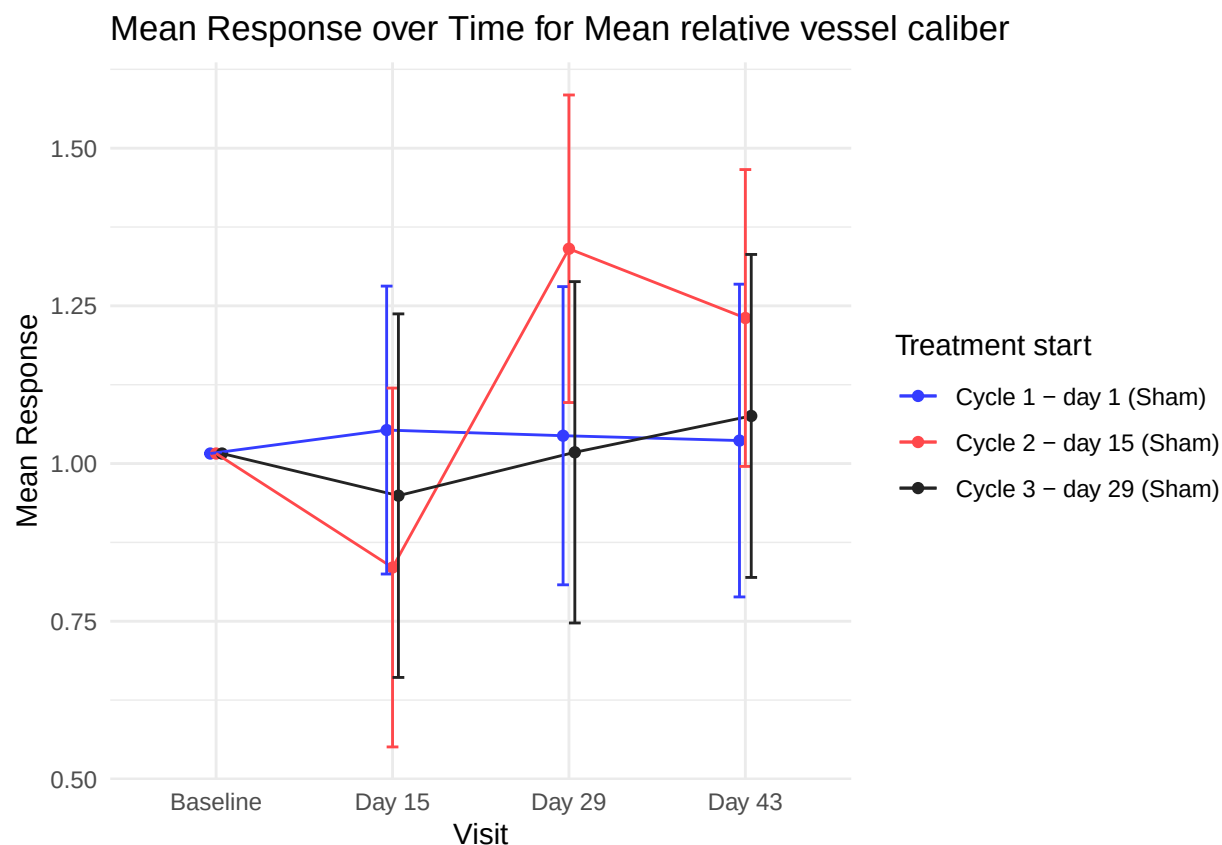
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.05 (-0.04 to 0.13)

Mean relative vessel caliber

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



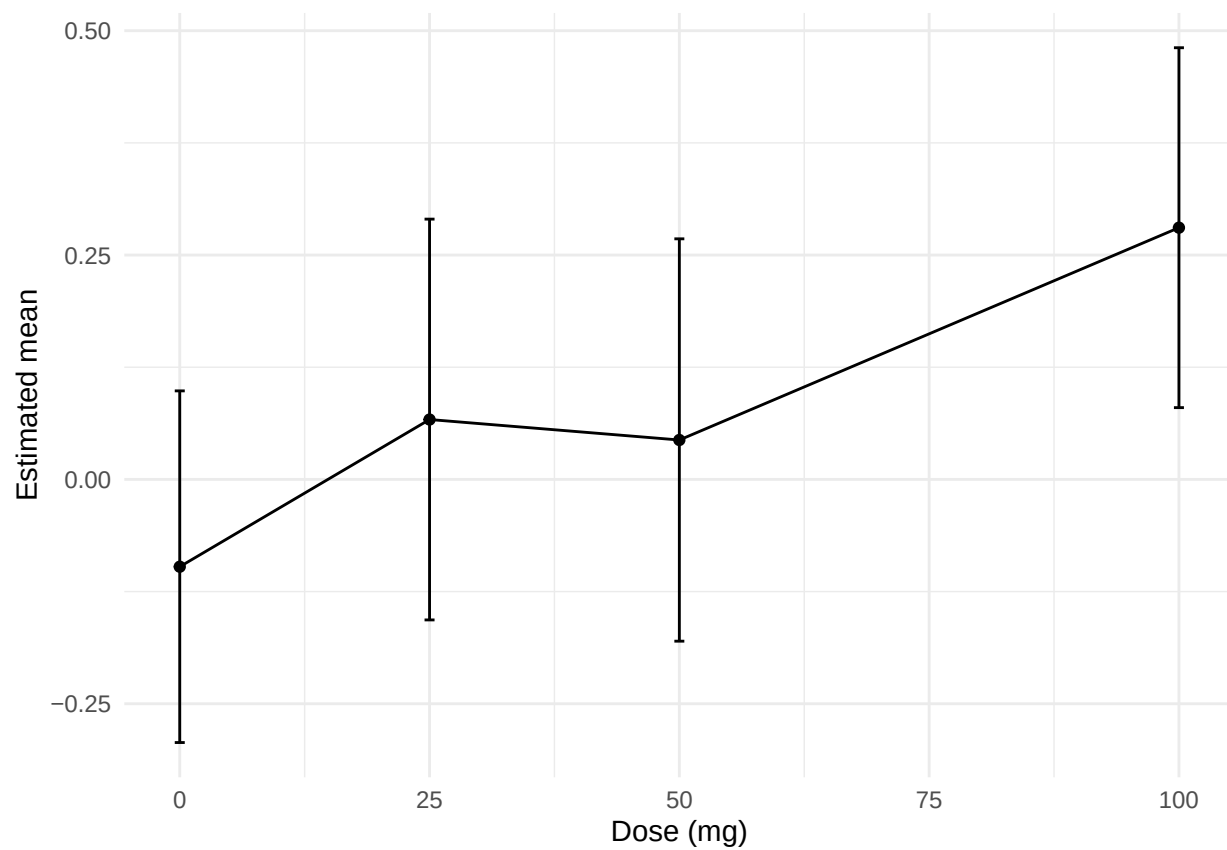
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.10 (-0.29 to 0.10)
25 mg	0.07 (-0.16 to 0.29)
50 mg	0.04 (-0.18 to 0.27)
100 mg	0.28 (0.08 to 0.48)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.16 (-0.16 to 0.49)
50 mg vs 0 mg	0.14 (-0.19 to 0.47)
100 mg vs 0 mg	0.38 (0.07 to 0.68)

Plot



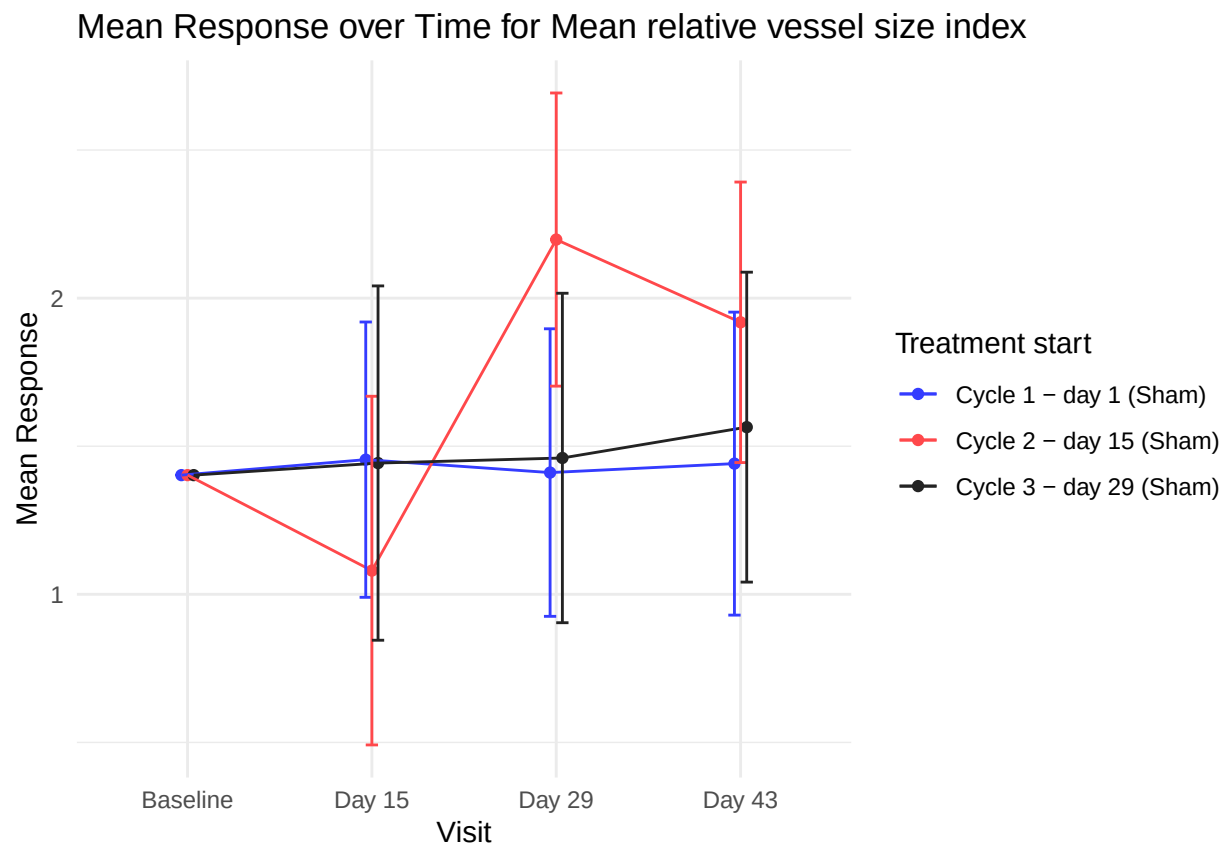
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.23 (0.02 to 0.44)

Mean relative vessel size index

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



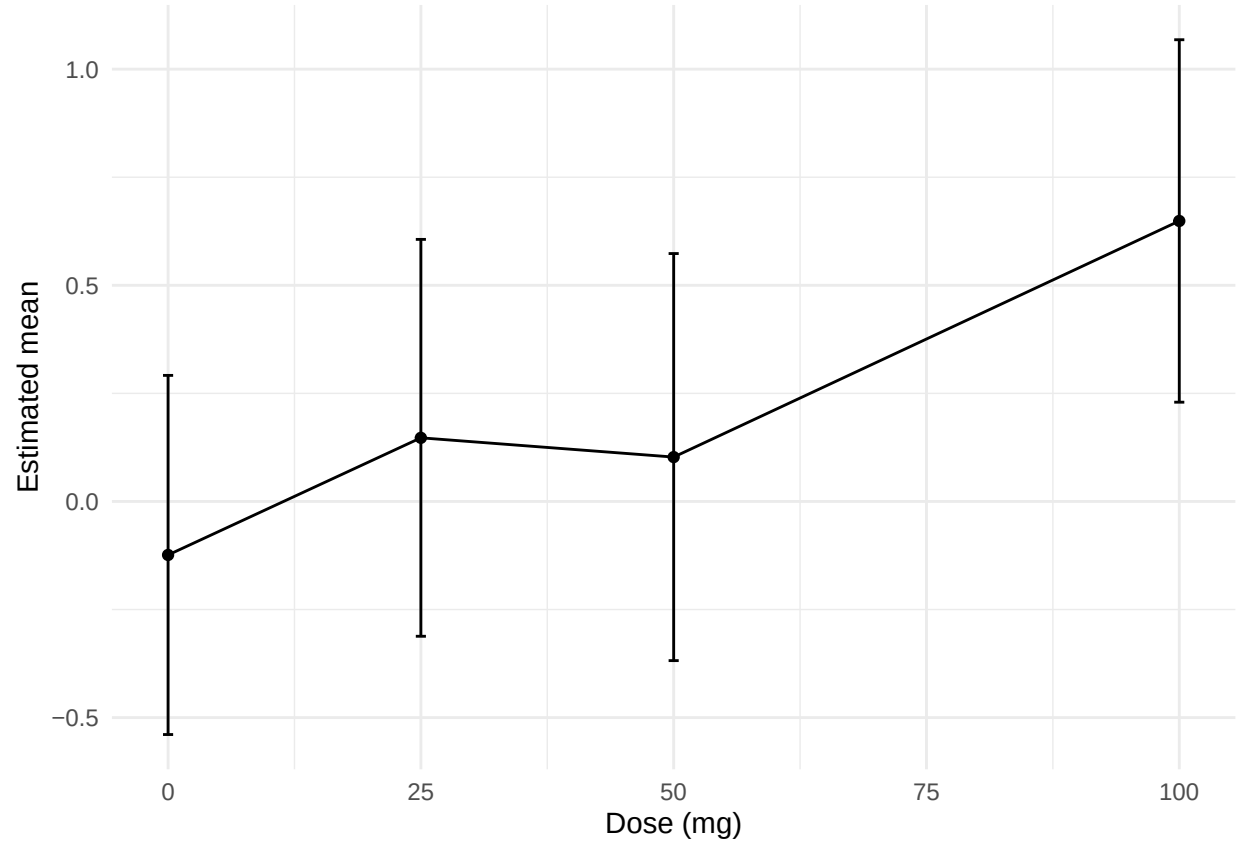
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.12 (-0.54 to 0.29)
25 mg	0.15 (-0.31 to 0.61)
50 mg	0.10 (-0.37 to 0.57)
100 mg	0.65 (0.23 to 1.07)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.27 (-0.41 to 0.96)
50 mg vs 0 mg	0.23 (-0.49 to 0.94)
100 mg vs 0 mg	0.77 (0.11 to 1.43)

Plot



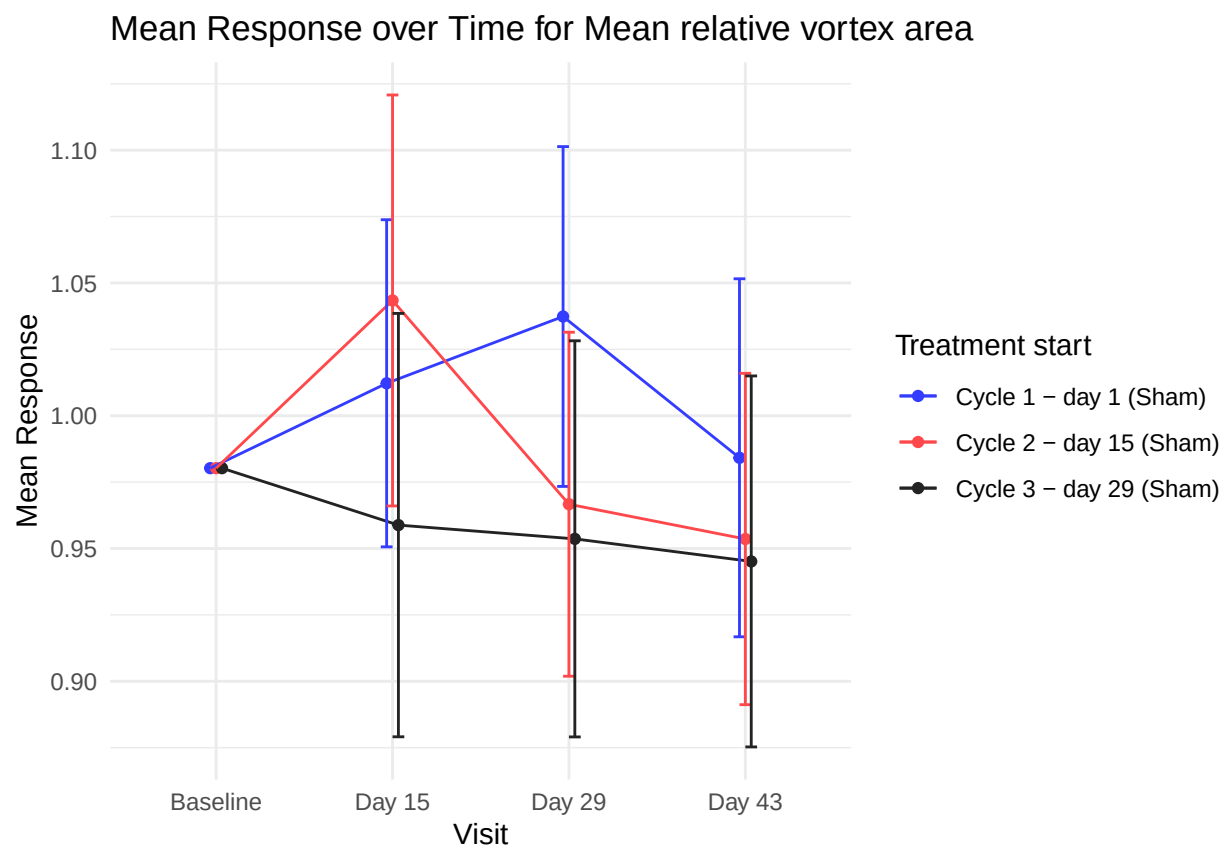
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.41 (-0.04 to 0.87)

Mean relative vortex area

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



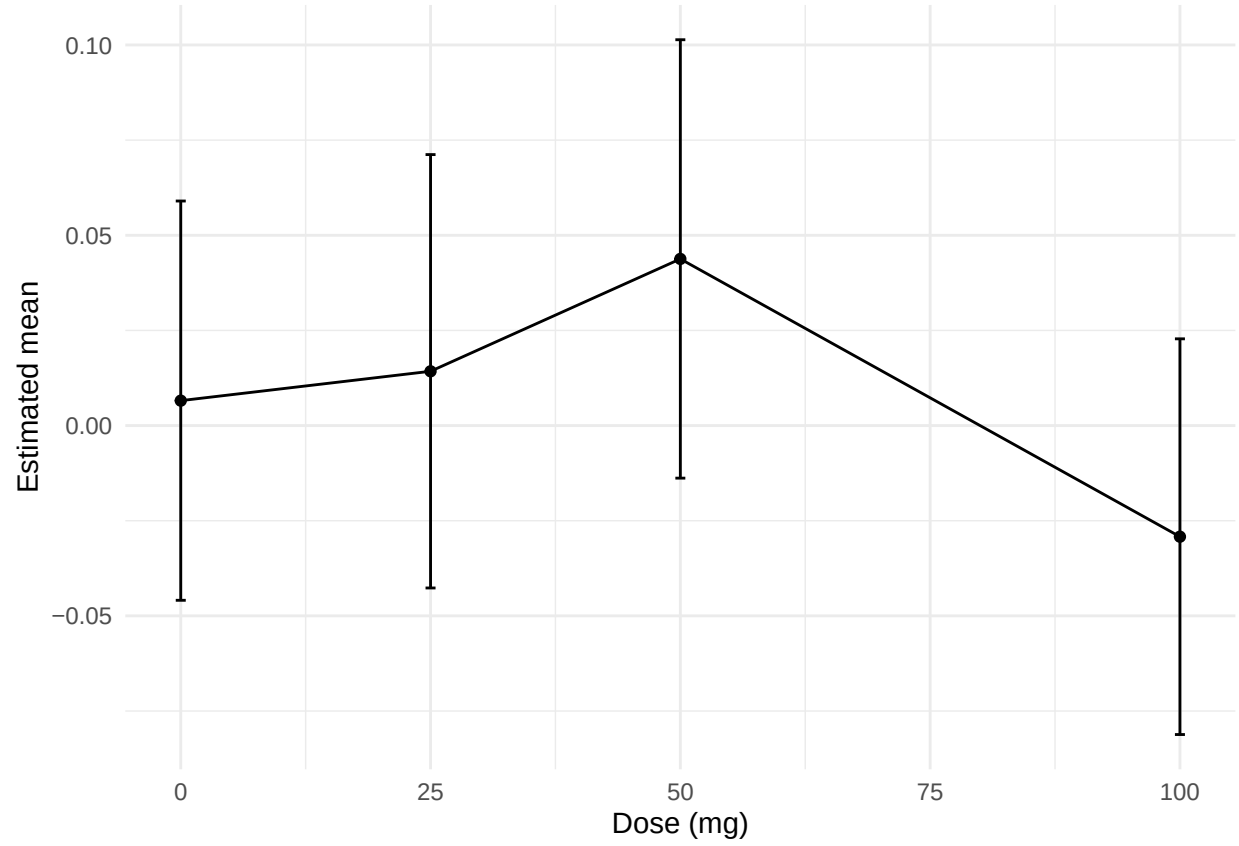
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.01 (-0.05 to 0.06)
25 mg	0.01 (-0.04 to 0.07)
50 mg	0.04 (-0.01 to 0.10)
100 mg	-0.03 (-0.08 to 0.02)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.01 (-0.08 to 0.10)
50 mg vs 0 mg	0.04 (-0.05 to 0.13)
100 mg vs 0 mg	-0.04 (-0.12 to 0.05)

Plot



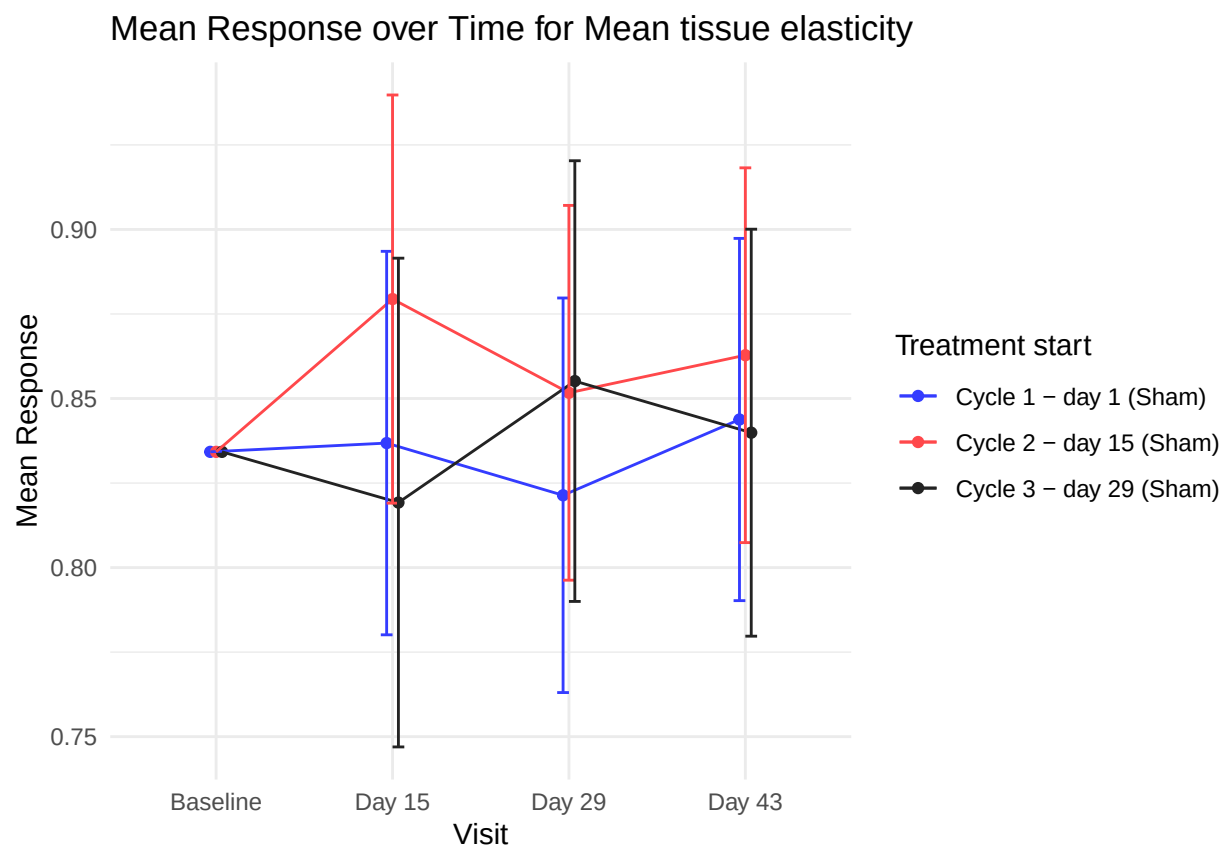
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.00 (-0.06 to 0.06)

Mean tissue elasticity

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



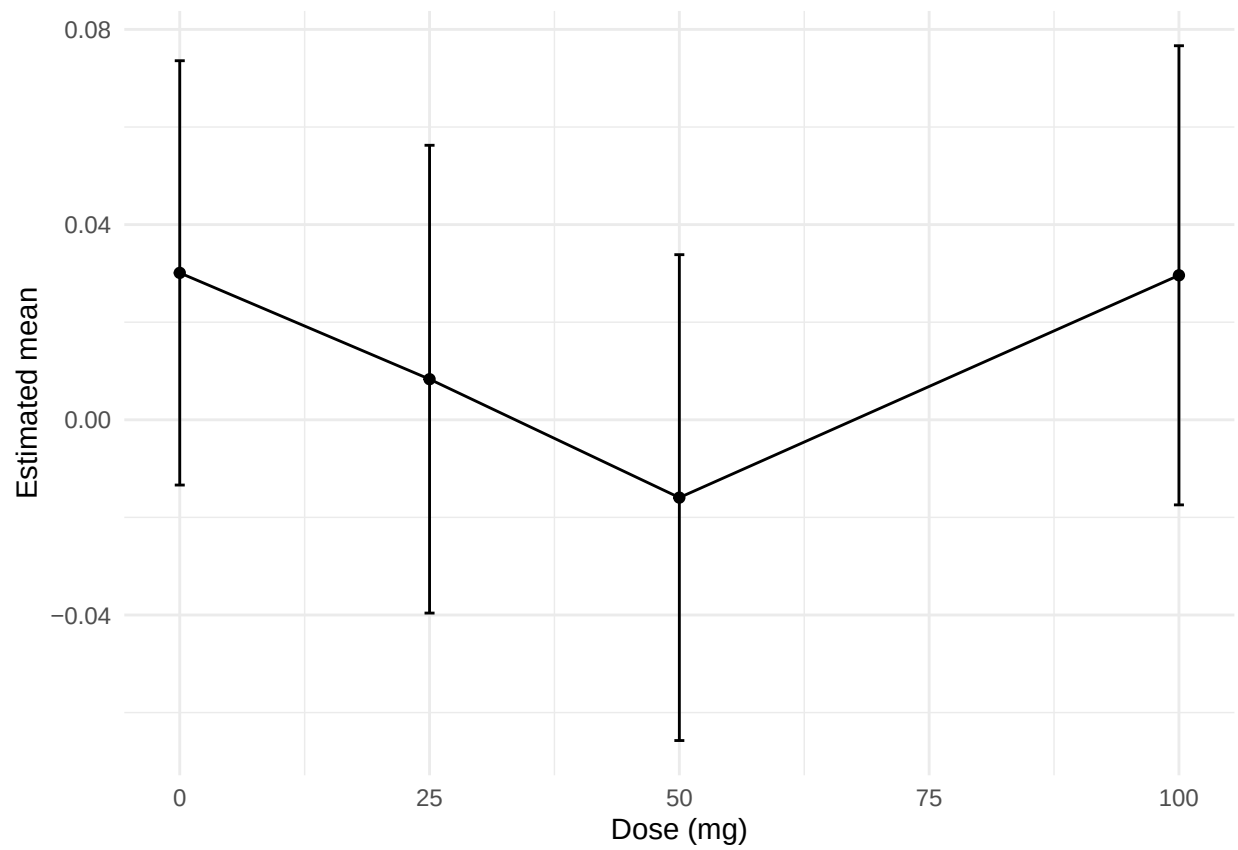
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.03 (-0.01 to 0.07)
25 mg	0.01 (-0.04 to 0.06)
50 mg	-0.02 (-0.07 to 0.03)
100 mg	0.03 (-0.02 to 0.08)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.02 (-0.10 to 0.05)
50 mg vs 0 mg	-0.05 (-0.12 to 0.03)
100 mg vs 0 mg	0.00 (-0.07 to 0.07)

Plot



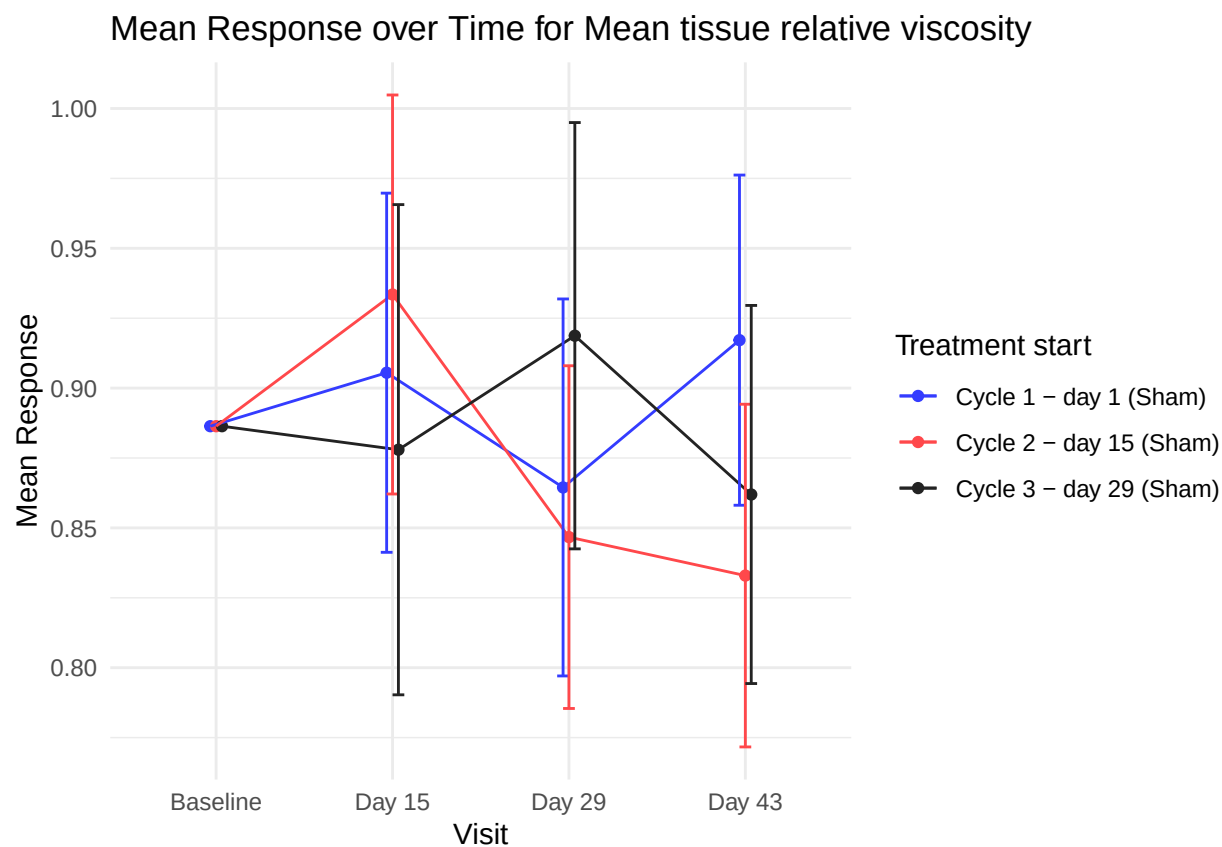
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.02 (-0.07 to 0.03)

Mean tissue relative viscosity

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



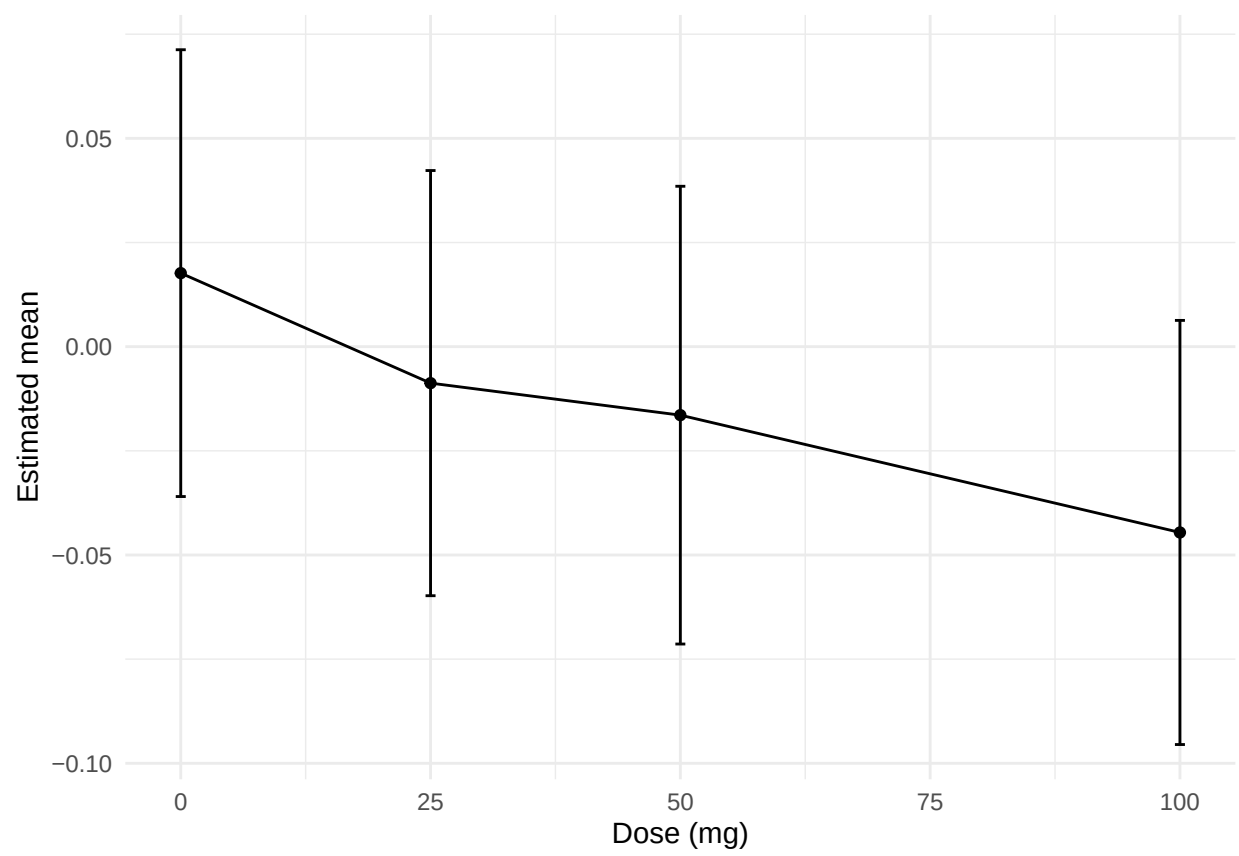
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.02 (-0.04 to 0.07)
25 mg	-0.01 (-0.06 to 0.04)
50 mg	-0.02 (-0.07 to 0.04)
100 mg	-0.04 (-0.10 to 0.01)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.03 (-0.12 to 0.06)
50 mg vs 0 mg	-0.03 (-0.13 to 0.06)
100 mg vs 0 mg	-0.06 (-0.15 to 0.03)

Plot



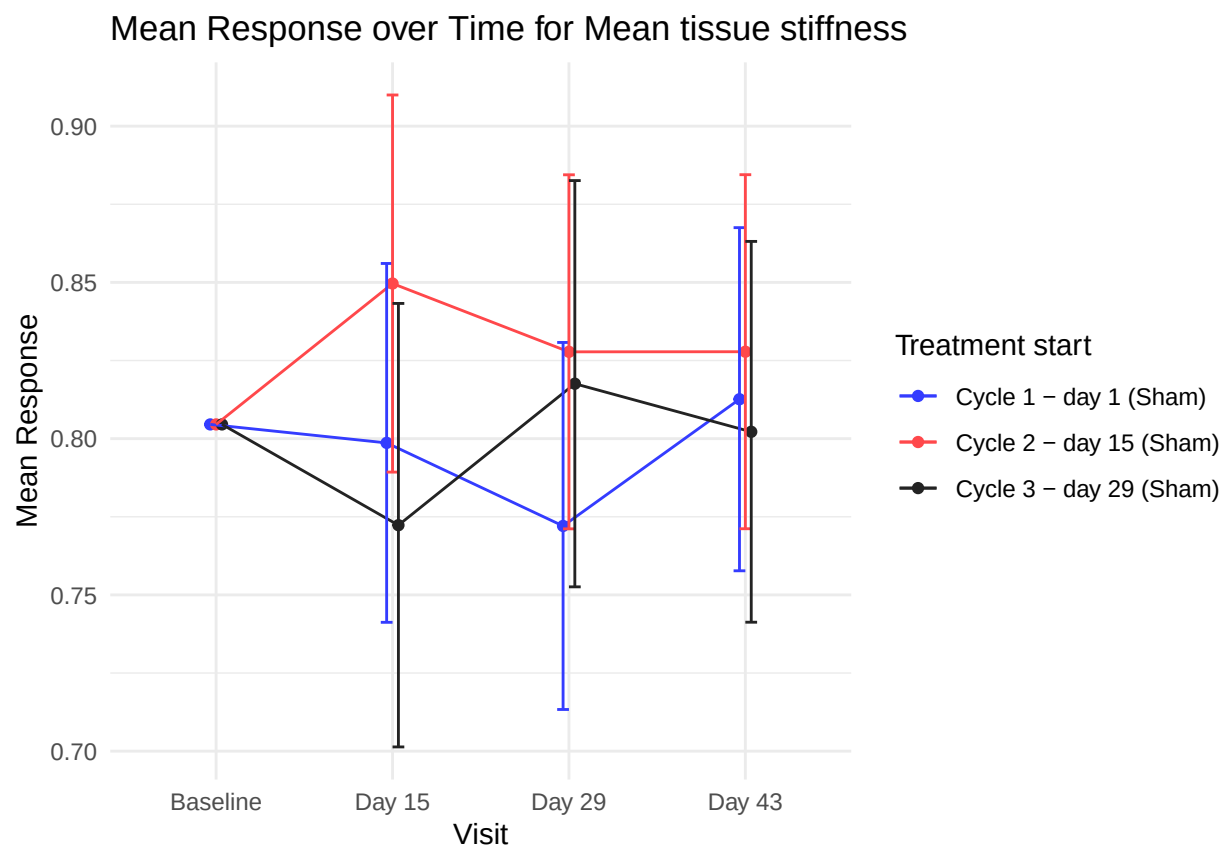
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.04 (-0.11 to 0.02)

Mean tissue stiffness

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



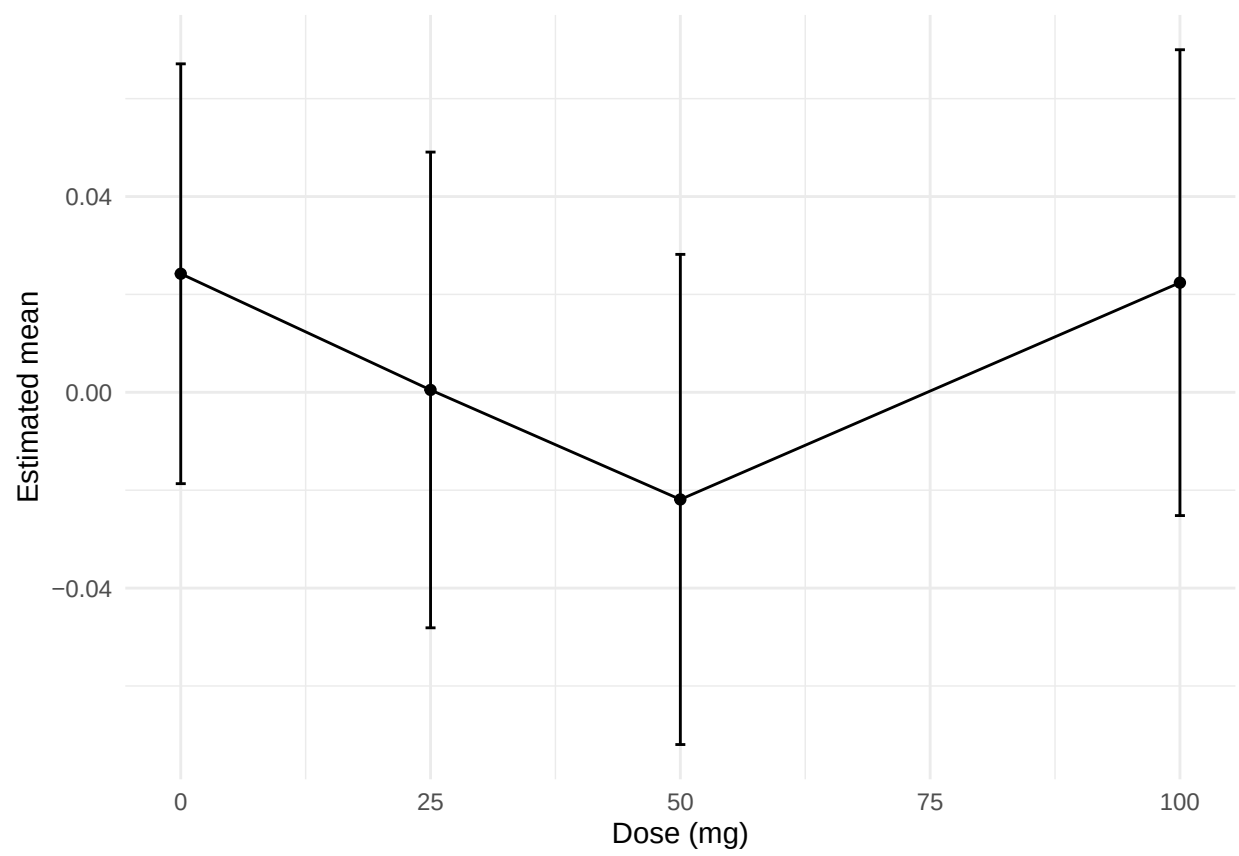
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.02 (-0.02 to 0.07)
25 mg	0.00 (-0.05 to 0.05)
50 mg	-0.02 (-0.07 to 0.03)
100 mg	0.02 (-0.03 to 0.07)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.02 (-0.10 to 0.05)
50 mg vs 0 mg	-0.05 (-0.12 to 0.03)
100 mg vs 0 mg	0.00 (-0.07 to 0.07)

Plot



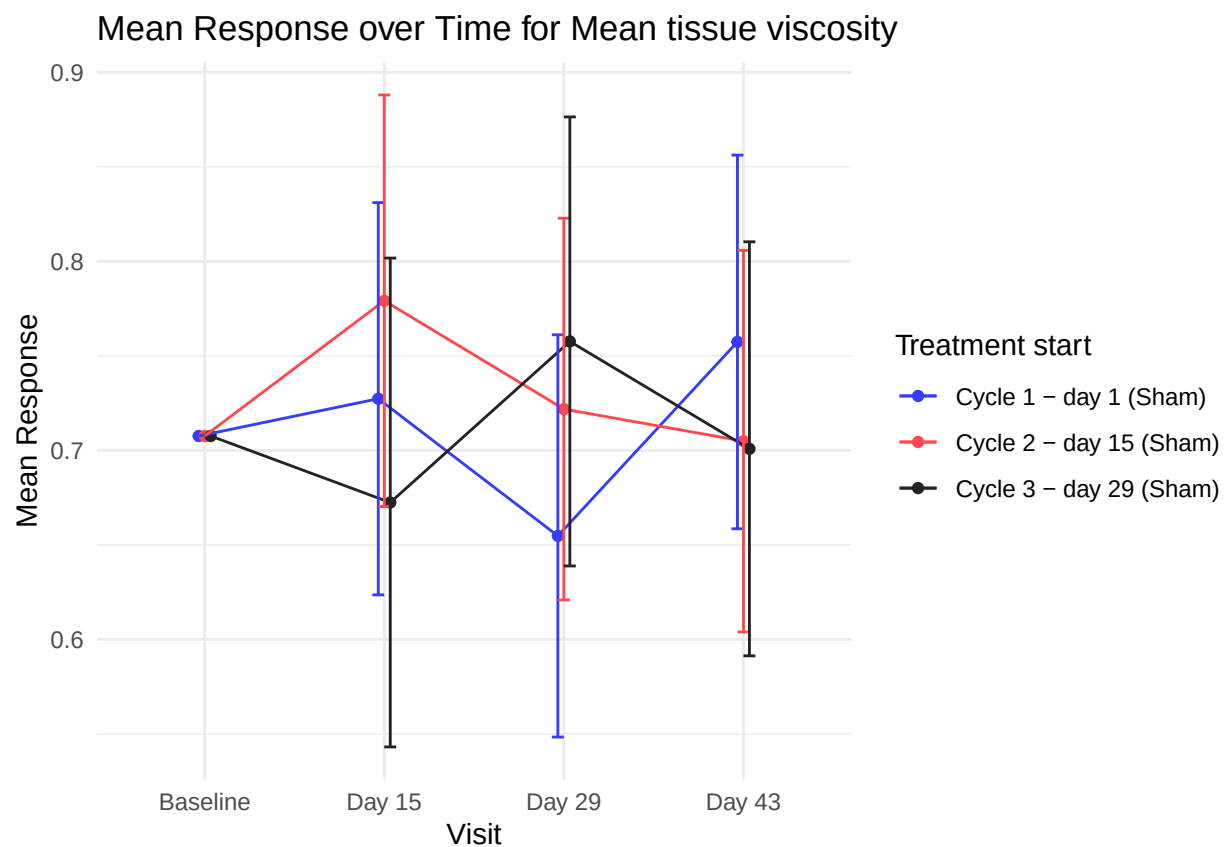
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.02 (-0.07 to 0.02)

Mean tissue viscosity

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



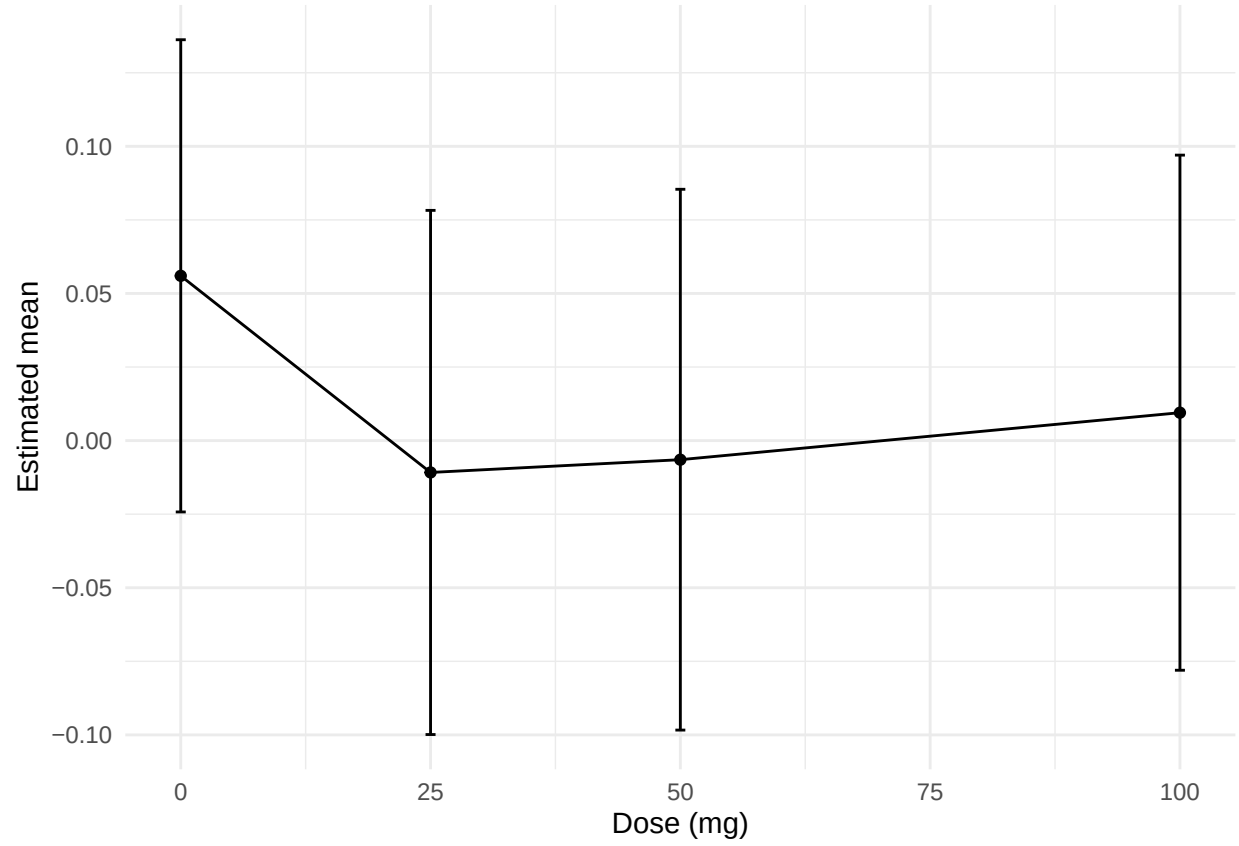
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.06 (-0.02 to 0.14)
25 mg	-0.01 (-0.10 to 0.08)
50 mg	-0.01 (-0.10 to 0.09)
100 mg	0.01 (-0.08 to 0.10)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.07 (-0.20 to 0.07)
50 mg vs 0 mg	-0.06 (-0.20 to 0.08)
100 mg vs 0 mg	-0.05 (-0.17 to 0.08)

Plot



Supplementary analysis: any dose vs 0 mg

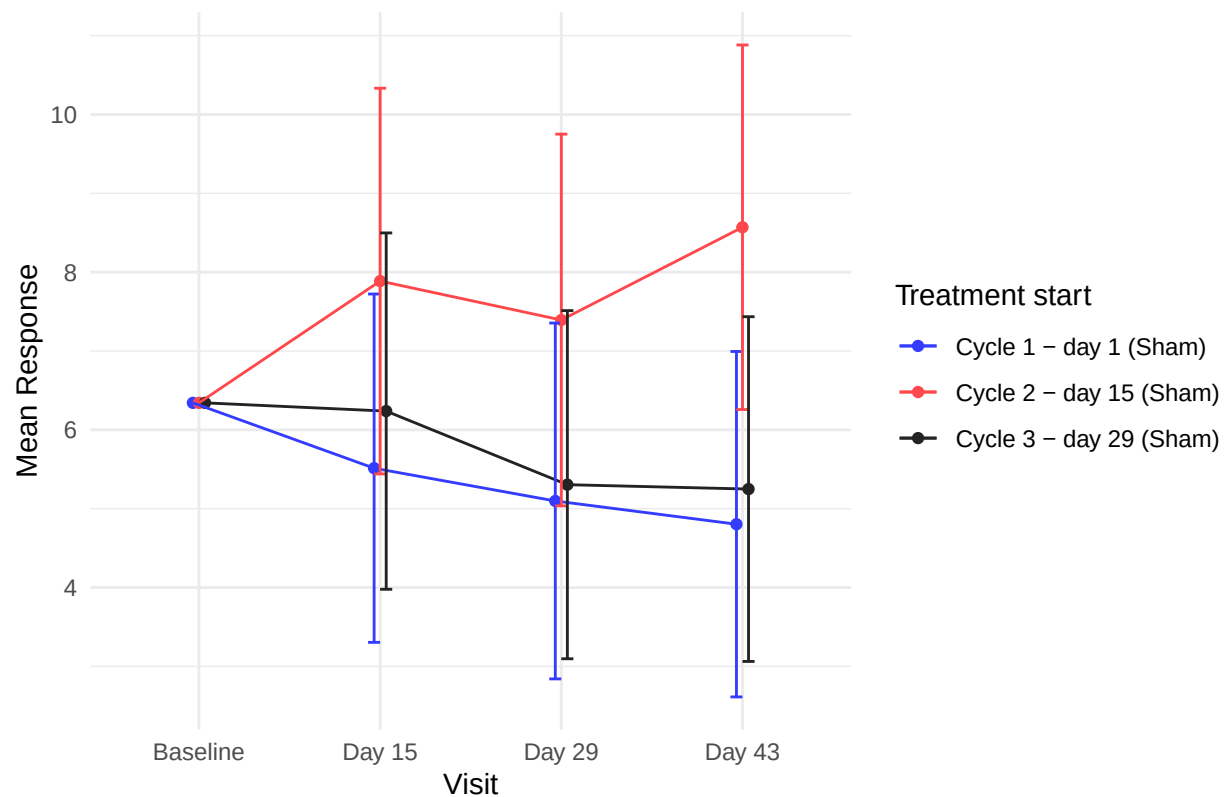
Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.06 (-0.14 to 0.03)

Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start

Mean Response over Time for Total volume of contrast agent enhancing les



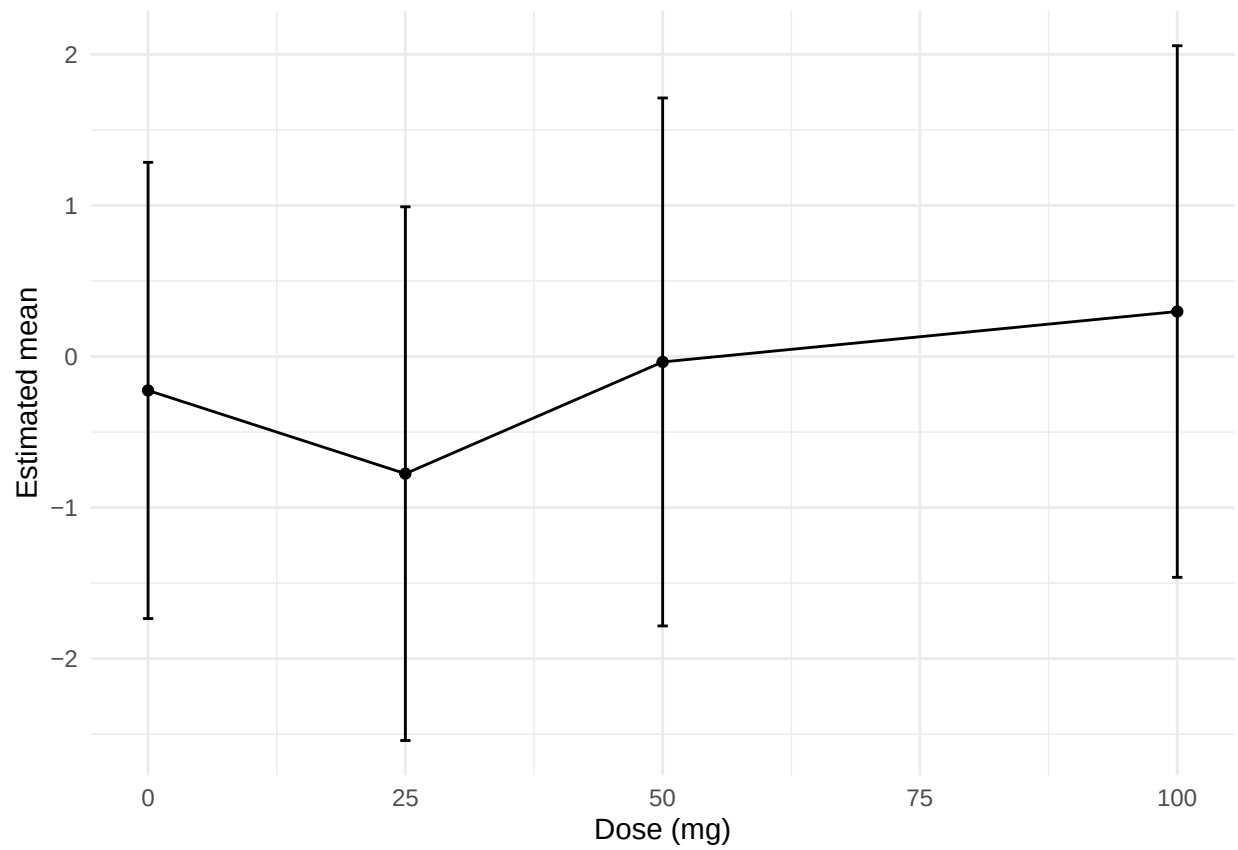
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.22 (-1.73 to 1.29)
25 mg	-0.78 (-2.54 to 0.99)
50 mg	-0.04 (-1.78 to 1.71)
100 mg	0.30 (-1.46 to 2.06)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.55 (-2.70 to 1.60)
50 mg vs 0 mg	0.19 (-2.01 to 2.39)
100 mg vs 0 mg	0.52 (-1.66 to 2.70)

Plot



Supplementary analysis: any dose vs 0 mg

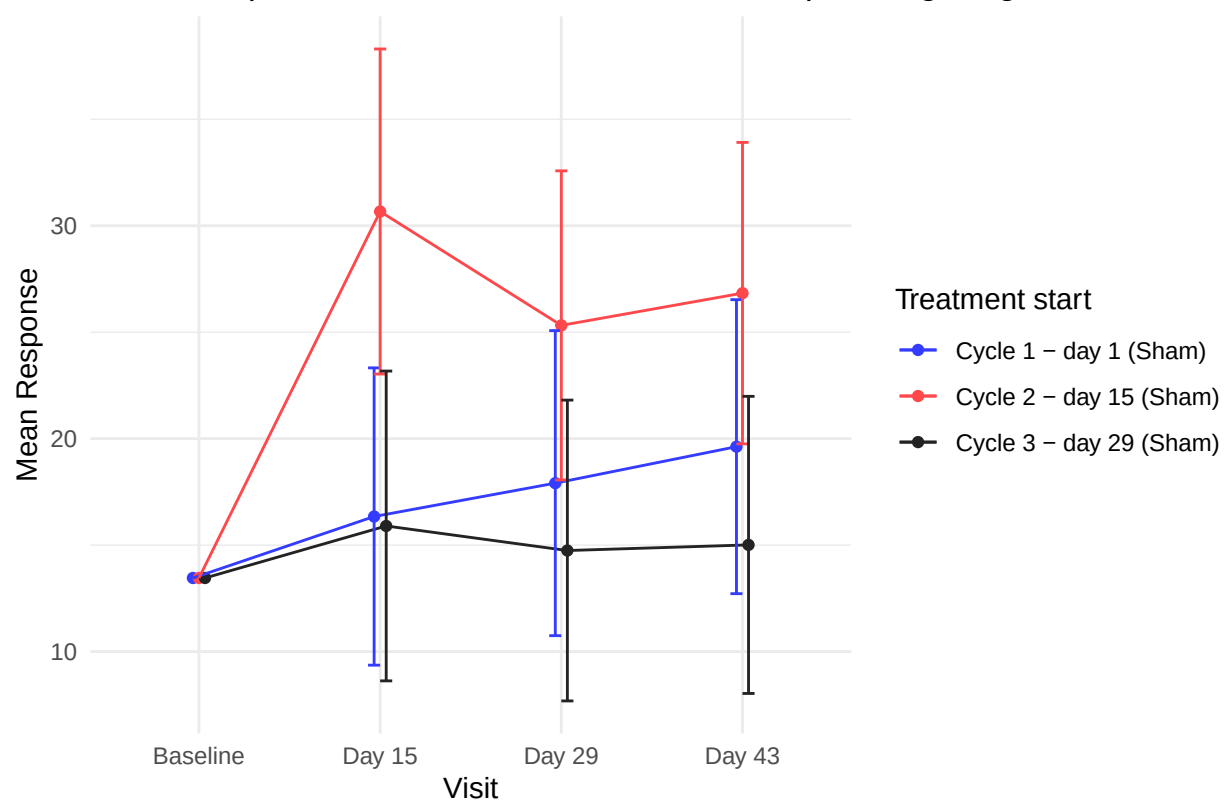
Comparison	Difference (95% CI)
Any dose vs 0 mg	0.04 (-1.27 to 1.36)

Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start

Mean Response over Time for Total volume of pathologic signal on T2/FLAIR



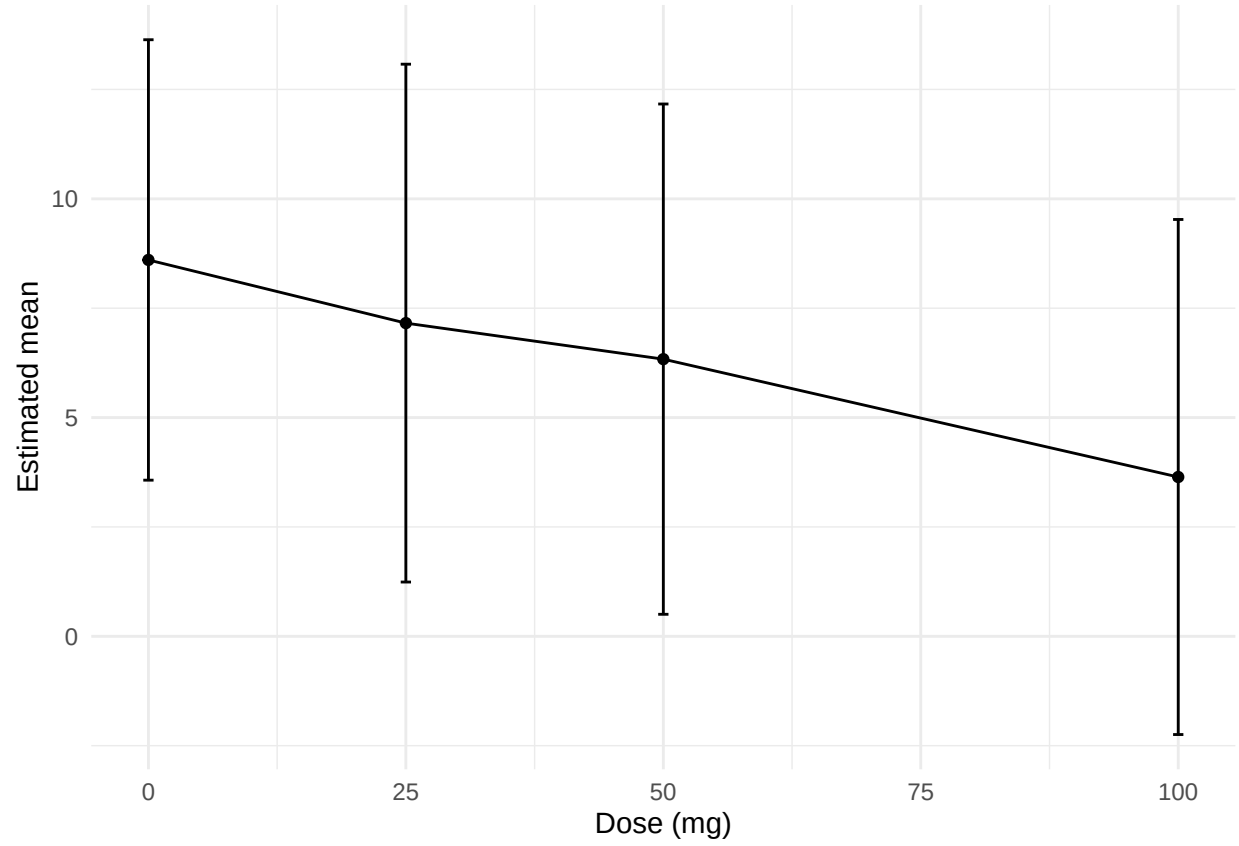
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	8.60 (3.57 to 13.64)
25 mg	7.16 (1.24 to 13.08)
50 mg	6.34 (0.50 to 12.17)
100 mg	3.64 (-2.25 to 9.53)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-1.44 (-8.97 to 6.09)
50 mg vs 0 mg	-2.27 (-9.94 to 5.41)
100 mg vs 0 mg	-4.96 (-12.56 to 2.64)

Plot



Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-2.91 (-7.54 to 1.73)

MRI Endpoints at Cycle 11 (Day 141)

For all MRI endpoints, we report estimated change from baseline at Cycle 11 (day 141), grouped by the extended treatment variable. The summaries include mean estimates, contrasts versus the reference group, and a plot with confidence intervals.

Mean DSC K2 leakage

A total of 47 subjects were included in the analysis for this endpoint.

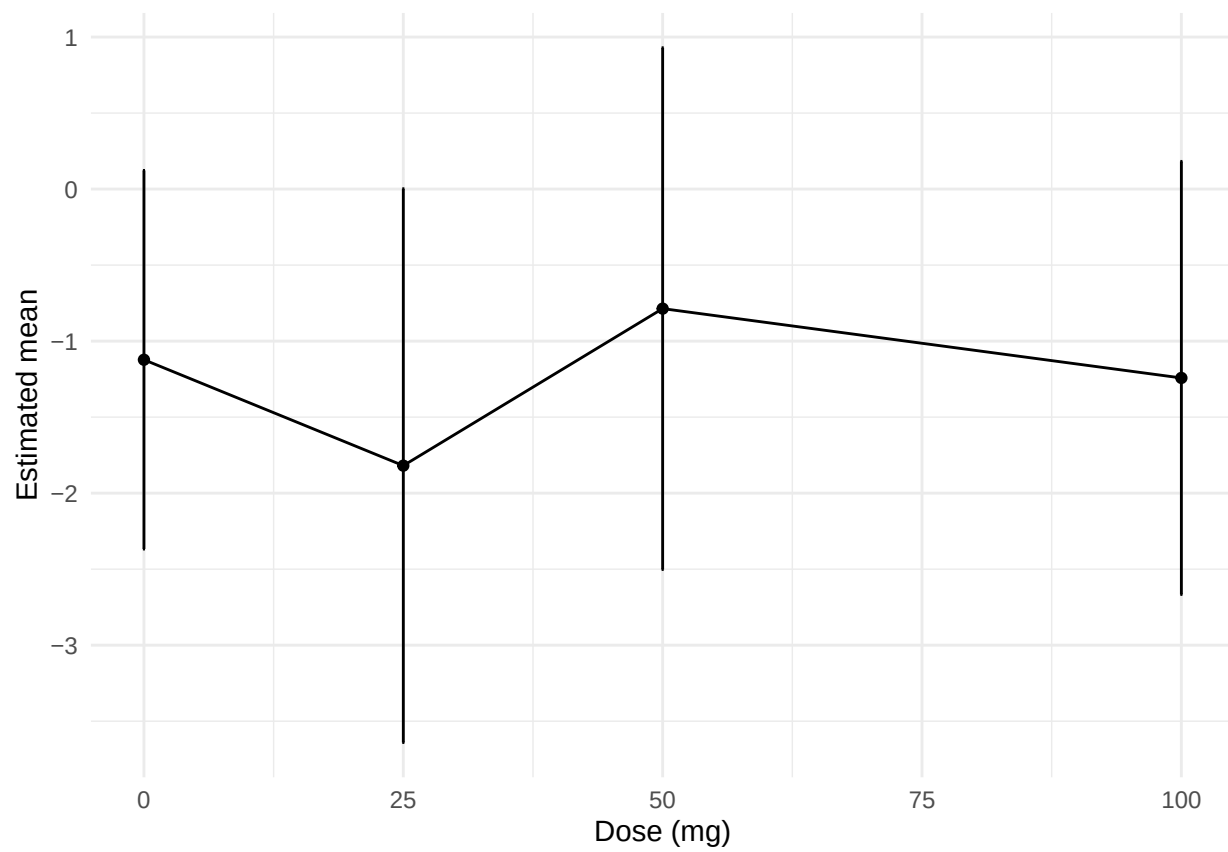
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-1.12 (-2.37 to 0.12)
losartan 25 mg	-1.82 (-3.64 to 0.00)
losartan 50 mg	-0.79 (-2.50 to 0.93)
losartan 100 mg	-1.24 (-2.67 to 0.18)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.70 (-3.38 to 1.99)
losartan 50 mg - losartan 0 mg	0.34 (-2.23 to 2.90)
losartan 100 mg - losartan 0 mg	-0.12 (-2.42 to 2.18)

Plot



Mean relative DSC extravasation

A total of 20 subjects were included in the analysis for this endpoint.

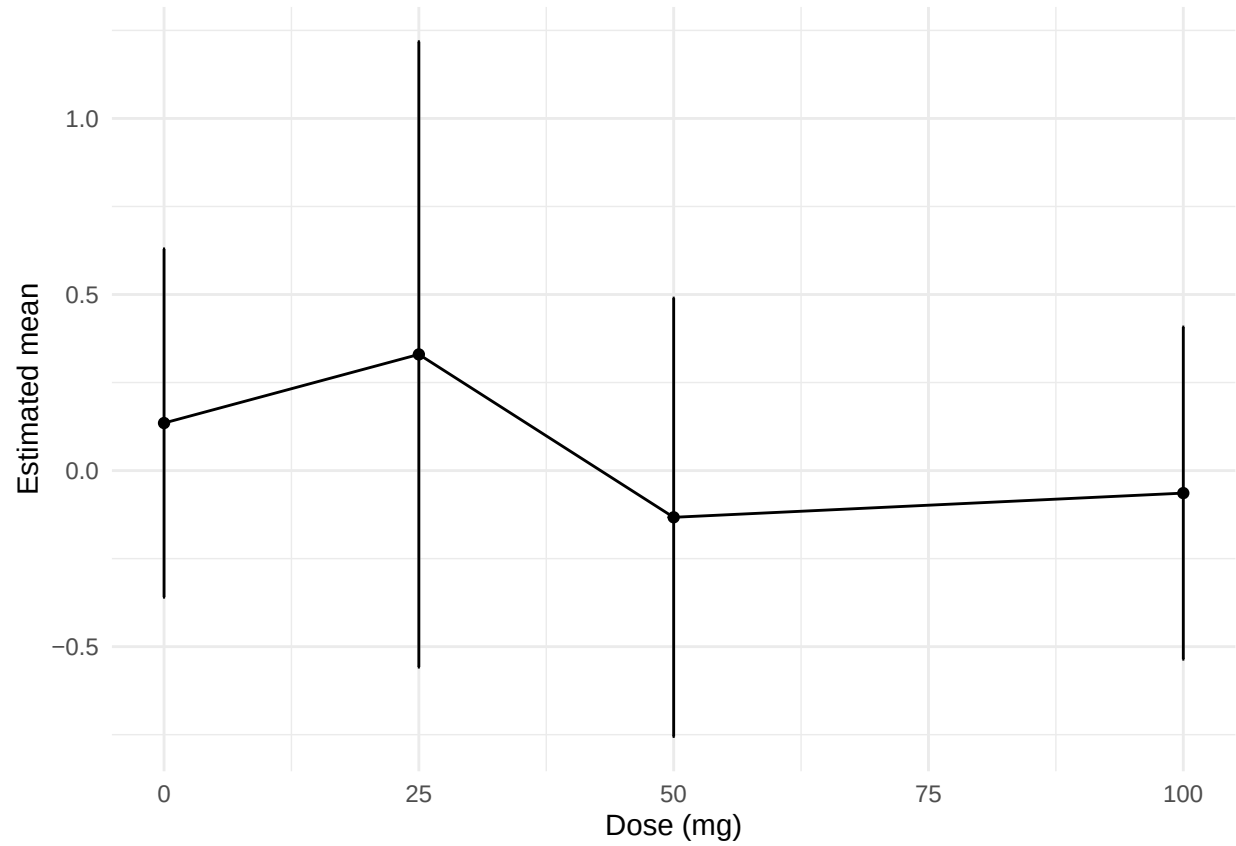
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.13 (-0.36 to 0.63)
losartan 25 mg	0.33 (-0.56 to 1.22)
losartan 50 mg	-0.13 (-0.76 to 0.49)
losartan 100 mg	-0.06 (-0.54 to 0.41)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.19 (-1.09 to 1.48)
losartan 50 mg - losartan 0 mg	-0.27 (-1.27 to 0.73)
losartan 100 mg - losartan 0 mg	-0.20 (-1.06 to 0.66)

Plot



Mean relative apparent diffusion coefficient

A total of 43 subjects were included in the analysis for this endpoint.

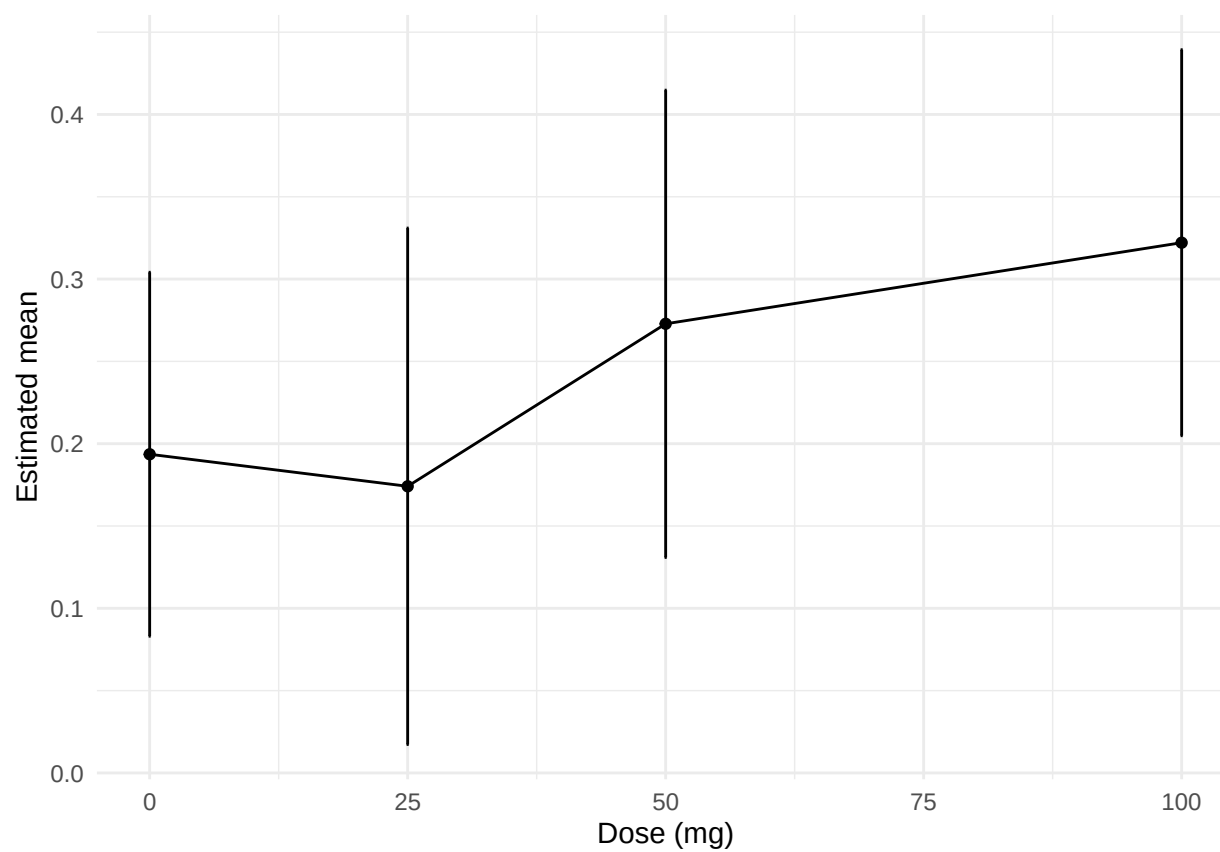
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.19 (0.08 to 0.30)
losartan 25 mg	0.17 (0.02 to 0.33)
losartan 50 mg	0.27 (0.13 to 0.41)
losartan 100 mg	0.32 (0.20 to 0.44)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.02 (-0.25 to 0.22)
losartan 50 mg - losartan 0 mg	0.08 (-0.14 to 0.29)
losartan 100 mg - losartan 0 mg	0.13 (-0.07 to 0.33)

Plot



Mean relative cerebral blood

A total of 47 subjects were included in the analysis for this endpoint.

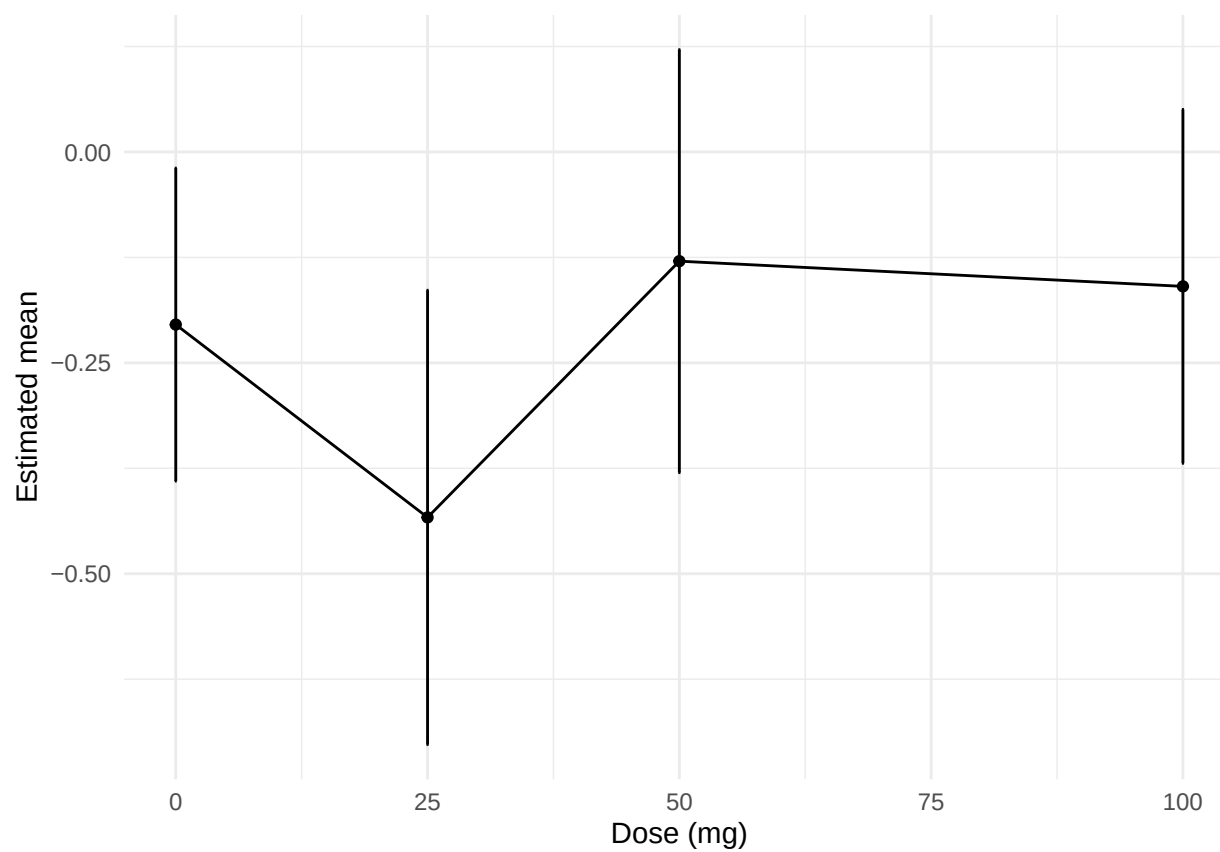
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.20 (-0.39 to -0.02)
losartan 25 mg	-0.43 (-0.70 to -0.16)
losartan 50 mg	-0.13 (-0.38 to 0.12)
losartan 100 mg	-0.16 (-0.37 to 0.05)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.23 (-0.63 to 0.17)
losartan 50 mg - losartan 0 mg	0.08 (-0.30 to 0.45)
losartan 100 mg - losartan 0 mg	0.05 (-0.30 to 0.39)

Plot



Mean relative cerebral blood flow (PTA-ROI)

A total of 47 subjects were included in the analysis for this endpoint.

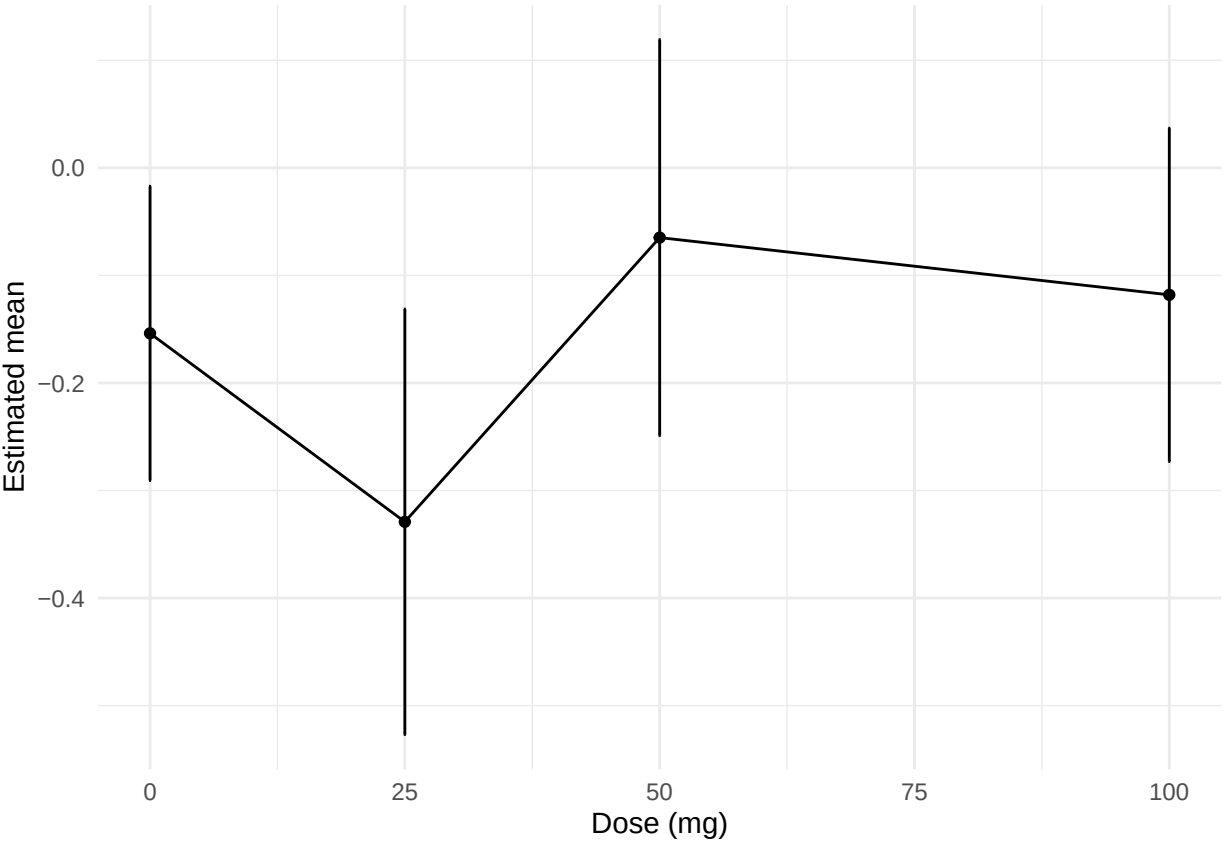
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.15 (-0.29 to -0.02)
losartan 25 mg	-0.33 (-0.53 to -0.13)
losartan 50 mg	-0.06 (-0.25 to 0.12)
losartan 100 mg	-0.12 (-0.27 to 0.04)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.18 (-0.47 to 0.12)
losartan 50 mg - losartan 0 mg	0.09 (-0.19 to 0.37)
losartan 100 mg - losartan 0 mg	0.04 (-0.22 to 0.29)

Plot



Mean relative cerebral blood flow (ROI2)

A total of 40 subjects were included in the analysis for this endpoint.

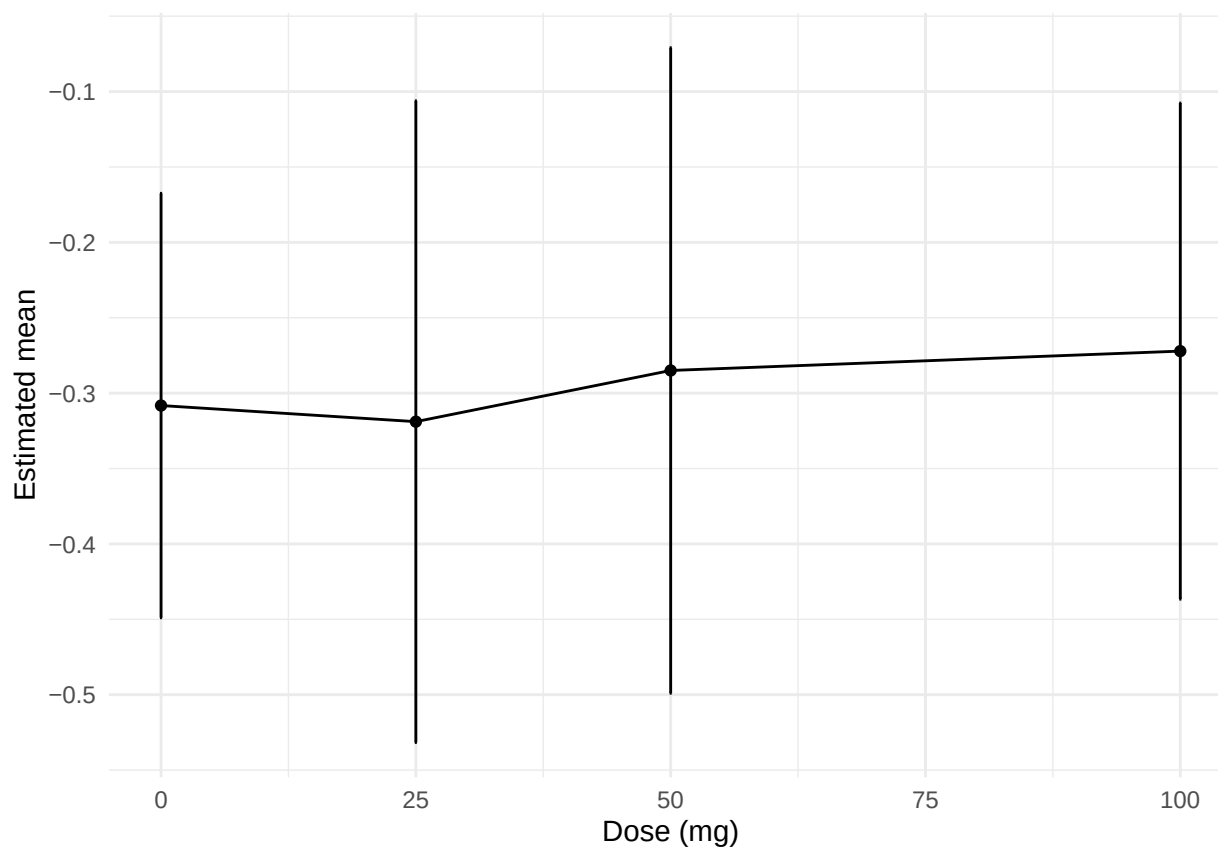
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.31 (-0.45 to -0.17)
losartan 25 mg	-0.32 (-0.53 to -0.11)
losartan 50 mg	-0.28 (-0.50 to -0.07)
losartan 100 mg	-0.27 (-0.44 to -0.11)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.01 (-0.32 to 0.30)
losartan 50 mg - losartan 0 mg	0.02 (-0.29 to 0.33)
losartan 100 mg - losartan 0 mg	0.04 (-0.23 to 0.30)

Plot



Mean relative direction value

A total of 20 subjects were included in the analysis for this endpoint.

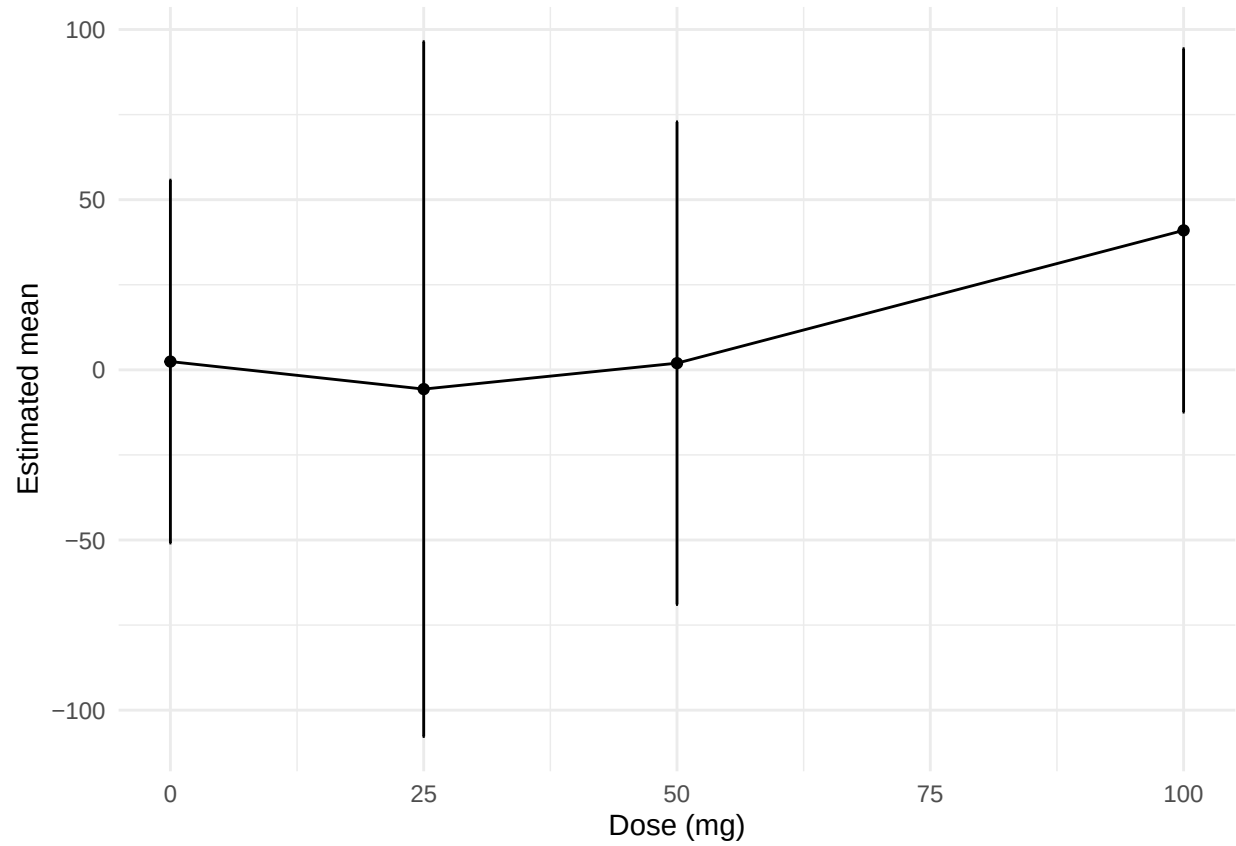
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	2.43 (-50.88 to 55.73)
losartan 25 mg	-5.66 (-107.75 to 96.42)
losartan 50 mg	1.96 (-68.94 to 72.86)
losartan 100 mg	40.97 (-12.42 to 94.37)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-8.09 (-149.82 to 133.64)
losartan 50 mg - losartan 0 mg	-0.47 (-110.03 to 109.09)
losartan 100 mg - losartan 0 mg	38.55 (-54.59 to 131.69)

Plot



Mean relative fractional anisotropy

A total of 40 subjects were included in the analysis for this endpoint.

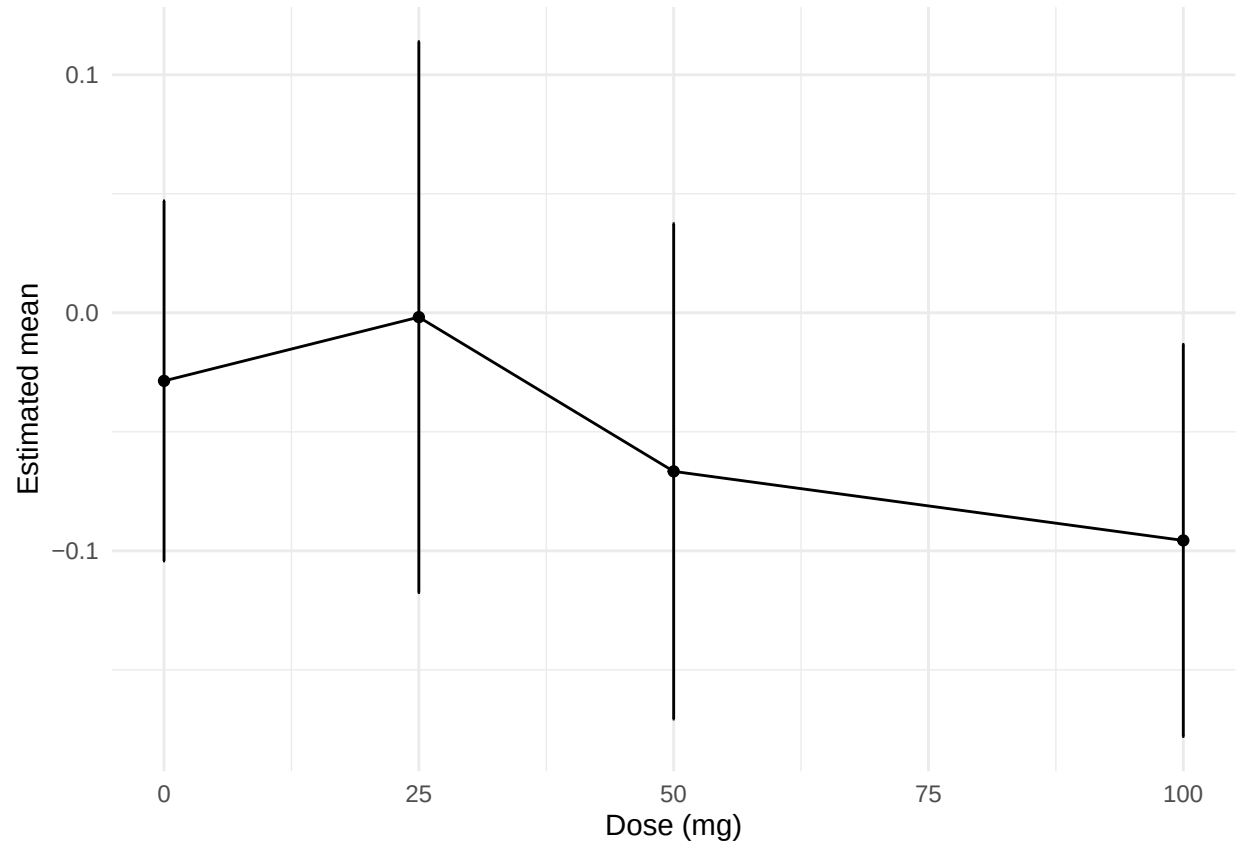
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.03 (-0.10 to 0.05)
losartan 25 mg	0.00 (-0.12 to 0.11)
losartan 50 mg	-0.07 (-0.17 to 0.04)
losartan 100 mg	-0.10 (-0.18 to -0.01)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.03 (-0.14 to 0.19)
losartan 50 mg - losartan 0 mg	-0.04 (-0.20 to 0.12)
losartan 100 mg - losartan 0 mg	-0.07 (-0.20 to 0.07)

Plot



Mean relative restricted spectrum imaging

A total of 40 subjects were included in the analysis for this endpoint.

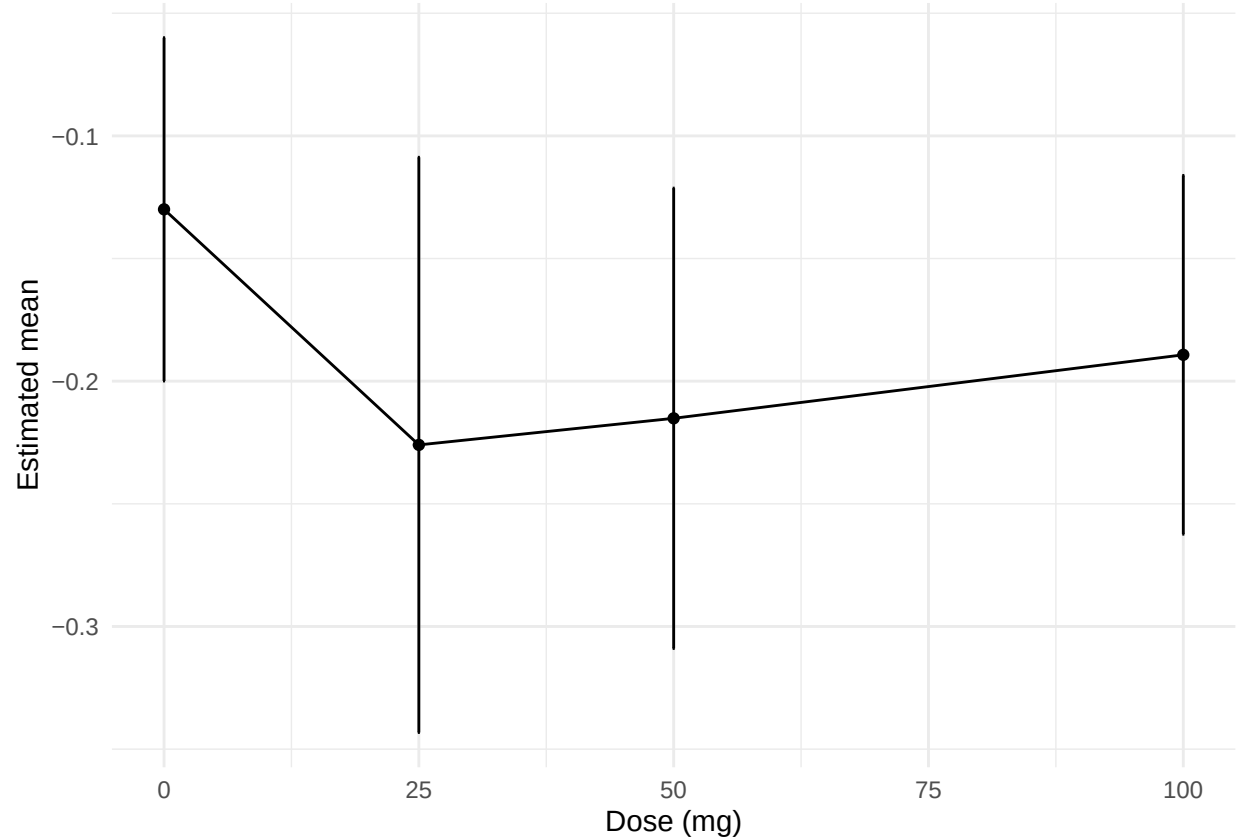
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.13 (-0.20 to -0.06)
losartan 25 mg	-0.23 (-0.34 to -0.11)
losartan 50 mg	-0.22 (-0.31 to -0.12)
losartan 100 mg	-0.19 (-0.26 to -0.12)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.10 (-0.26 to 0.07)
losartan 50 mg - losartan 0 mg	-0.09 (-0.23 to 0.06)
losartan 100 mg - losartan 0 mg	-0.06 (-0.18 to 0.06)

Plot



Mean relative solid stress (PTA-ROI)

A total of 29 subjects were included in the analysis for this endpoint.

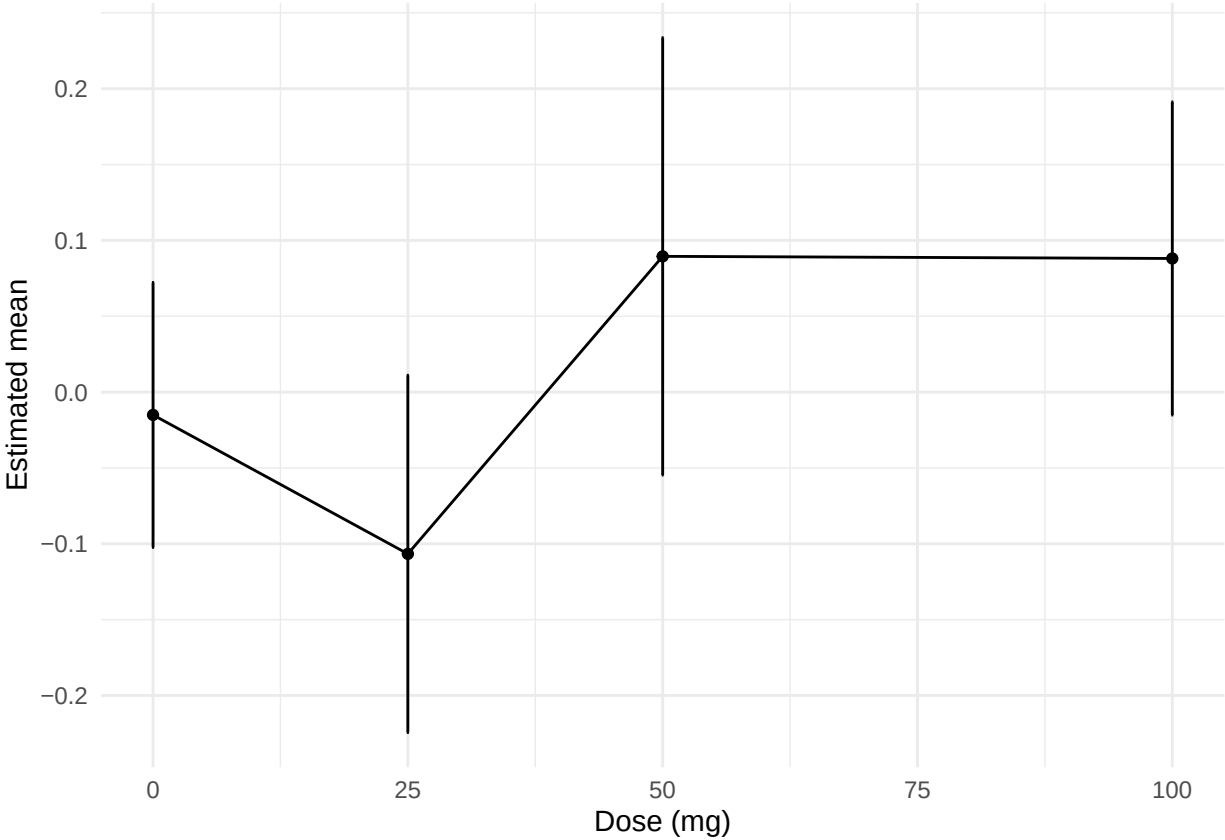
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.02 (-0.10 to 0.07)
losartan 25 mg	-0.11 (-0.22 to 0.01)
losartan 50 mg	0.09 (-0.05 to 0.23)
losartan 100 mg	0.09 (-0.02 to 0.19)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.09 (-0.27 to 0.09)
losartan 50 mg - losartan 0 mg	0.10 (-0.10 to 0.31)
losartan 100 mg - losartan 0 mg	0.10 (-0.06 to 0.27)

Plot



Mean relative solid stress (ROI2)

A total of 23 subjects were included in the analysis for this endpoint.

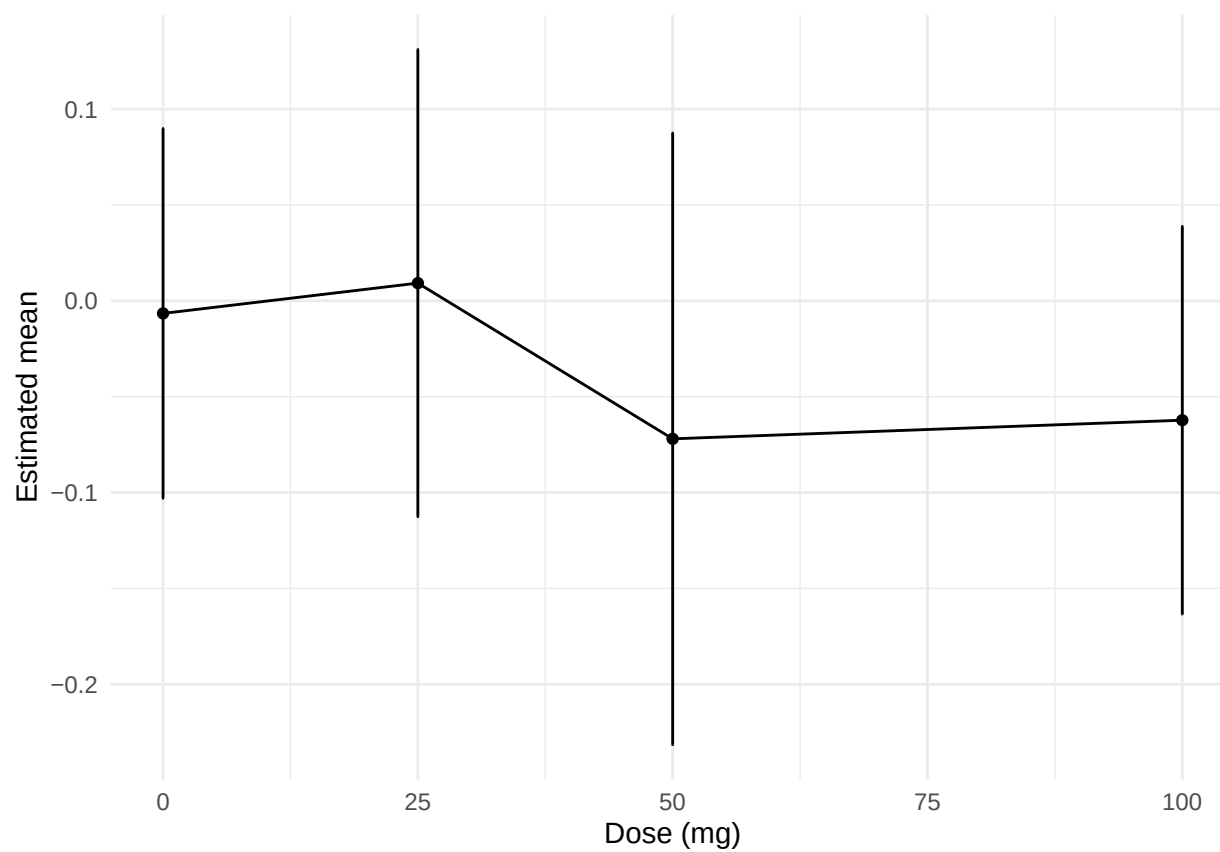
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.01 (-0.10 to 0.09)
losartan 25 mg	0.01 (-0.11 to 0.13)
losartan 50 mg	-0.07 (-0.23 to 0.09)
losartan 100 mg	-0.06 (-0.16 to 0.04)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.02 (-0.18 to 0.21)
losartan 50 mg - losartan 0 mg	-0.07 (-0.30 to 0.17)
losartan 100 mg - losartan 0 mg	-0.06 (-0.22 to 0.11)

Plot



Mean relative transit time

A total of 47 subjects were included in the analysis for this endpoint.

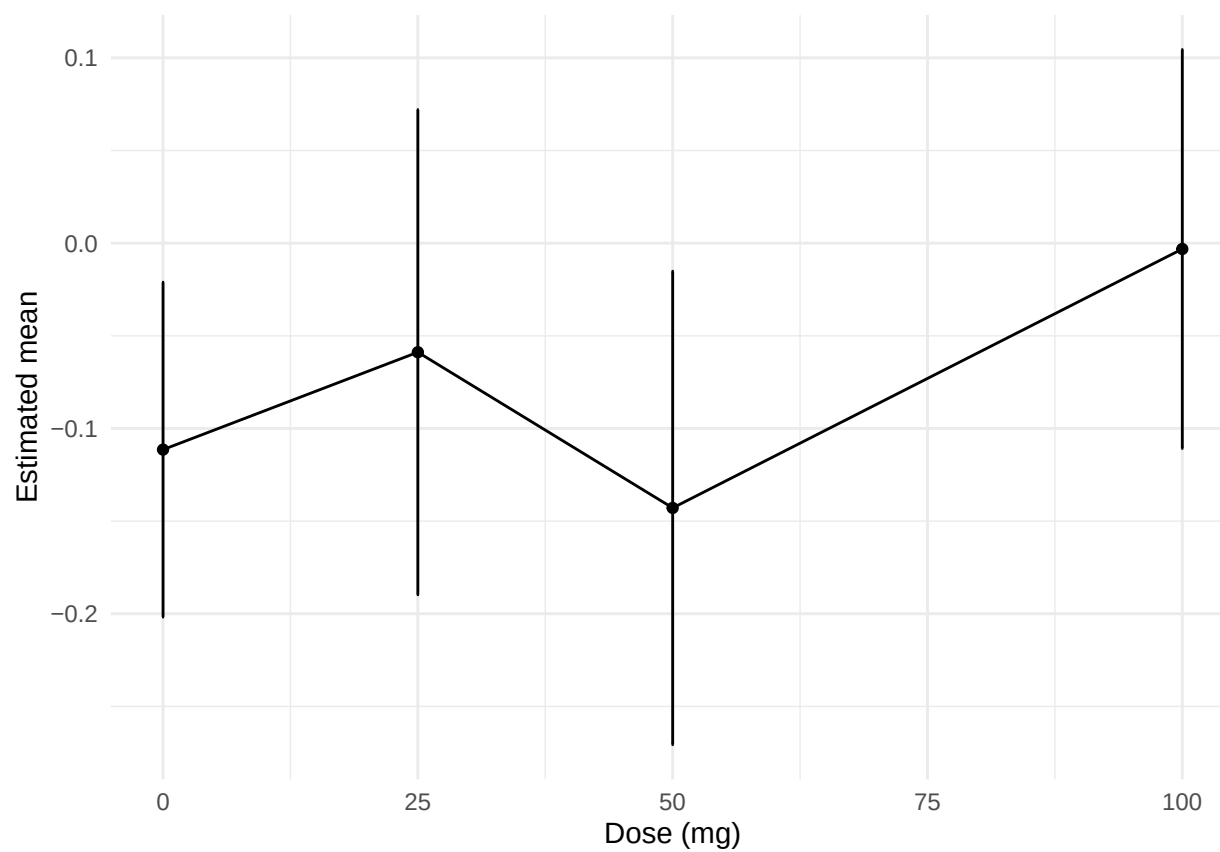
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.11 (-0.20 to -0.02)
losartan 25 mg	-0.06 (-0.19 to 0.07)
losartan 50 mg	-0.14 (-0.27 to -0.02)
losartan 100 mg	0.00 (-0.11 to 0.10)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.05 (-0.14 to 0.25)
losartan 50 mg - losartan 0 mg	-0.03 (-0.22 to 0.15)
losartan 100 mg - losartan 0 mg	0.11 (-0.07 to 0.28)

Plot



Mean relative vessel caliber

A total of 20 subjects were included in the analysis for this endpoint.

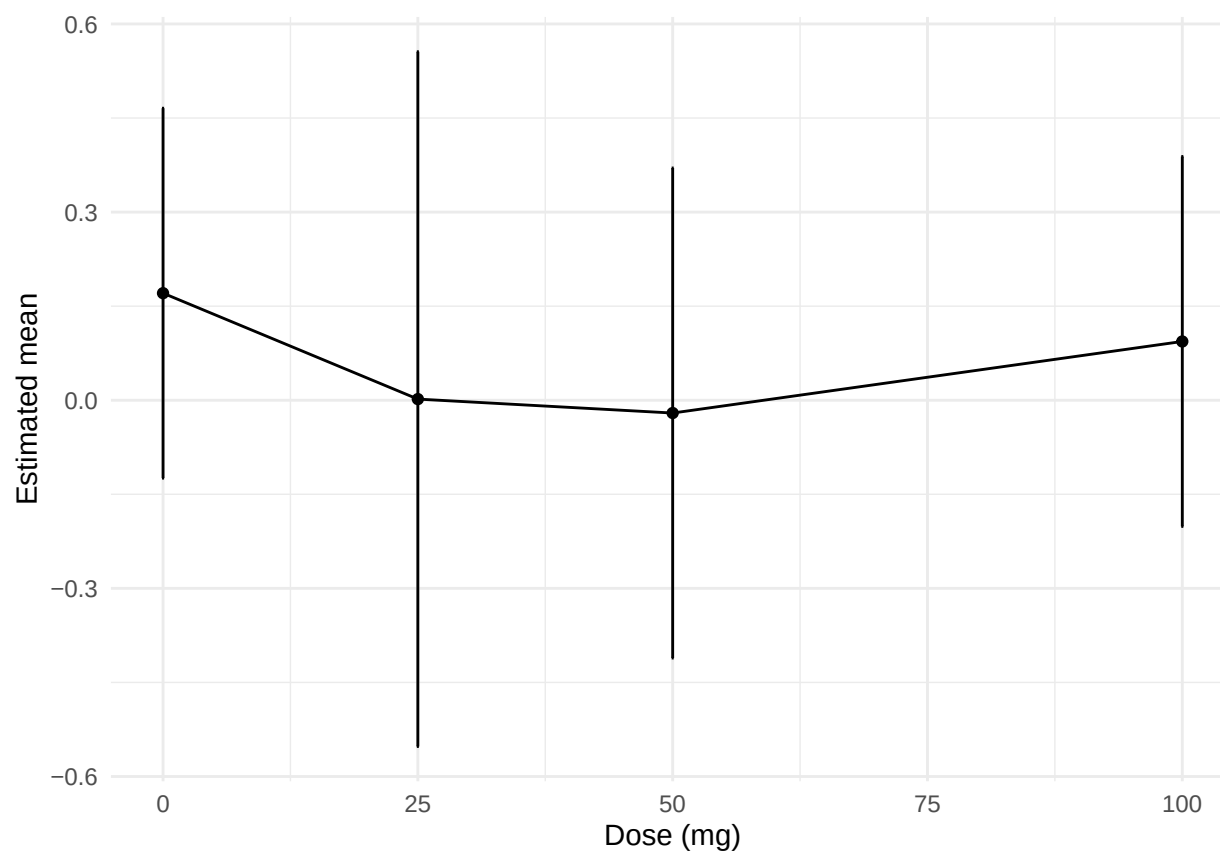
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.17 (-0.12 to 0.47)
losartan 25 mg	0.00 (-0.55 to 0.56)
losartan 50 mg	-0.02 (-0.41 to 0.37)
losartan 100 mg	0.09 (-0.20 to 0.39)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.17 (-0.94 to 0.60)
losartan 50 mg - losartan 0 mg	-0.19 (-0.80 to 0.41)
losartan 100 mg - losartan 0 mg	-0.08 (-0.59 to 0.44)

Plot



Mean relative vessel size index

A total of 20 subjects were included in the analysis for this endpoint.

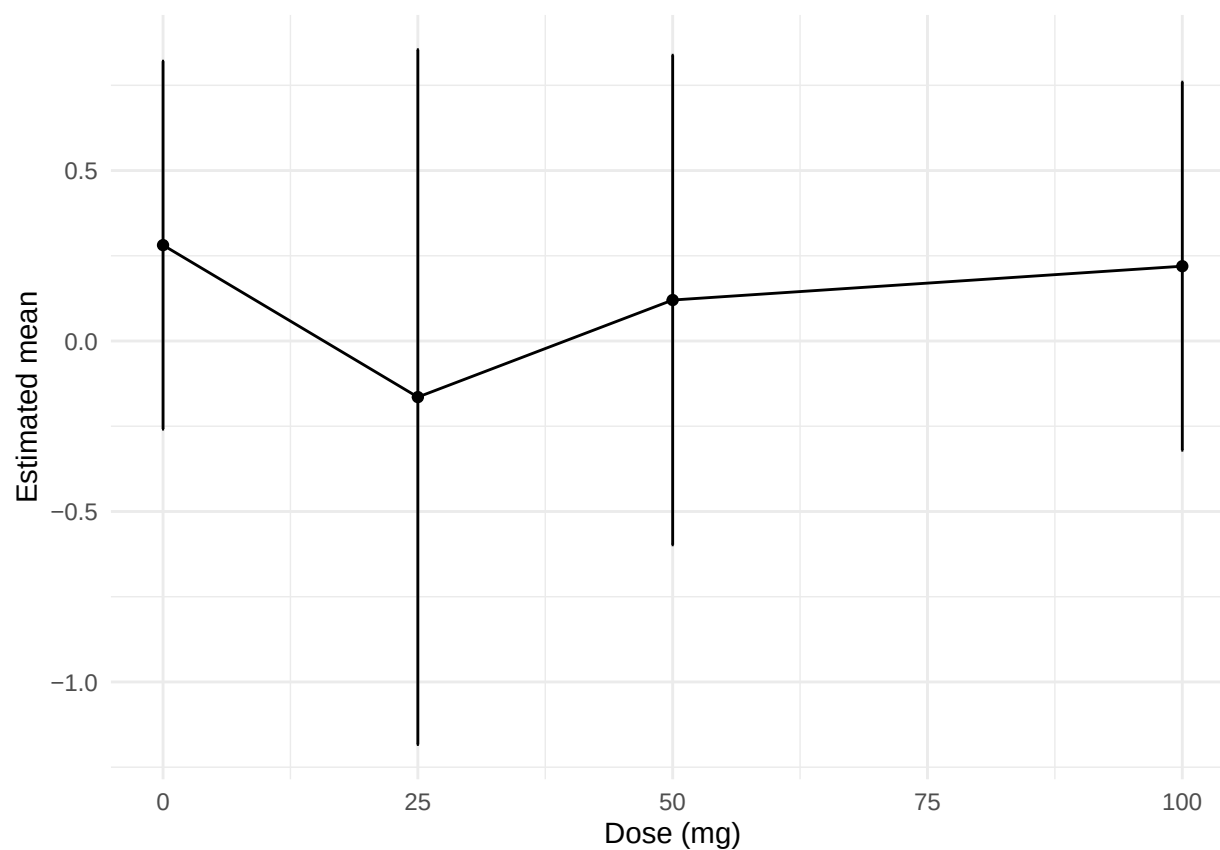
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.28 (-0.26 to 0.82)
losartan 25 mg	-0.16 (-1.18 to 0.85)
losartan 50 mg	0.12 (-0.60 to 0.84)
losartan 100 mg	0.22 (-0.32 to 0.76)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.45 (-1.87 to 0.98)
losartan 50 mg - losartan 0 mg	-0.16 (-1.27 to 0.95)
losartan 100 mg - losartan 0 mg	-0.06 (-1.00 to 0.88)

Plot



Mean relative vortex area

A total of 20 subjects were included in the analysis for this endpoint.

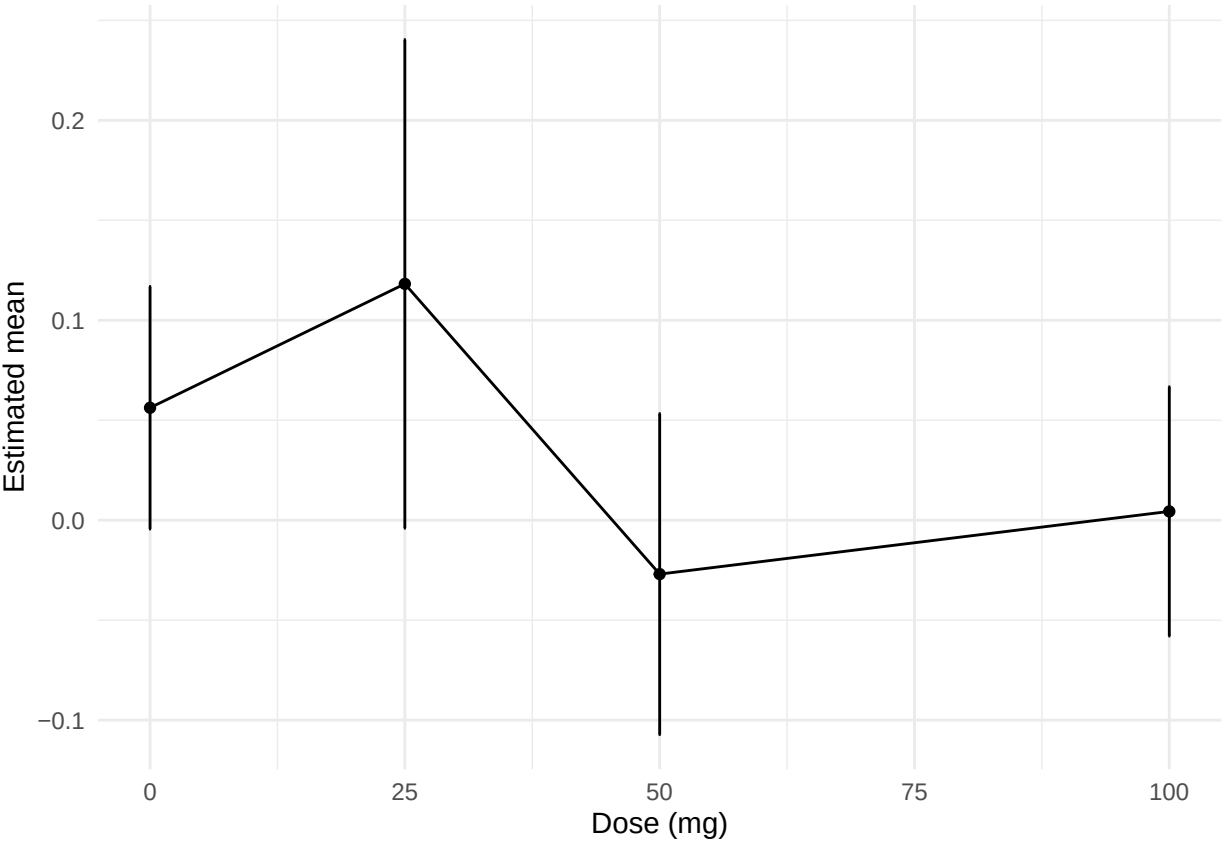
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.06 (0.00 to 0.12)
losartan 25 mg	0.12 (0.00 to 0.24)
losartan 50 mg	-0.03 (-0.11 to 0.05)
losartan 100 mg	0.00 (-0.06 to 0.07)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.06 (-0.11 to 0.23)
losartan 50 mg - losartan 0 mg	-0.08 (-0.21 to 0.04)
losartan 100 mg - losartan 0 mg	-0.05 (-0.16 to 0.06)

Plot



Mean tissue elasticity

A total of 29 subjects were included in the analysis for this endpoint.

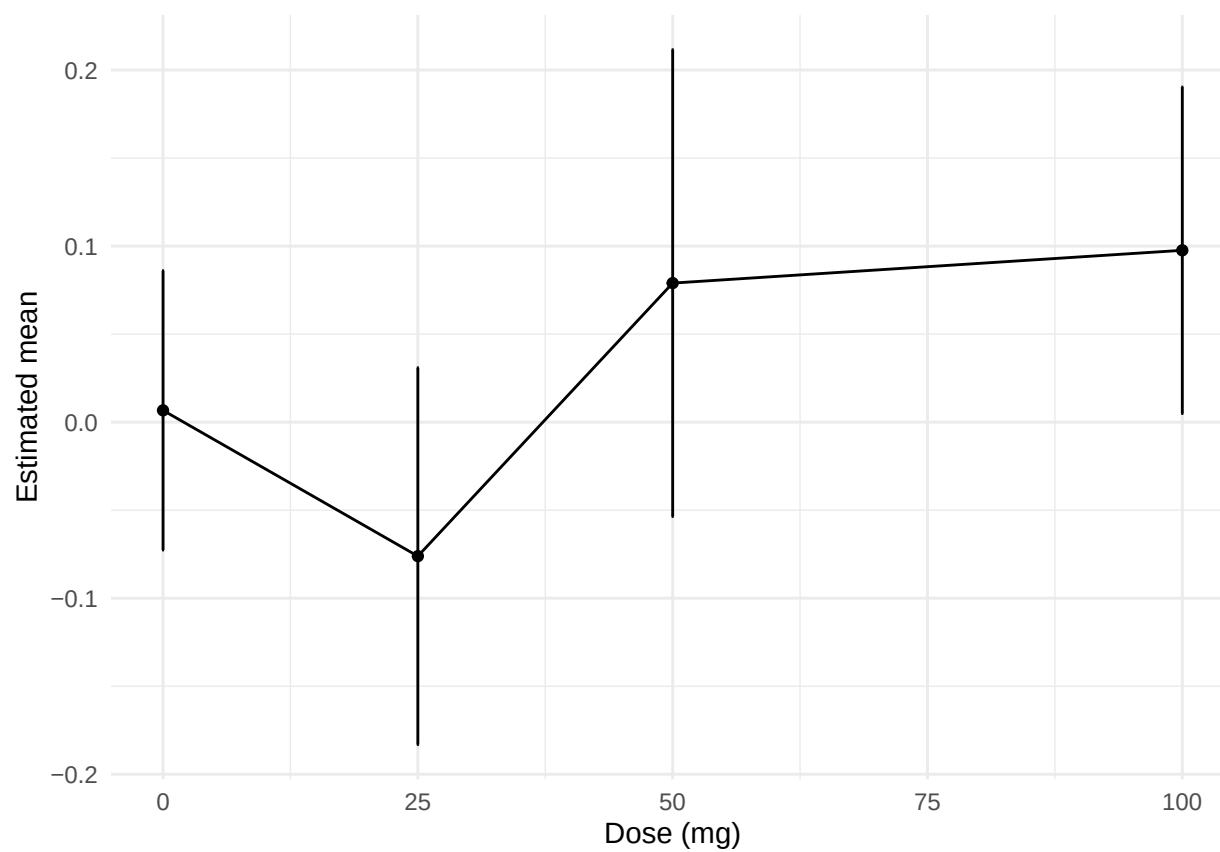
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.01 (-0.07 to 0.09)
losartan 25 mg	-0.08 (-0.18 to 0.03)
losartan 50 mg	0.08 (-0.05 to 0.21)
losartan 100 mg	0.10 (0.00 to 0.19)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.08 (-0.24 to 0.08)
losartan 50 mg - losartan 0 mg	0.07 (-0.12 to 0.26)
losartan 100 mg - losartan 0 mg	0.09 (-0.06 to 0.24)

Plot



Mean tissue relative viscosity

A total of 29 subjects were included in the analysis for this endpoint.

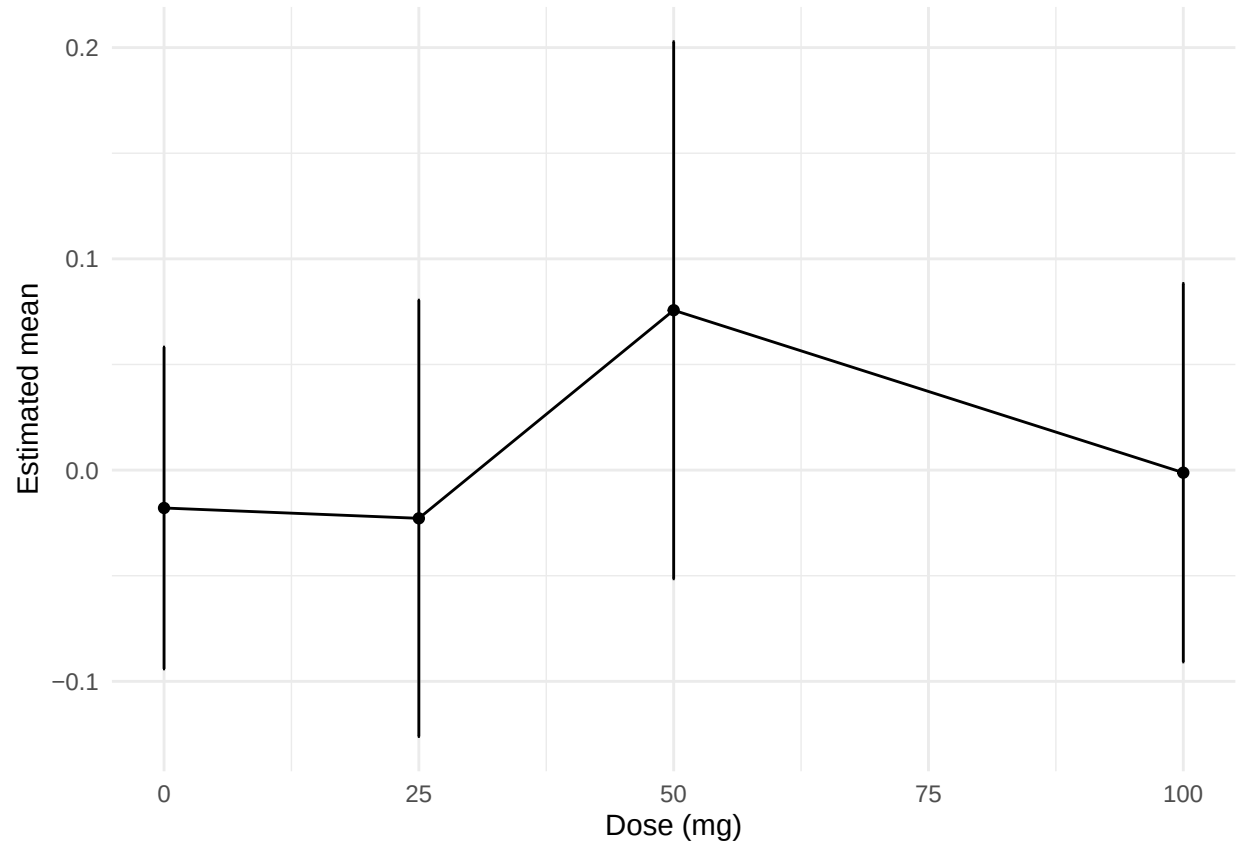
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.02 (-0.09 to 0.06)
losartan 25 mg	-0.02 (-0.13 to 0.08)
losartan 50 mg	0.08 (-0.05 to 0.20)
losartan 100 mg	0.00 (-0.09 to 0.09)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.00 (-0.16 to 0.15)
losartan 50 mg - losartan 0 mg	0.09 (-0.09 to 0.28)
losartan 100 mg - losartan 0 mg	0.02 (-0.13 to 0.16)

Plot



Mean tissue stiffness

A total of 29 subjects were included in the analysis for this endpoint.

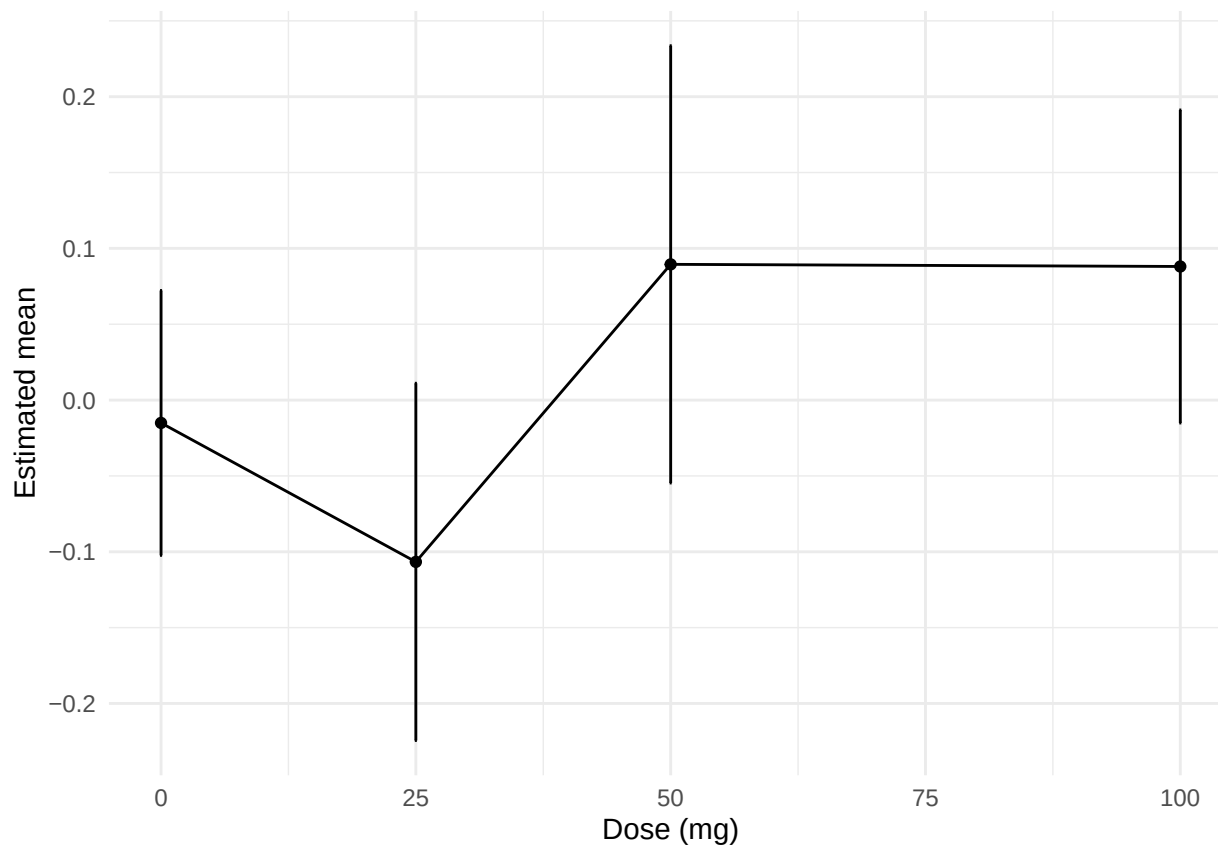
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.02 (-0.10 to 0.07)
losartan 25 mg	-0.11 (-0.22 to 0.01)
losartan 50 mg	0.09 (-0.05 to 0.23)
losartan 100 mg	0.09 (-0.02 to 0.19)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.09 (-0.27 to 0.09)
losartan 50 mg - losartan 0 mg	0.10 (-0.10 to 0.31)
losartan 100 mg - losartan 0 mg	0.10 (-0.06 to 0.27)

Plot



Mean tissue viscosity

A total of 29 subjects were included in the analysis for this endpoint.

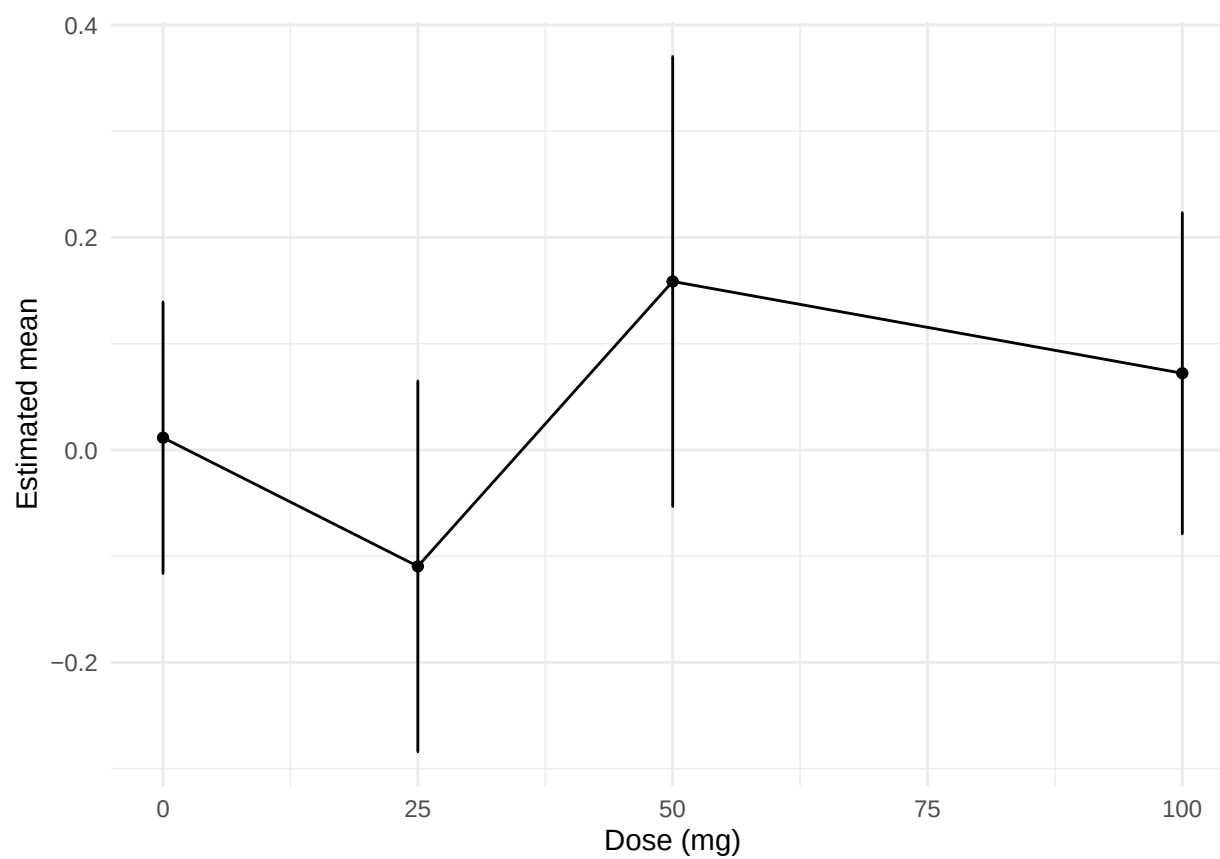
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.01 (-0.12 to 0.14)
losartan 25 mg	-0.11 (-0.28 to 0.06)
losartan 50 mg	0.16 (-0.05 to 0.37)
losartan 100 mg	0.07 (-0.08 to 0.22)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.12 (-0.38 to 0.14)
losartan 50 mg - losartan 0 mg	0.15 (-0.15 to 0.45)
losartan 100 mg - losartan 0 mg	0.06 (-0.18 to 0.30)

Plot



Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3

A total of 47 subjects were included in the analysis for this endpoint.

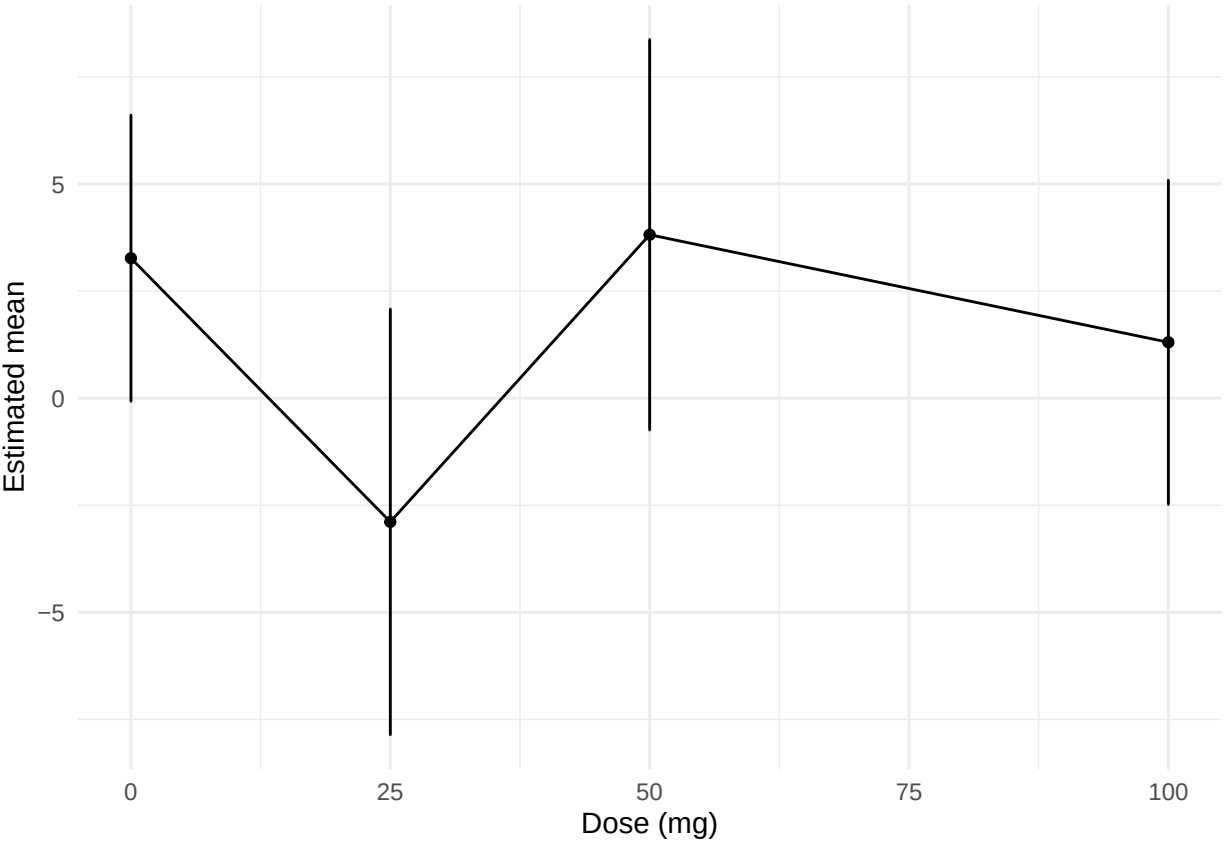
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	3.27 (-0.07 to 6.60)
losartan 25 mg	-2.89 (-7.85 to 2.08)
losartan 50 mg	3.81 (-0.74 to 8.37)
losartan 100 mg	1.30 (-2.47 to 5.08)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-6.15 (-13.52 to 1.21)
losartan 50 mg - losartan 0 mg	0.55 (-6.27 to 7.37)
losartan 100 mg - losartan 0 mg	-1.96 (-8.09 to 4.17)

Plot



Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3

A total of 47 subjects were included in the analysis for this endpoint.

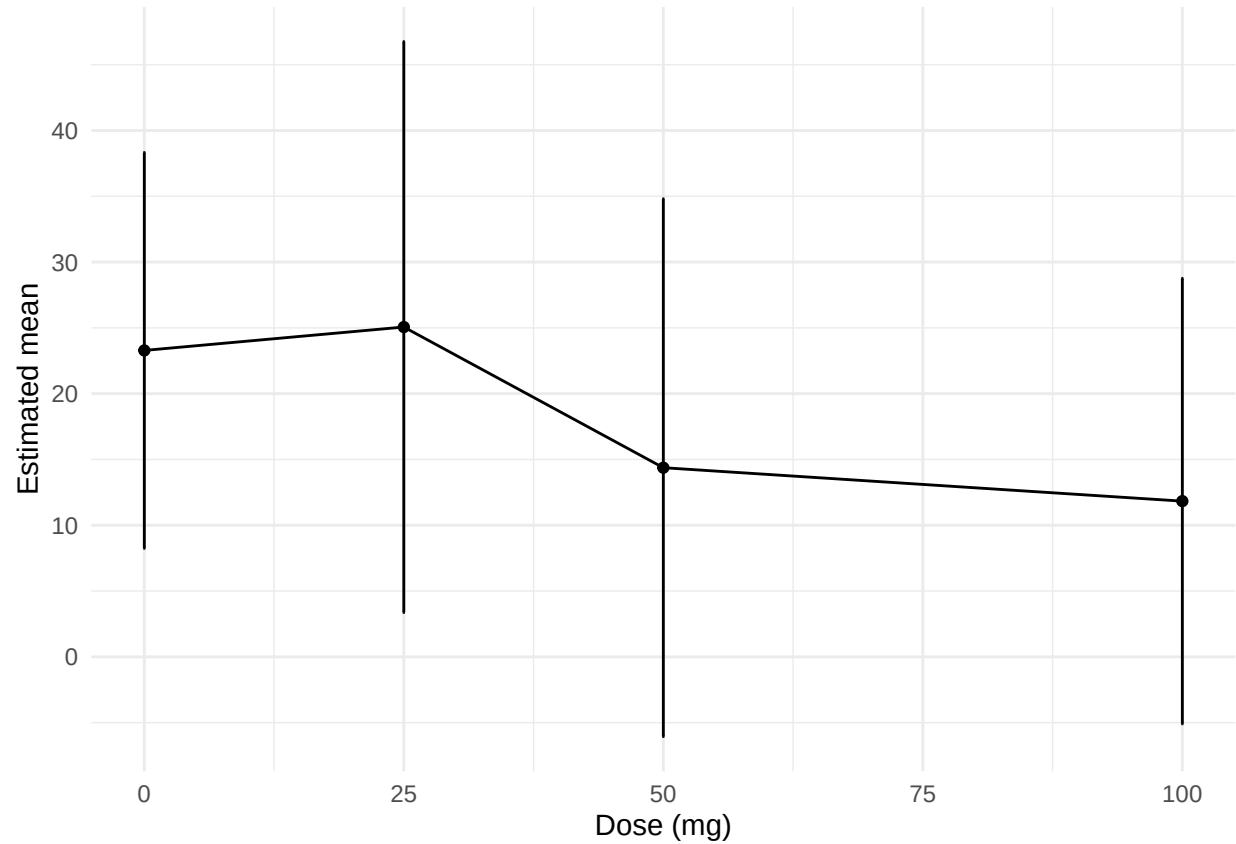
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	23.29 (8.25 to 38.32)
losartan 25 mg	25.06 (3.38 to 46.75)
losartan 50 mg	14.37 (-6.05 to 34.80)
losartan 100 mg	11.83 (-5.09 to 28.75)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	1.78 (-30.55 to 34.11)
losartan 50 mg - losartan 0 mg	-8.91 (-39.97 to 22.15)
losartan 100 mg - losartan 0 mg	-11.45 (-39.02 to 16.11)

Plot



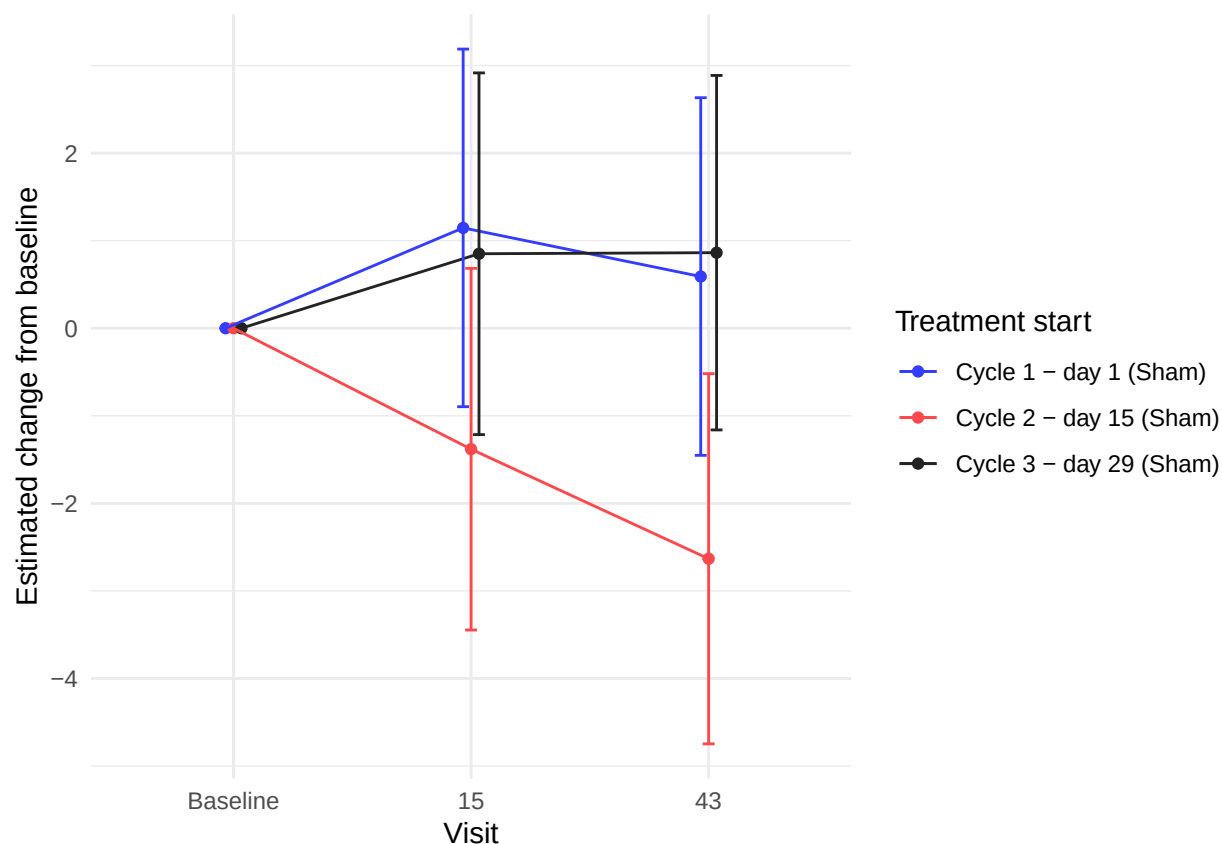
Neurologic Performance

Change from baseline in KPS, ECOG, and NANO scores at Days 15/43 and model-based estimates at Days 155/239 are reported below for the newly diagnosed glioblastoma cohort. Note that since the day 29 is not measured for any of the measures, the dose-response analyses are based on Days 15 and 43 only for the early timepoints. There might therefore be a limited ability to detect dose-response relationships at these early visits because we are lacking one contrast.

Karnofsky Performance Scale

A total of 54 subjects were included in the analysis for this endpoint.

Line plot by treatment start



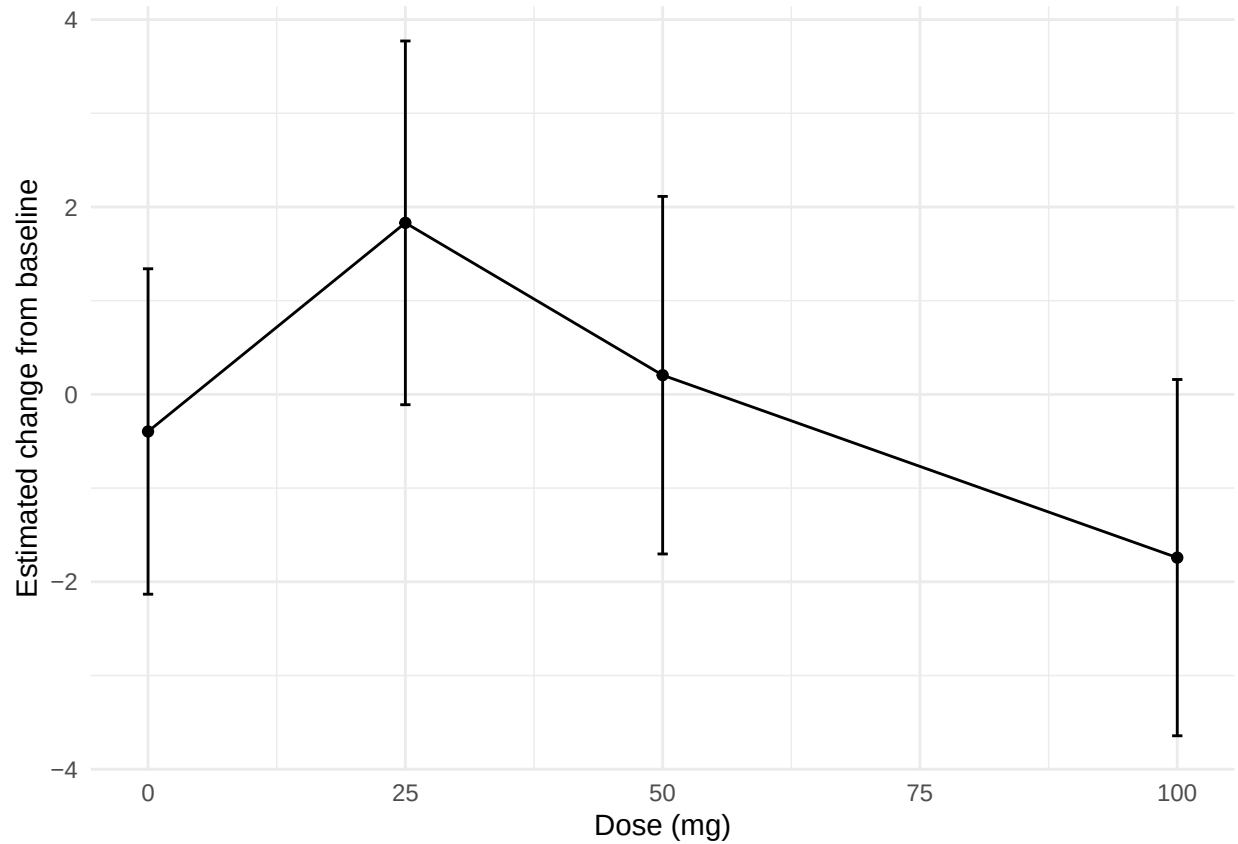
Days 15/43 (change from baseline)

Dose (mg)	Estimated change from baseline (95% CI)
0	-0.40 (-2.13 to 1.34)
25	1.83 (-0.11 to 3.77)
50	0.21 (-1.70 to 2.11)
100	-1.74 (-3.64 to 0.16)

Differences vs 0 mg

Comparison	Difference (95% CI)
losartan 25 mg vs losartan 0 mg	2.23 (-0.81 to 5.27)
losartan 50 mg vs losartan 0 mg	0.60 (-2.44 to 3.64)
losartan 100 mg vs losartan 0 mg	-1.35 (-4.35 to 1.66)

Plot



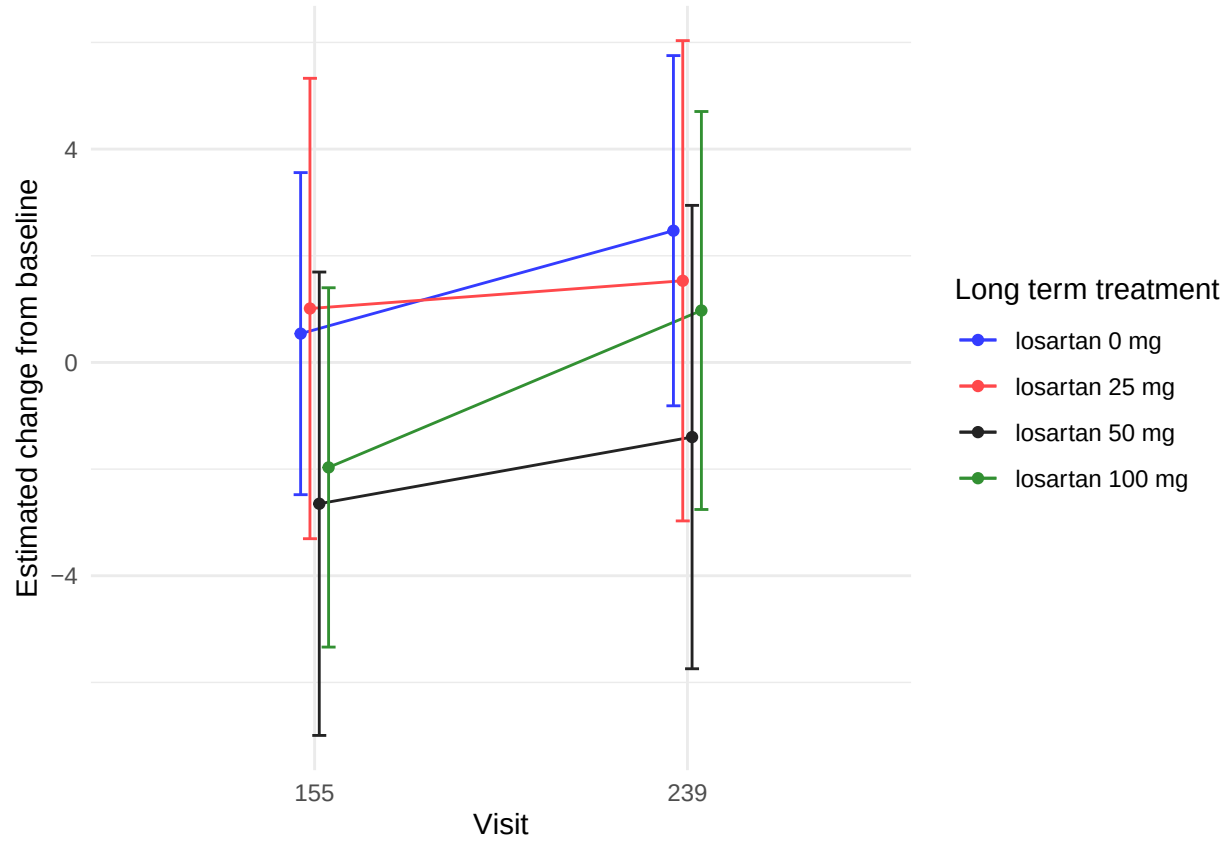
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	0.54 (-2.48 to 3.56)
155	losartan 25 mg	1.01 (-3.31 to 5.33)
155	losartan 50 mg	-2.65 (-6.99 to 1.69)
155	losartan 100 mg	-1.97 (-5.34 to 1.40)
239	losartan 0 mg	2.47 (-0.81 to 5.75)
239	losartan 25 mg	1.53 (-2.97 to 6.03)
239	losartan 50 mg	-1.40 (-5.74 to 2.94)
239	losartan 100 mg	0.97 (-2.76 to 4.70)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	0.47 (-5.88 to 6.83)
155	losartan 50 mg - losartan 0 mg	-3.19 (-9.61 to 3.23)
155	losartan 100 mg - losartan 0 mg	-2.51 (-7.97 to 2.96)
239	losartan 25 mg - losartan 0 mg	-0.94 (-7.66 to 5.78)
239	losartan 50 mg - losartan 0 mg	-3.87 (-10.47 to 2.73)
239	losartan 100 mg - losartan 0 mg	-1.50 (-7.50 to 4.50)

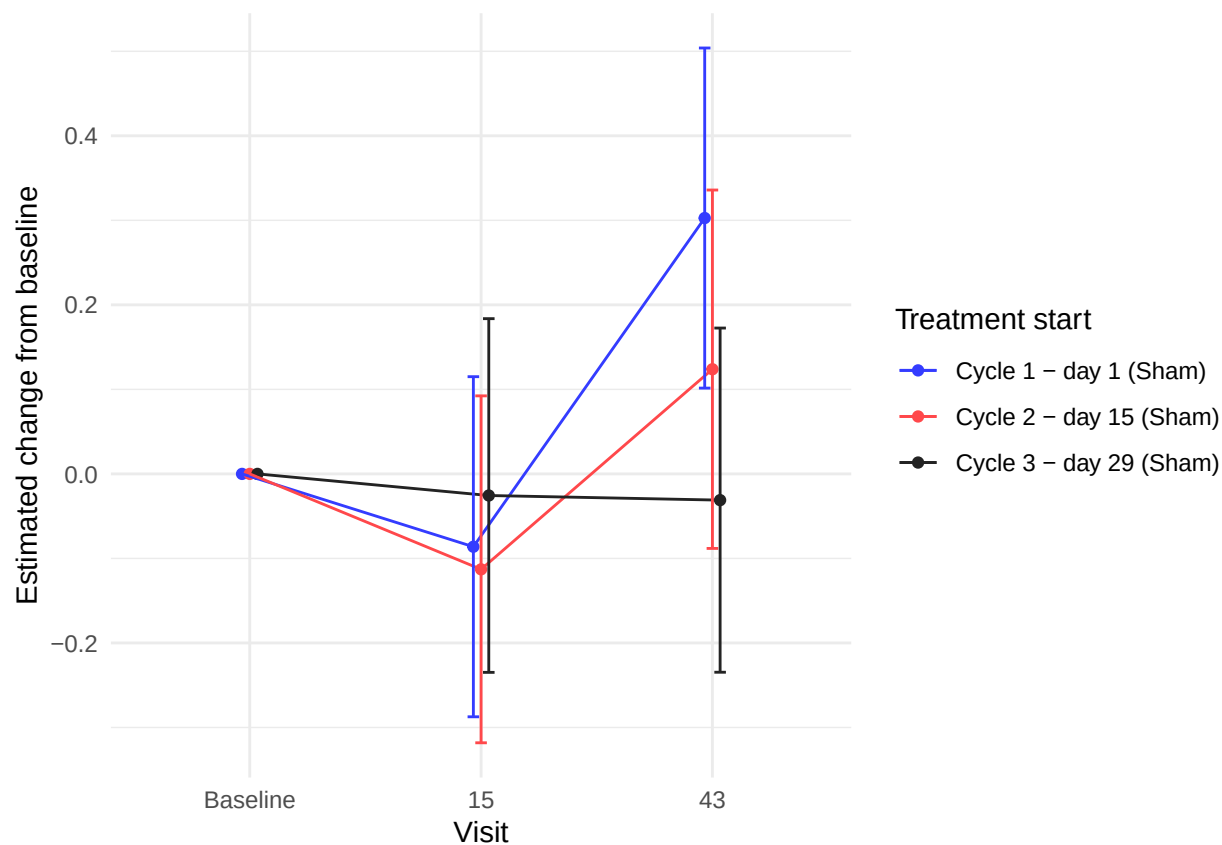
Plot



ECOG Score

A total of 54 subjects were included in the analysis for this endpoint.

Line plot by treatment start



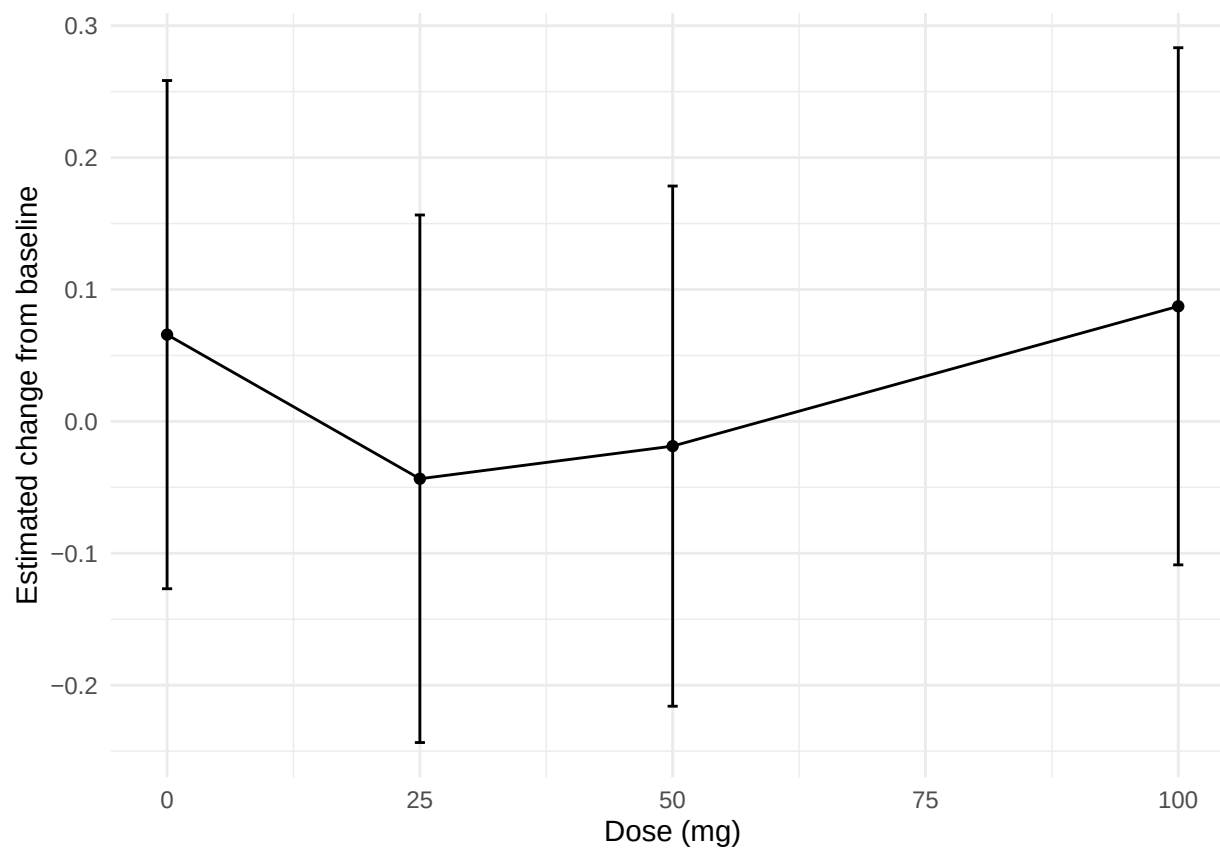
Days 15/43 (change from baseline)

Dose (mg)	Estimated change from baseline (95% CI)
0	0.07 (-0.13 to 0.26)
25	-0.04 (-0.24 to 0.16)
50	-0.02 (-0.22 to 0.18)
100	0.09 (-0.11 to 0.28)

Differences vs 0 mg

Comparison	Difference (95% CI)
losartan 25 mg vs losartan 0 mg	-0.11 (-0.47 to 0.25)
losartan 50 mg vs losartan 0 mg	-0.08 (-0.44 to 0.27)
losartan 100 mg vs losartan 0 mg	0.02 (-0.34 to 0.38)

Plot



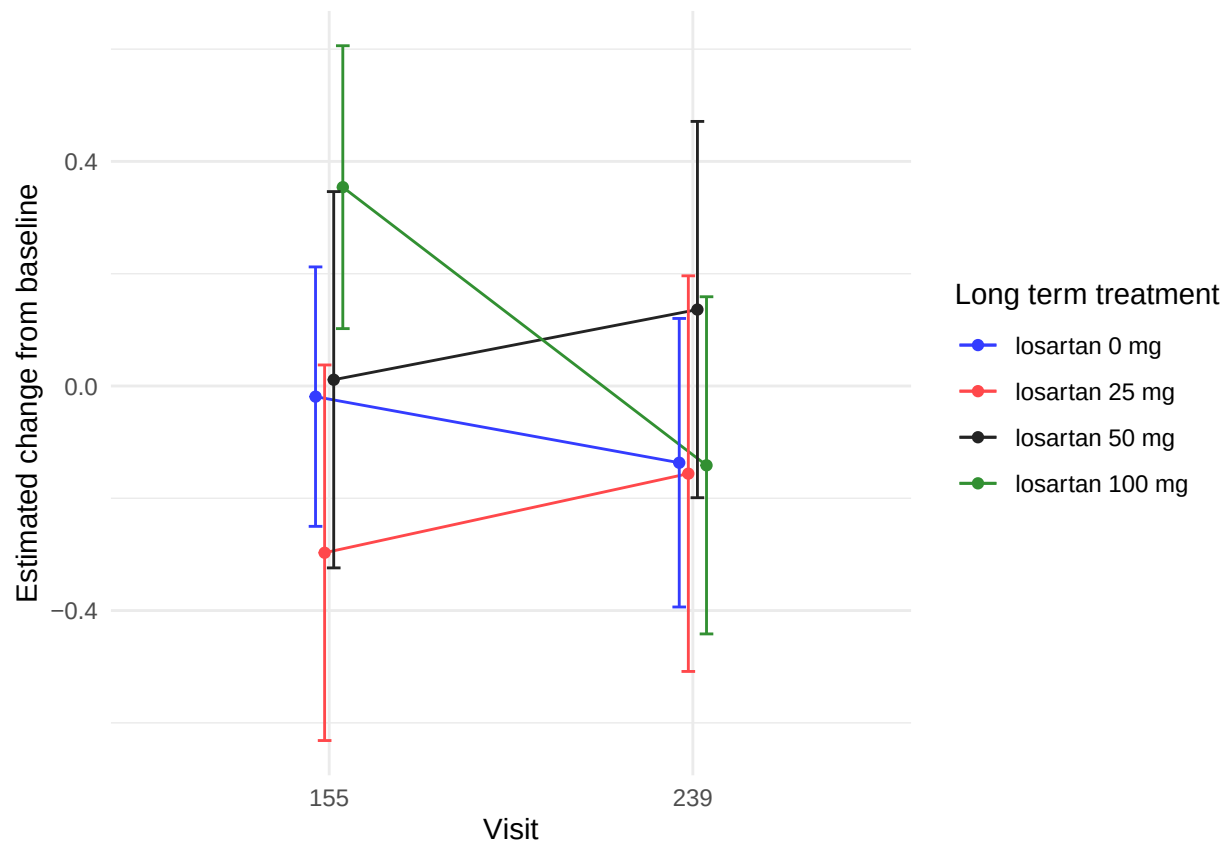
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	-0.02 (-0.25 to 0.21)
155	losartan 25 mg	-0.30 (-0.63 to 0.04)
155	losartan 50 mg	0.01 (-0.32 to 0.35)
155	losartan 100 mg	0.35 (0.10 to 0.61)
239	losartan 0 mg	-0.14 (-0.39 to 0.12)
239	losartan 25 mg	-0.16 (-0.51 to 0.20)
239	losartan 50 mg	0.14 (-0.20 to 0.47)
239	losartan 100 mg	-0.14 (-0.44 to 0.16)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	-0.28 (-0.77 to 0.21)
155	losartan 50 mg - losartan 0 mg	0.03 (-0.46 to 0.52)
155	losartan 100 mg - losartan 0 mg	0.37 (-0.04 to 0.79)
239	losartan 25 mg - losartan 0 mg	-0.02 (-0.54 to 0.51)
239	losartan 50 mg - losartan 0 mg	0.27 (-0.24 to 0.78)
239	losartan 100 mg - losartan 0 mg	0.00 (-0.48 to 0.47)

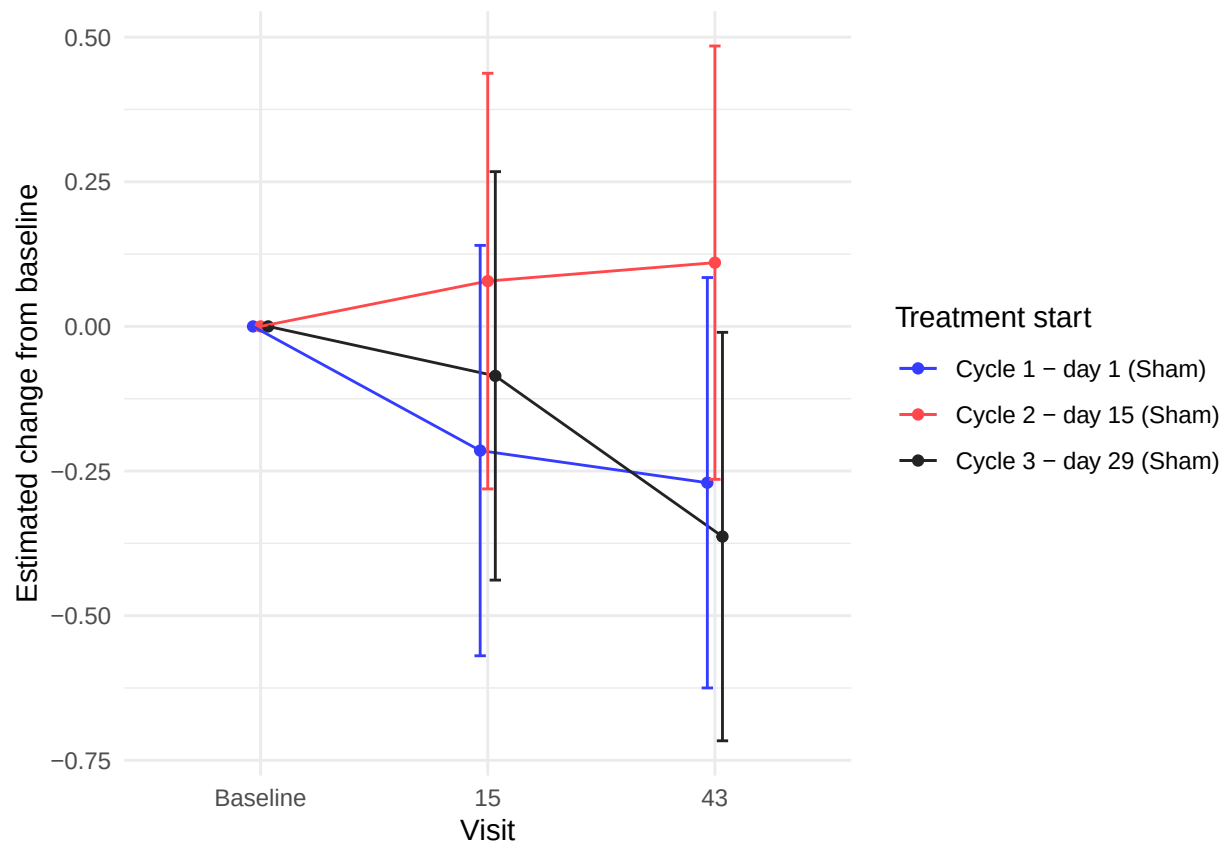
Plot



Total NANO score

A total of 54 subjects were included in the analysis for this endpoint.

Line plot by treatment start



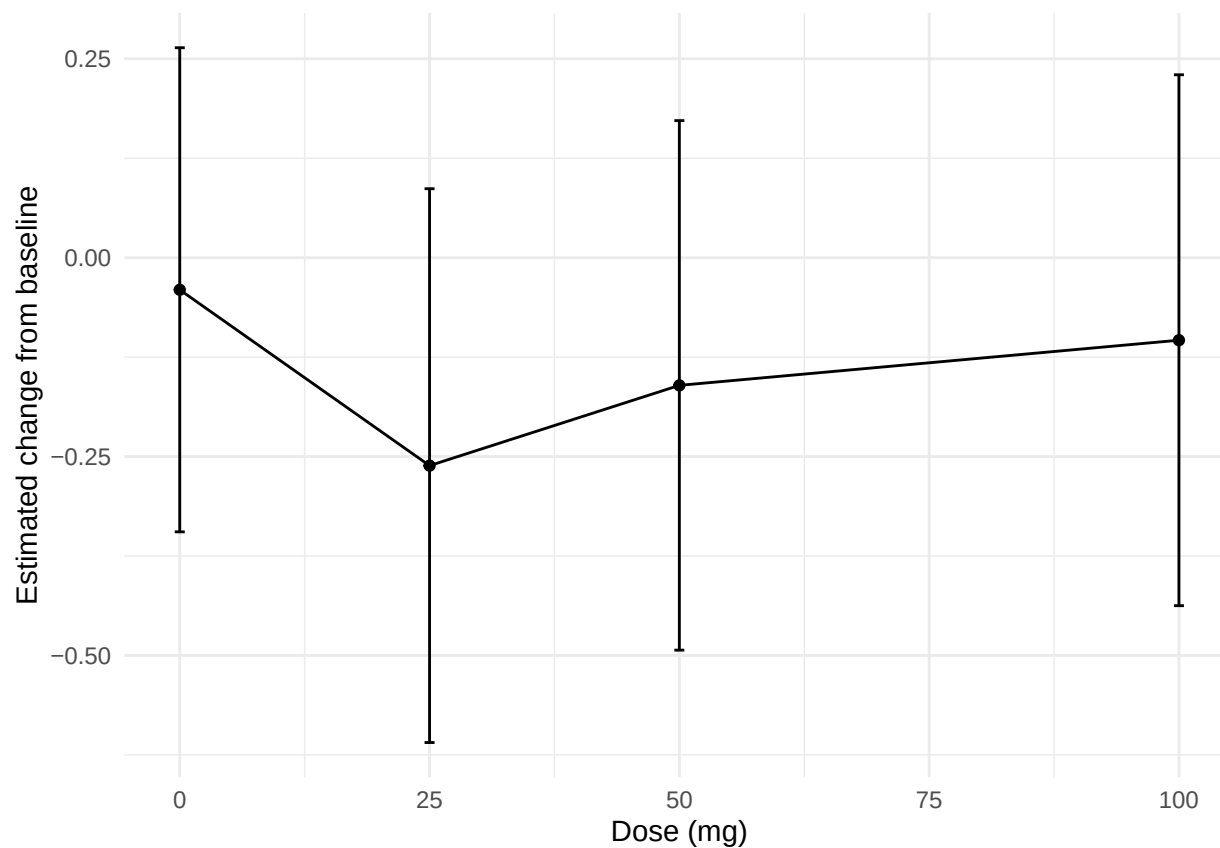
Days 15/43 (change from baseline)

Dose (mg)	Estimated change from baseline (95% CI)
0	-0.04 (-0.34 to 0.26)
25	-0.26 (-0.61 to 0.09)
50	-0.16 (-0.49 to 0.17)
100	-0.10 (-0.44 to 0.23)

Differences vs 0 mg

Comparison	Difference (95% CI)
losartan 25 mg vs losartan 0 mg	-0.22 (-0.77 to 0.33)
losartan 50 mg vs losartan 0 mg	-0.12 (-0.66 to 0.42)
losartan 100 mg vs losartan 0 mg	-0.06 (-0.60 to 0.48)

Plot



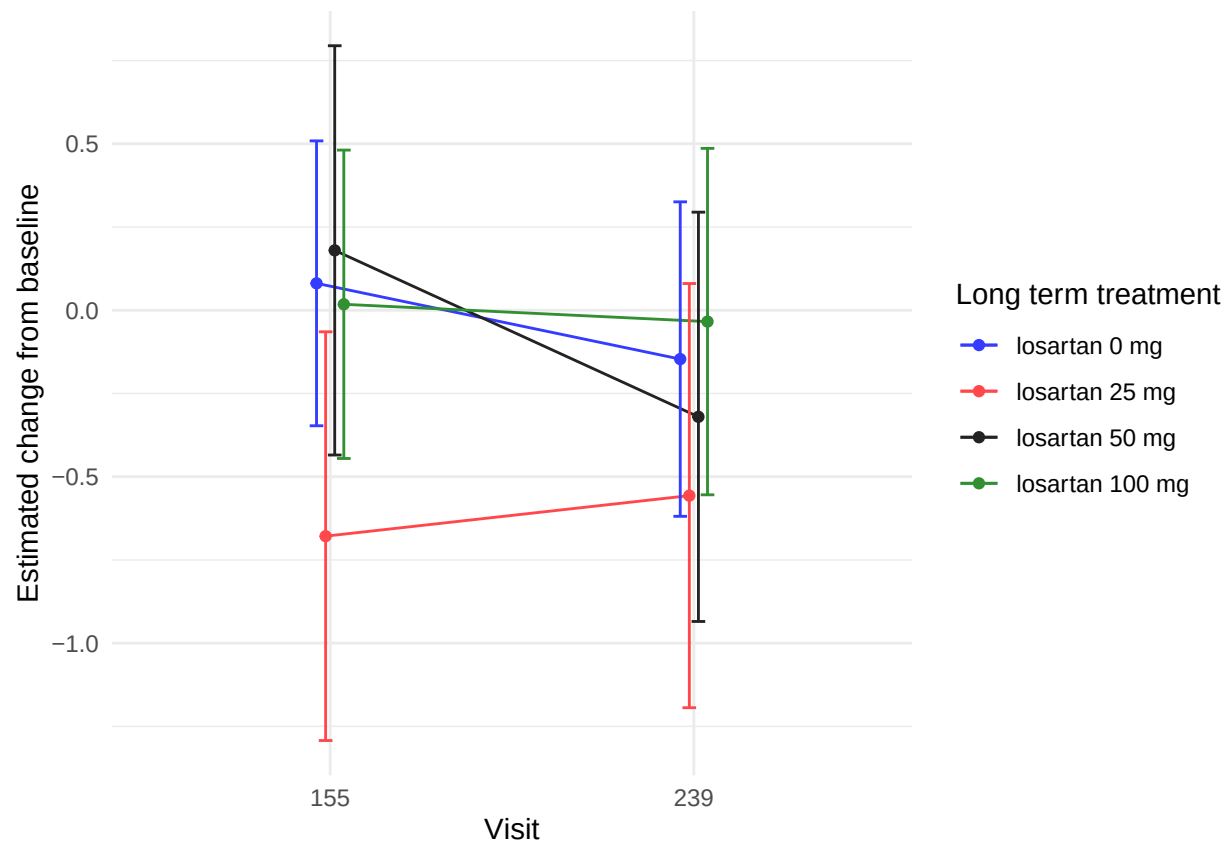
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	0.08 (-0.35 to 0.51)
155	losartan 25 mg	-0.68 (-1.29 to -0.06)
155	losartan 50 mg	0.18 (-0.43 to 0.79)
155	losartan 100 mg	0.02 (-0.45 to 0.48)
239	losartan 0 mg	-0.15 (-0.62 to 0.33)
239	losartan 25 mg	-0.56 (-1.19 to 0.08)
239	losartan 50 mg	-0.32 (-0.93 to 0.29)
239	losartan 100 mg	-0.03 (-0.55 to 0.49)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	-0.76 (-1.66 to 0.14)
155	losartan 50 mg - losartan 0 mg	0.10 (-0.81 to 1.00)
155	losartan 100 mg - losartan 0 mg	-0.06 (-0.82 to 0.70)
239	losartan 25 mg - losartan 0 mg	-0.41 (-1.37 to 0.55)
239	losartan 50 mg - losartan 0 mg	-0.17 (-1.11 to 0.76)
239	losartan 100 mg - losartan 0 mg	0.11 (-0.73 to 0.96)

Plot



Quality of Life Outcomes

Model based estimates of change from baseline in QLQ-C30 and QLQ-BN20 scores at Days 155/239 are reported below for the newly diagnosed glioblastoma cohort. Note that even if the QoL outcomes were collected at Day 43 it is not possible to make a meaningful analysis of the effect of losartan as all participants received the treatment at that time point.

EORTC QLQ-C30 score

A total of 19 subjects were included in the analysis for this endpoint.

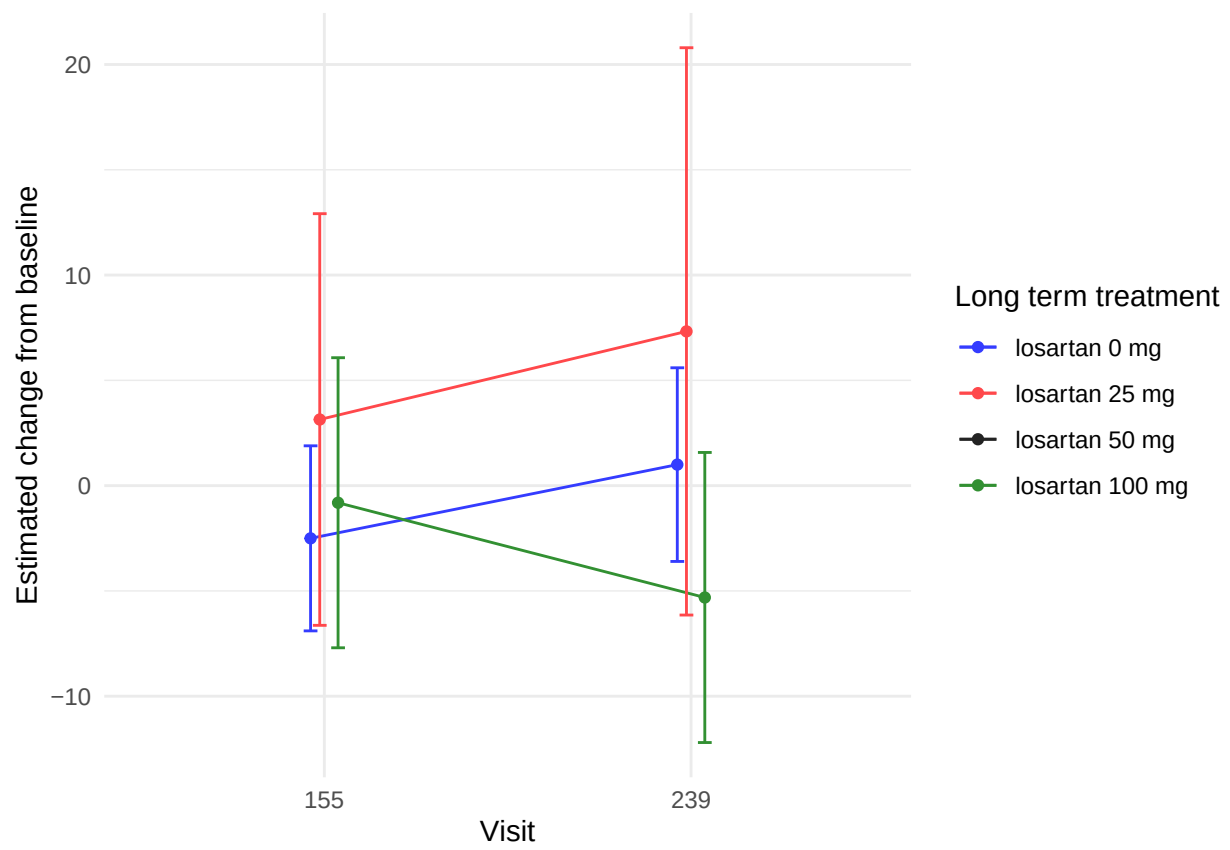
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	-2.50 (-6.90 to 1.89)
155	losartan 25 mg	3.14 (-6.63 to 12.91)
155	losartan 50 mg	NA (NA to NA)
155	losartan 100 mg	-0.81 (-7.70 to 6.08)
239	losartan 0 mg	1.00 (-3.60 to 5.60)
239	losartan 25 mg	7.33 (-6.15 to 20.80)
239	losartan 50 mg	NA (NA to NA)
239	losartan 100 mg	-5.31 (-12.20 to 1.58)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	5.64 (-6.62 to 17.90)
155	losartan 50 mg - losartan 0 mg	NA (NA to NA)
155	losartan 100 mg - losartan 0 mg	1.69 (-7.69 to 11.07)
239	losartan 25 mg - losartan 0 mg	6.33 (-9.87 to 22.53)
239	losartan 50 mg - losartan 0 mg	NA (NA to NA)
239	losartan 100 mg - losartan 0 mg	-6.31 (-15.81 to 3.19)

Plot



EORTC QLQ-BN20 score

A total of 18 subjects were included in the analysis for this endpoint.

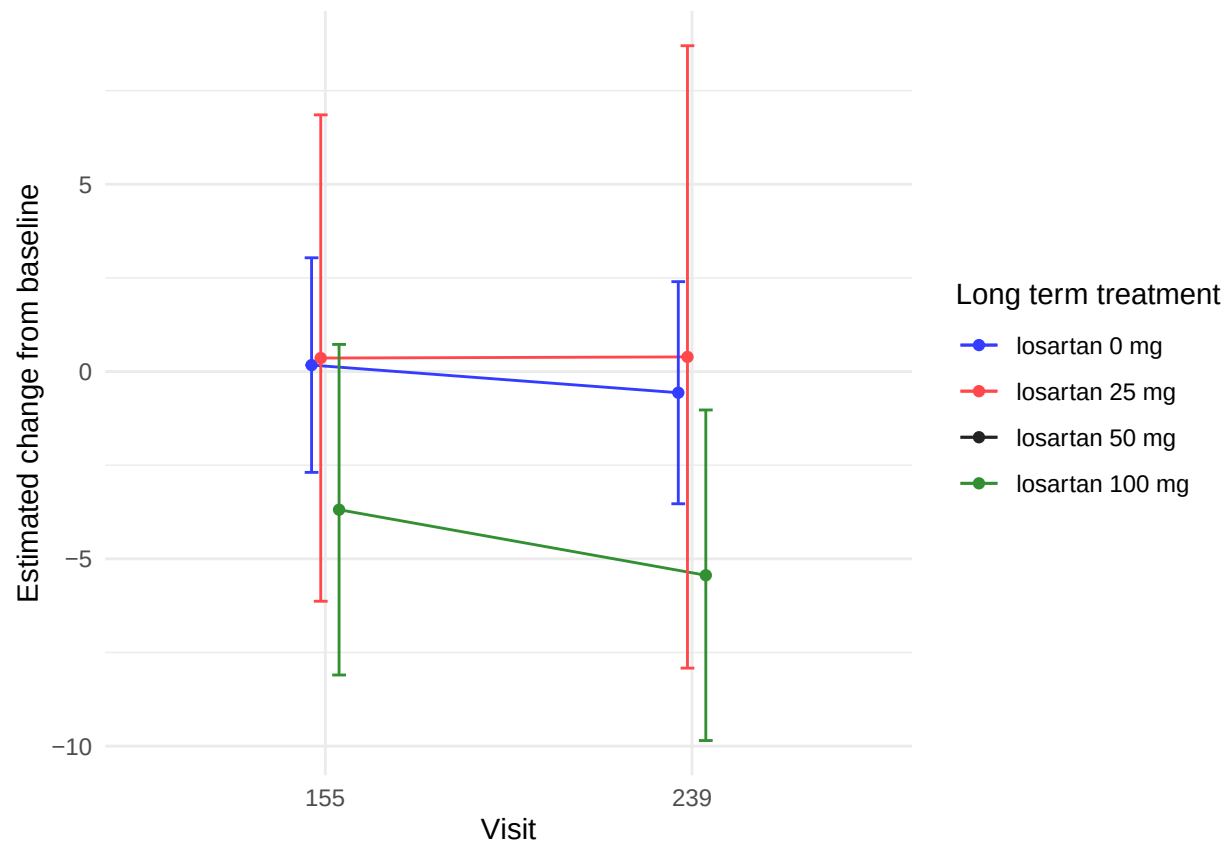
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	0.17 (-2.69 to 3.04)
155	losartan 25 mg	0.36 (-6.13 to 6.85)
155	losartan 50 mg	NA (NA to NA)
155	losartan 100 mg	-3.69 (-8.10 to 0.72)
239	losartan 0 mg	-0.57 (-3.53 to 2.40)
239	losartan 25 mg	0.39 (-7.92 to 8.70)
239	losartan 50 mg	NA (NA to NA)
239	losartan 100 mg	-5.44 (-9.85 to -1.03)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	0.19 (-7.96 to 8.33)
155	losartan 50 mg - losartan 0 mg	NA (NA to NA)
155	losartan 100 mg - losartan 0 mg	-3.86 (-9.87 to 2.15)
239	losartan 25 mg - losartan 0 mg	0.96 (-9.13 to 11.04)
239	losartan 50 mg - losartan 0 mg	NA (NA to NA)
239	losartan 100 mg - losartan 0 mg	-4.87 (-10.95 to 1.20)

Plot



Steroid Use

Binary steroid use is analysed with longitudinal logistic regression and population-averaged marginal risk differences for early (Days 15/29/43) and late (Days 155/239) visits in the newly diagnosed glioblastoma cohort.

Steroid use

A total of 54 subjects were included in the analysis.

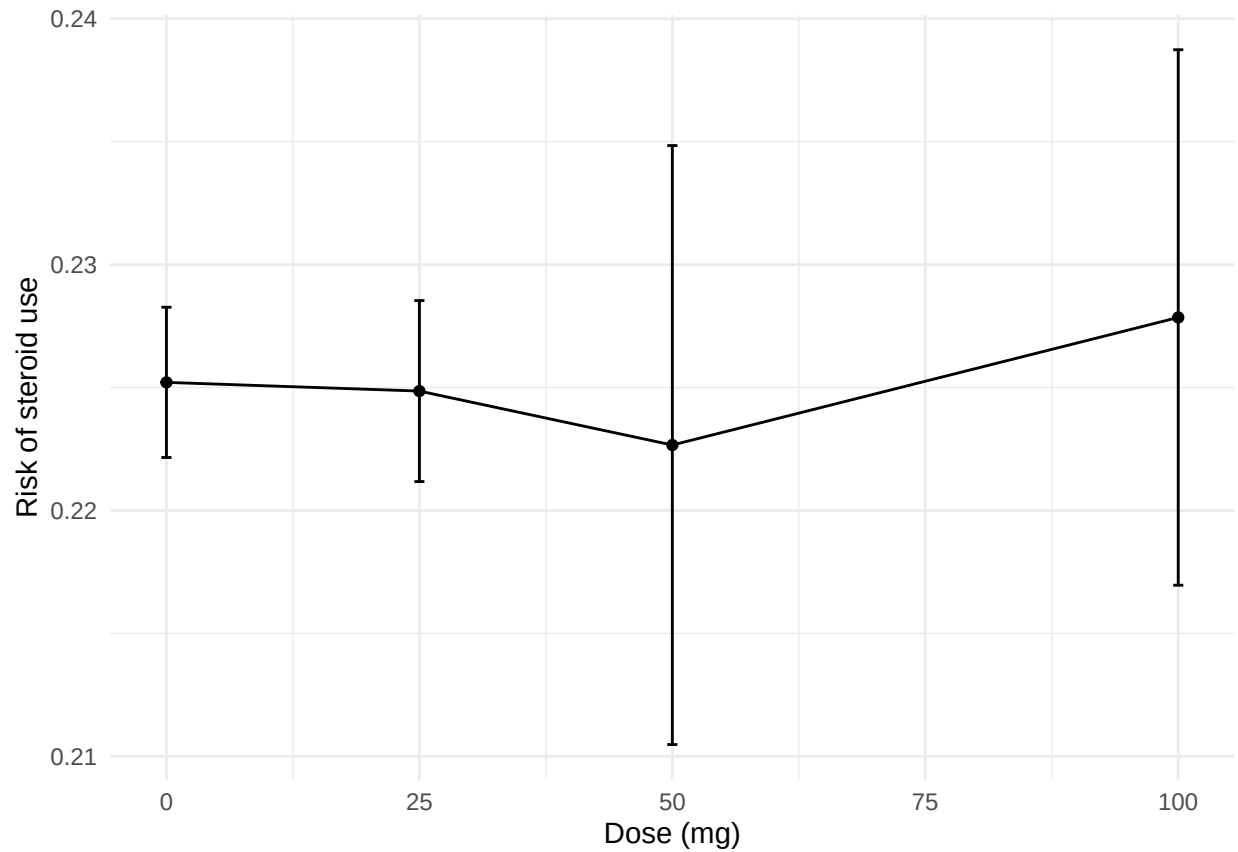
Days 15/29/43 (by dose)

Treatment	Risk (95% CI)
losartan 0 mg	0.225 (0.222 to 0.228)
losartan 25 mg	0.225 (0.221 to 0.229)
losartan 50 mg	0.223 (0.210 to 0.235)
losartan 100 mg	0.228 (0.217 to 0.239)

Risk differences vs 0 mg

Comparison	Risk difference (95% CI)
100 - 0	0.003 (-0.008 to 0.013)
25 - 0	0.000 (-0.004 to 0.004)
50 - 0	-0.003 (-0.015 to 0.010)

Plot



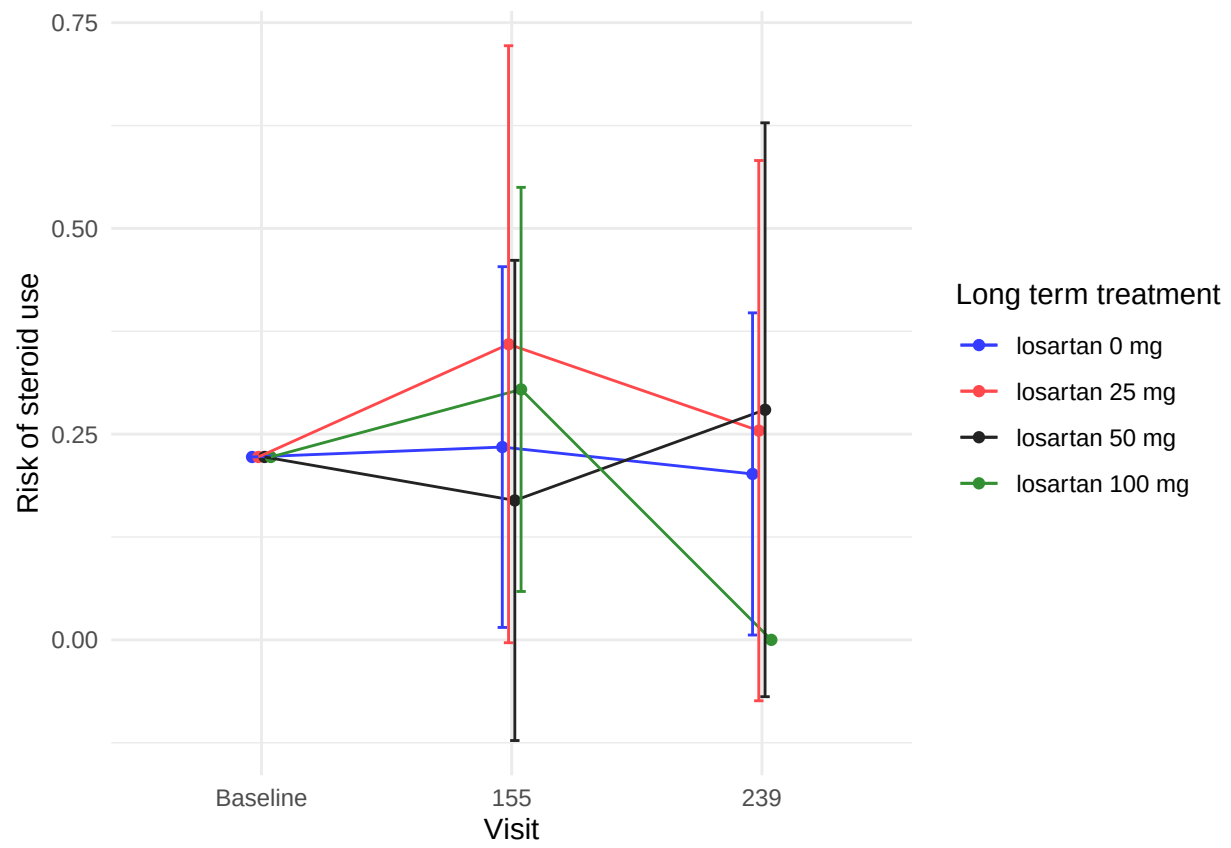
Days 155/239 (population-averaged risk)

Visit	Treatment	Risk (95% CI)
155	losartan 0 mg	0.234 (0.015 to 0.453)
239	losartan 0 mg	0.202 (0.006 to 0.397)
155	losartan 25 mg	0.359 (-0.004 to 0.722)
239	losartan 25 mg	0.254 (-0.074 to 0.582)
155	losartan 50 mg	0.169 (-0.122 to 0.461)
239	losartan 50 mg	0.280 (-0.069 to 0.628)
155	losartan 100 mg	0.304 (0.059 to 0.550)
239	losartan 100 mg	0.000 (0.000 to 0.000)

Risk differences vs reference

Visit	Comparison	Risk difference (95% CI)
155	losartan 100 mg - losartan 0 mg	0.066 (-0.246 to 0.379)
155	losartan 25 mg - losartan 0 mg	0.117 (-0.278 to 0.513)
155	losartan 50 mg - losartan 0 mg	-0.063 (-0.409 to 0.283)
239	losartan 100 mg - losartan 0 mg	-0.219 (-0.412 to -0.026)
239	losartan 25 mg - losartan 0 mg	0.051 (-0.318 to 0.420)
239	losartan 50 mg - losartan 0 mg	0.075 (-0.303 to 0.454)

Plot



RANO Responses

Ordinal RANO outcomes are analysed using proportional odds models within the configured cohort. For State of disease according to the RANO criteria, visits at Days 141 and 225 are included; for Best overall radiographic response, only Day 225 is included. Proportional odds ratios and confidence intervals are reported below.

State of disease according to the RANO criteria

Visit 141

Comparison	Propoprtional Odds Ratio (95% CI)
losartan 25 vs 0 mg	4.313 (0.423 to 57.601)
losartan 50 vs 0 mg	2.198 (0.255 to 21.220)
losartan 100 vs 0 mg	0.363 (0.015 to 3.839)

Visit 225

Comparison	Propoprtional Odds Ratio (95% CI)
losartan 25 vs 0 mg	0.500 (0.014 to 17.472)
losartan 50 vs 0 mg	0.500 (0.032 to 6.959)
losartan 100 vs 0 mg	0.500 (0.014 to 17.472)

Best overall radiographic response

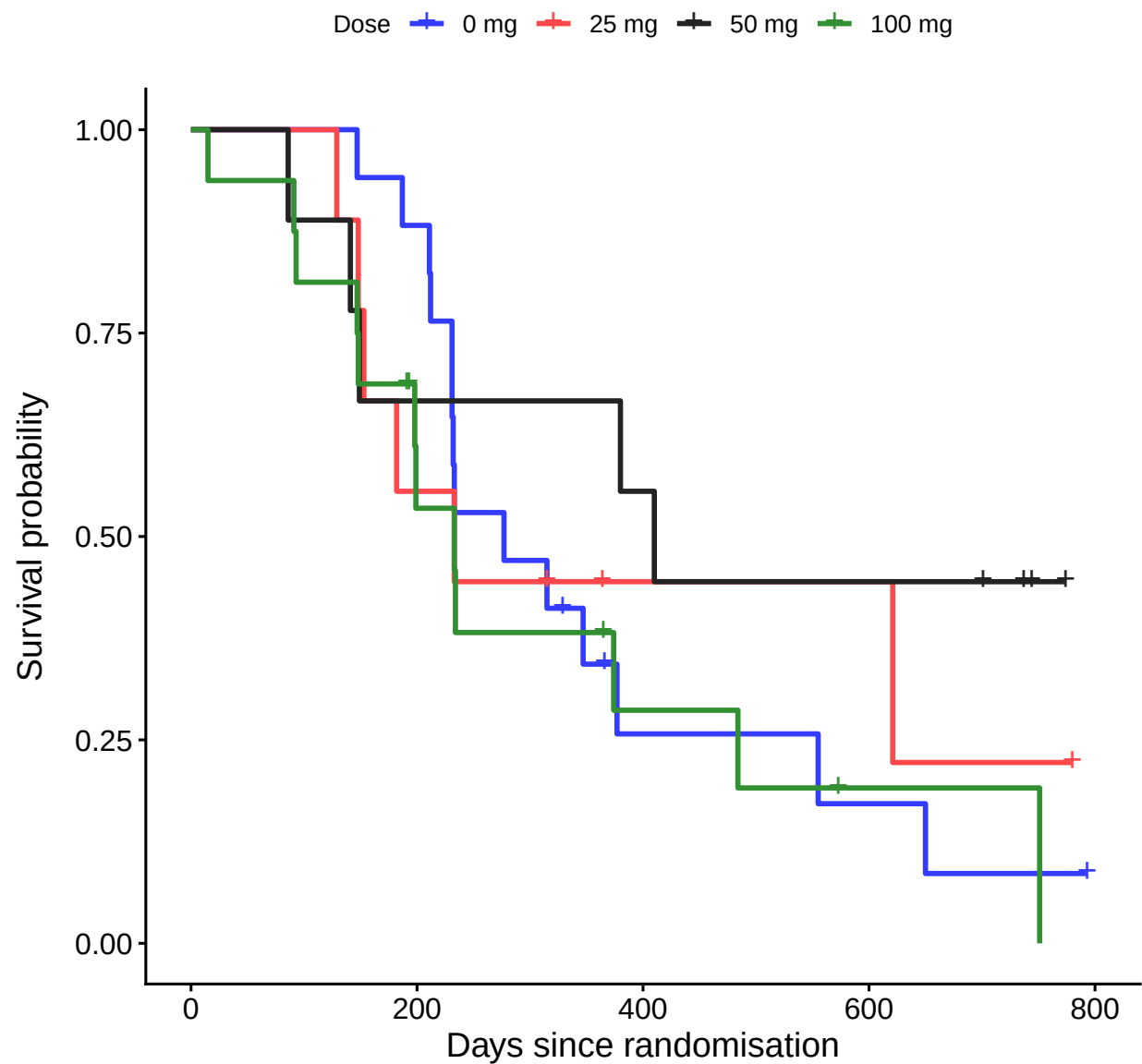
Visit 225

Comparison	Propoprtional Odds Ratio (95% CI)
losartan 25 mg vs 0 mg	0.200 (CI not available)
losartan 50 mg vs 0 mg	0.600 (CI not available)
losartan 100 mg vs 0 mg	3978.715 (CI not available)

Time-to-Event Endpoints

Kaplan-Meier curves are shown with the at-risk table below. Overall log-rank p-values are reported for each endpoint, followed by Cox regression hazard ratios versus 0 mg.

Progression-free survival



Number at risk

Dose	0	200	400	600	800
0 mg	17	15	3	2	0
25 mg	9	5	2	2	0
50 mg	9	6	5	4	0
100 mg	16	7	3	1	0

Days since randomisation

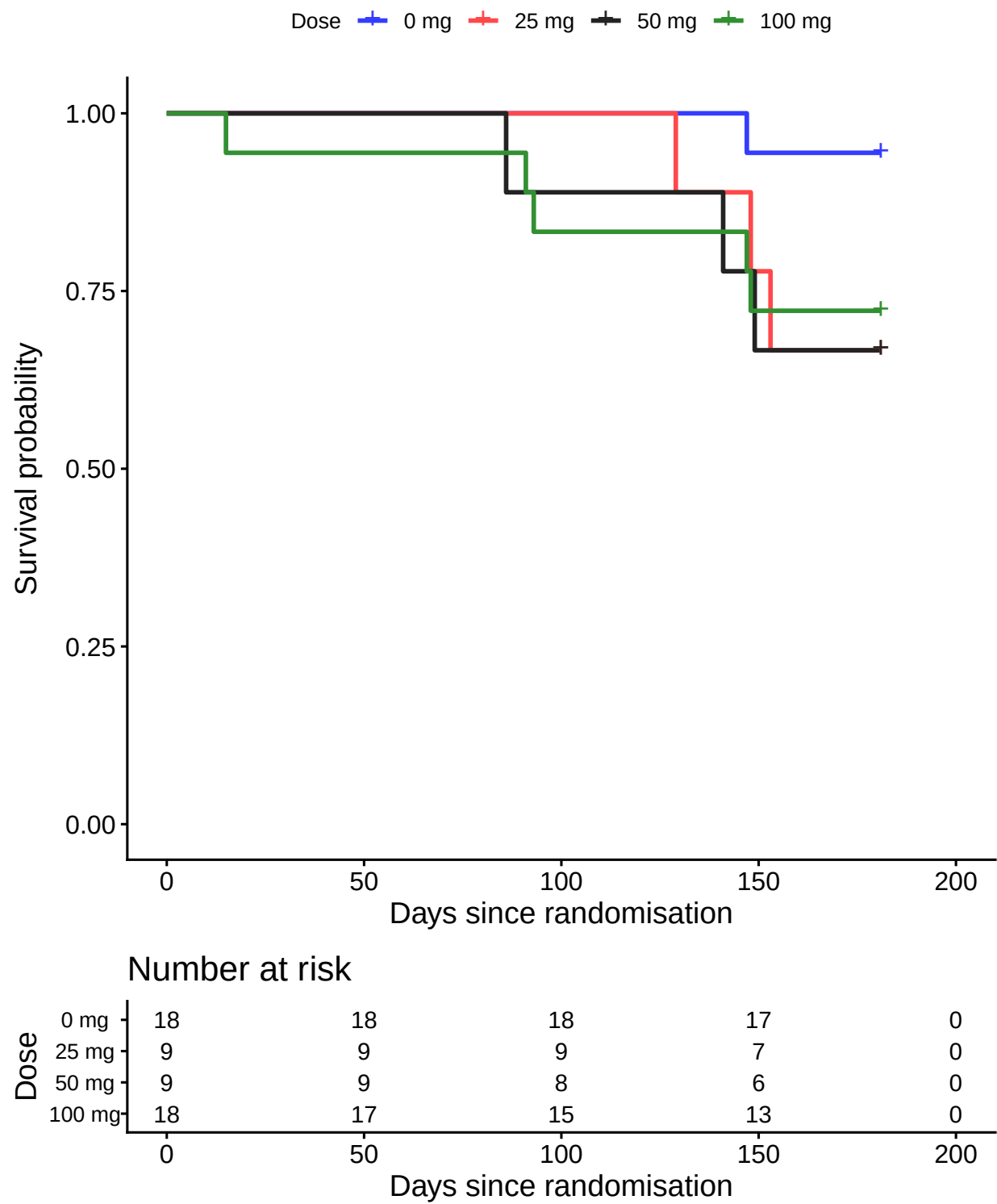
Log-rank p-value

Overall log-rank p-value: 0.422

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	0.88 (0.34 to 2.30)	0.792
50 mg vs 0 mg	0.50 (0.18 to 1.42)	0.195
100 mg vs 0 mg	1.21 (0.56 to 2.62)	0.629

6-month progression-free survival



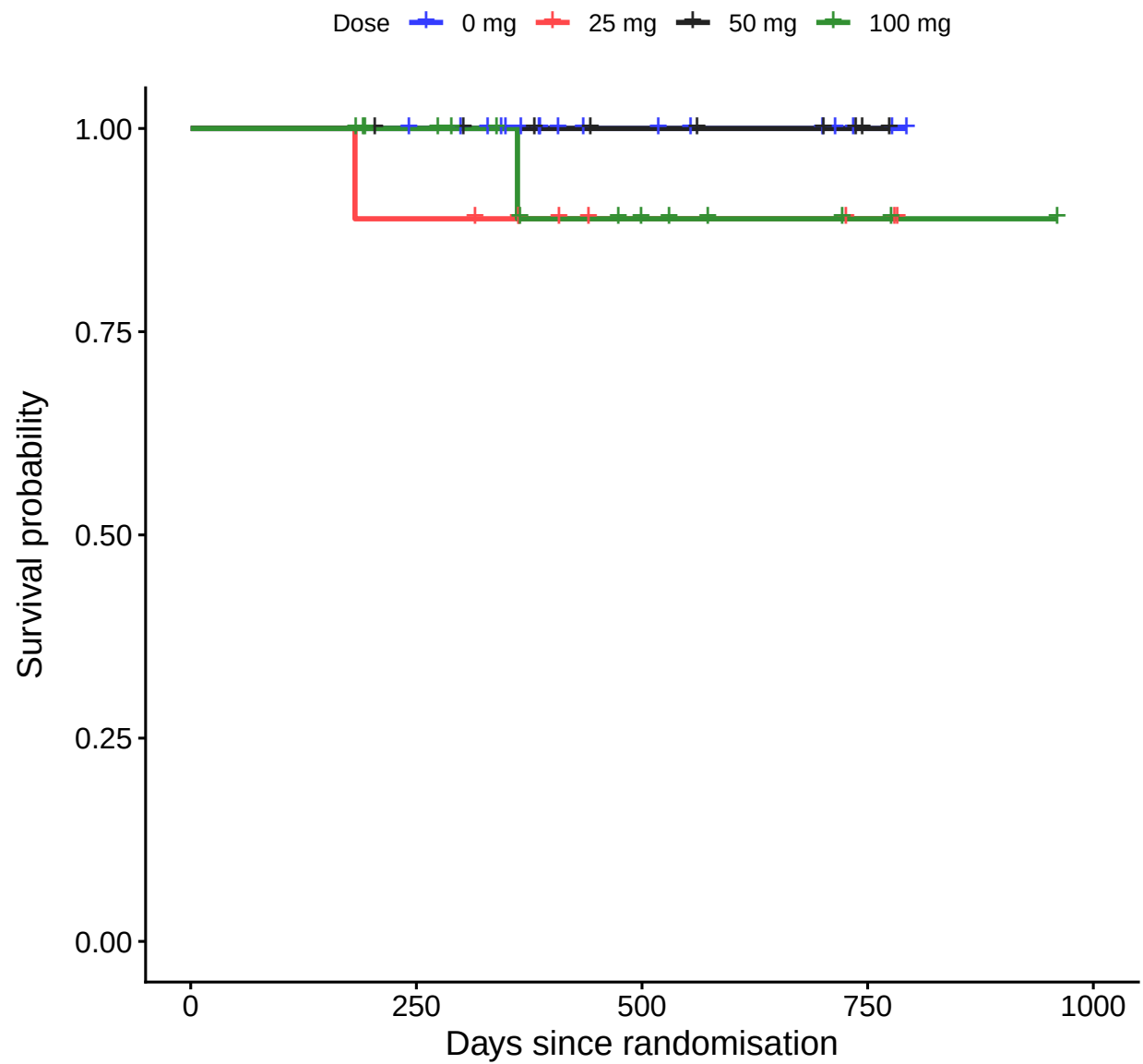
Log-rank p-value

Overall log-rank p-value: 0.239

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	6.48 (0.67 to 62.32)	0.106
50 mg vs 0 mg	7.03 (0.73 to 67.61)	0.091
100 mg vs 0 mg	5.89 (0.69 to 50.40)	0.106

Overall survival



Number at risk

Dose	200	400	600	800	1000
0 mg	17	9	5	0	0
25 mg	8	5	3	0	0
50 mg	9	6	4	0	0
100 mg	12	7	3	1	0

Days since randomisation

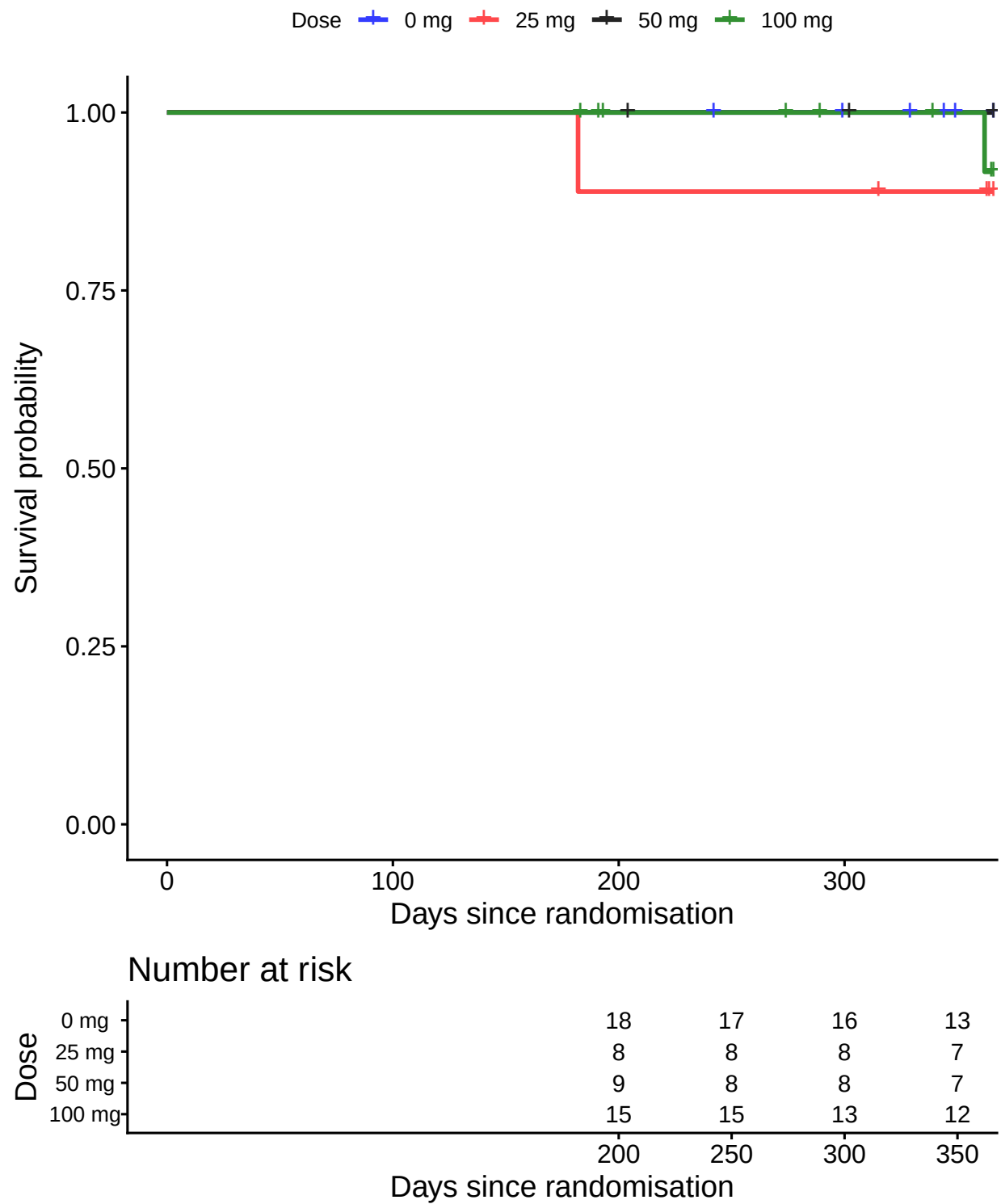
Log-rank p-value

Overall log-rank p-value: 0.489

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	926789447.29 (0.00 to Inf)	0.999
50 mg vs 0 mg	1.00 (0.00 to Inf)	1.000
100 mg vs 0 mg	633117722.45 (0.00 to Inf)	0.999

Overall survival at 12 months



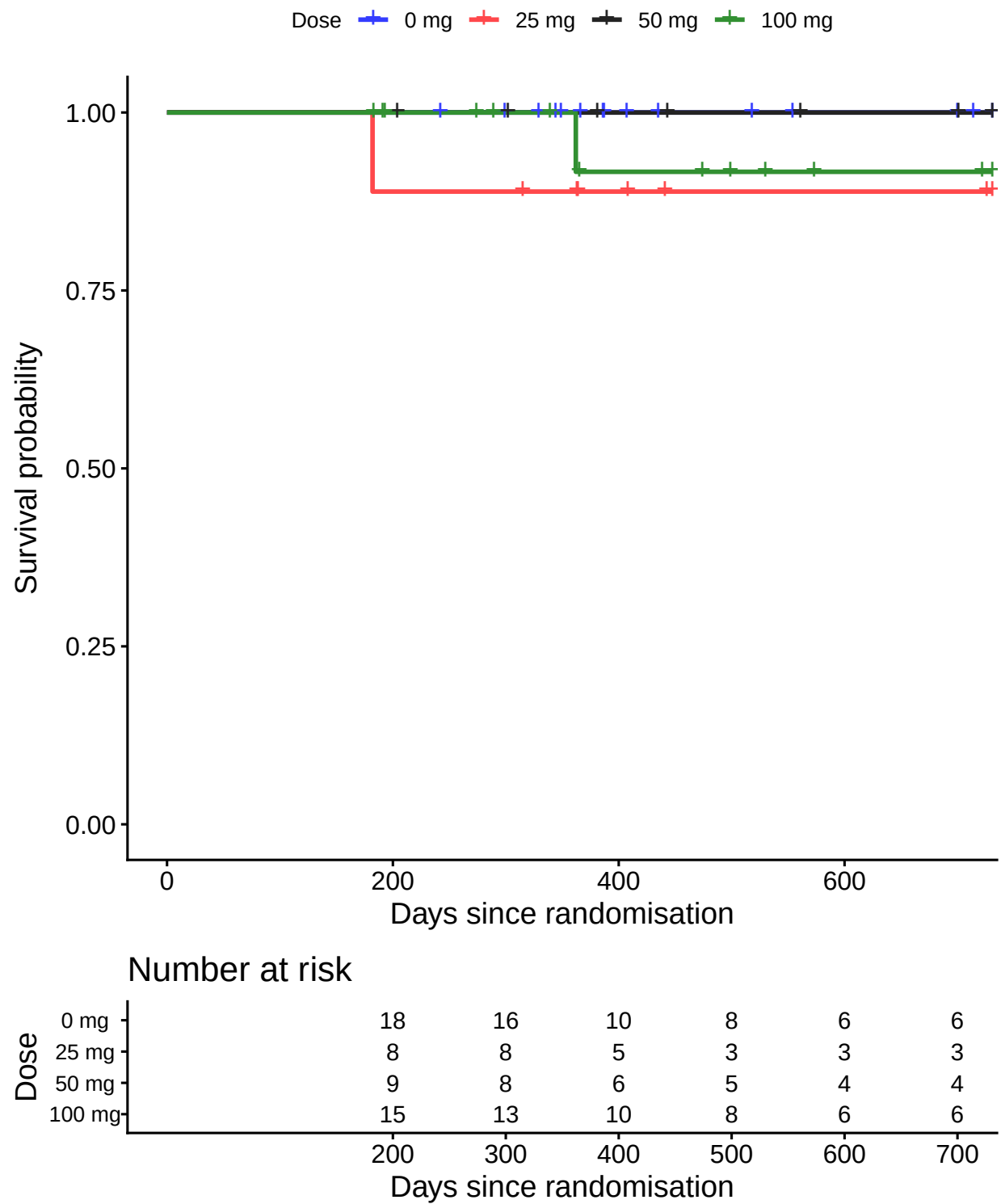
Log-rank p-value

Overall log-rank p-value: 0.485

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	1129678775.16 (0.00 to Inf)	0.999
50 mg vs 0 mg	1.00 (0.00 to Inf)	1.000
100 mg vs 0 mg	610096267.86 (0.00 to Inf)	0.999

Overall survival at 24 months



Log-rank p-value

Overall log-rank p-value: 0.485

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	1129678773.66 (0.00 to Inf)	0.999
50 mg vs 0 mg	1.00 (0.00 to Inf)	1.000
100 mg vs 0 mg	610096267.05 (0.00 to Inf)	0.999